

DOKUMENTASI ASAT PRAKTIK
KONSENTRASI KEAHLIAN TKJ
TAHUN 2026



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TEKNIK JARINGAN KOMPUTER DAN TELEKOMUNIKASI

SMK WIRKAMA BOGOR

KATA PENGANTAR

Puji syukur kehadiran Allah SWT karena atas rahmat dan karunia-Nya sehingga laporan dokumentasi proyek **Asesmen Sumatif Akhir Tahun (ASAT) Praktik Konsentrasi Keahlian TKJ Tahun 2026** ini dapat diselesaikan dengan baik.

Laporan ini disusun sebagai dokumentasi hasil pelaksanaan proyek praktik yang meliputi perancangan jaringan komputer, konfigurasi perangkat Mikrotik, implementasi VLAN, hotspot, firewall, manajemen bandwidth, serta implementasi layanan cloud menggunakan Amazon Web Services (AWS). Selain itu, laporan ini juga memuat proses deploy web server, konfigurasi database MariaDB, instalasi WordPress, serta penerapan keamanan layanan jaringan. Penyusunan laporan ini bertujuan untuk memenuhi tugas dan penilaian ASAT Praktik sekaligus sebagai bentuk penerapan kompetensi peserta didik dalam bidang jaringan komputer dan cloud computing sesuai dengan kebutuhan dunia kerja dan industri.

Penulis menyadari bahwa dalam penyusunan laporan ini masih terdapat kekurangan. Oleh karena itu, penulis mengharapkan kritik dan saran yang membangun demi perbaikan dan pengembangan di masa yang akan datang.

Penulis juga mengucapkan terima kasih kepada:

1. Allah SWT atas segala rahmat dan karunia-Nya.
2. Orang tua yang selalu memberikan doa dan dukungan.
3. Guru pembimbing yang telah memberikan arahan dan bimbingan.
4. Teman-teman yang telah membantu selama proses praktik dan penyusunan laporan.

Semoga laporan ini dapat memberikan manfaat serta menjadi bukti kompetensi dan pengalaman praktik di bidang Teknologi Jaringan Komputer dan Telekomunikasi.

Bogor, 2026

(Nama Siswa)

PENDAHULUAN

Perkembangan teknologi jaringan komputer dan cloud computing saat ini menuntut sumber daya manusia di bidang Teknologi Jaringan Komputer dan Telekomunikasi (TJKT) untuk memiliki kemampuan yang tidak hanya memahami teori, tetapi juga mampu mengimplementasikan langsung konfigurasi jaringan dan layanan server secara nyata. Oleh karena itu, kegiatan Asesmen Sumatif Akhir Tahun (ASAT) Praktik ini dilaksanakan sebagai bentuk evaluasi kompetensi peserta didik dalam menerapkan pengetahuan dan keterampilan di bidang administrasi jaringan komputer.

Projek praktik ini mengintegrasikan beberapa kompetensi utama, antara lain perencanaan kebutuhan jaringan, desain topologi jaringan, subnetting menggunakan metode VLSM, konfigurasi perangkat Mikrotik, VLAN, hotspot login, firewall, manajemen bandwidth, serta implementasi layanan cloud computing menggunakan Amazon Web Services (AWS). Selain itu, peserta didik juga dituntut mampu melakukan deploy web server, konfigurasi database MariaDB, instalasi WordPress, serta penerapan keamanan layanan jaringan (hardening service).

Dalam projek ini, peserta didik berperan sebagai seorang Network Administrator di PT TKJ Network Solution yang bertugas merancang dan membangun infrastruktur jaringan sesuai kebutuhan perusahaan. Infrastruktur tersebut harus mampu mendukung komunikasi antar divisi, akses internet, layanan hotspot, serta layanan website berbasis cloud yang dapat diakses secara publik.

Melalui kegiatan praktik ini diharapkan peserta didik mampu:

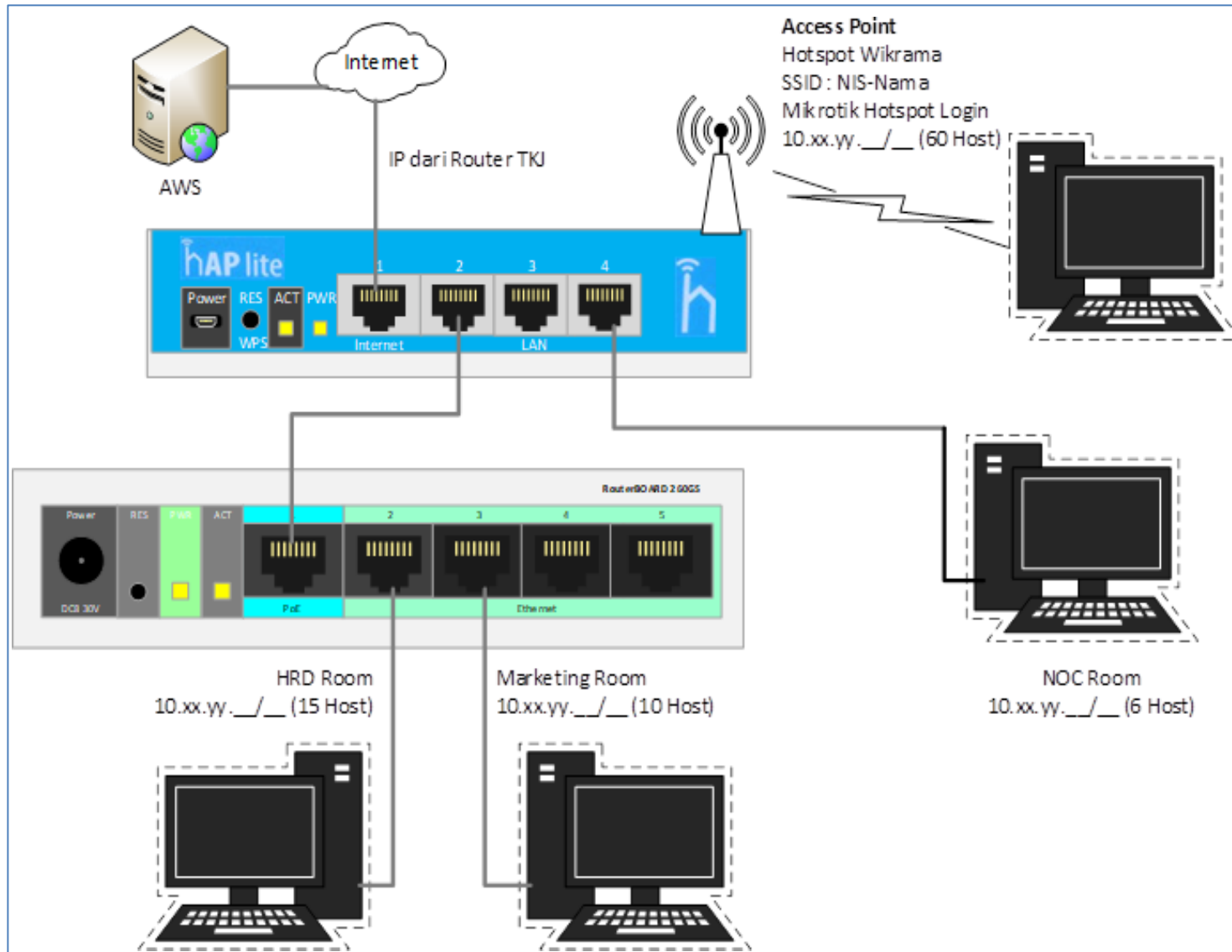
- Menganalisis kebutuhan jaringan komputer
- Merancang topologi jaringan sesuai kebutuhan perusahaan
- Melakukan konfigurasi jaringan menggunakan Mikrotik
- Mengimplementasikan layanan cloud server menggunakan AWS EC2
- Menerapkan keamanan jaringan dan server
- Melakukan pengujian dan dokumentasi hasil konfigurasi

Laporan dokumentasi ini dibuat sebagai bentuk pertanggungjawaban atas seluruh proses perancangan, implementasi, pengujian, dan analisis jaringan yang telah dilakukan selama kegiatan praktik berlangsung. Selain itu, laporan ini diharapkan dapat menjadi bukti pencapaian kompetensi

peserta didik dalam bidang jaringan komputer dan cloud computing sesuai kebutuhan dunia kerja dan industri saat ini.

RENCANA KEBUTUHAN

1. DESAIN TOPOLOGI JARINGAN



2. DAFTAR PERANGKAT JARINGAN

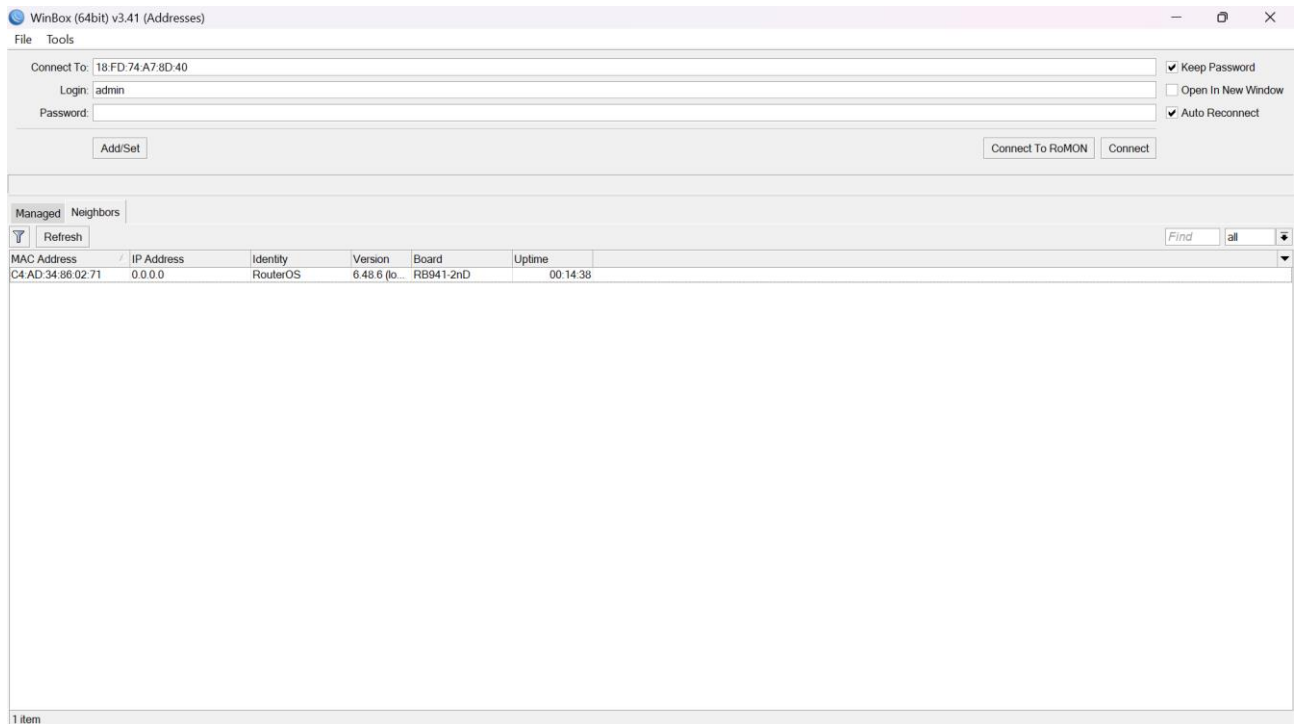
No	Nama Perangkat	Spesifikasi	Jumlah	Harga	Total	Keterangan
1	MikroTik Router hAP Lite	RB941-2nD, 4 Port Ethernet, Wireless 2.4 GHz	1 unit	Rp395.000	Rp395.000	Gateway Utama & Access Point Hotspot
2	MikroTik Switch	Switch RB260GS, 5 port Gigabit Ethernet	1 unit	Rp750.000	Rp750.000	Distribusi VLAN (HRD, Marketing, NOC)
3	Kabel LAN UTP	Cat5e / Cat6	2 Roll / Box	Rp500.000	Rp1.000.000	Kabel Penghubung Antar Perangkat & PC
4	LAN Converter	USB to RJ45 Ethernet Adapter	1 unit	Rp120.000	Rp120.000	Alat Konfigurasi Tambahan untuk Laptop
5	Konektor RJ45	Isi 50-100 pcs per pack	1 pack	Rp55.000	Rp55.000	Ujung Kabel LAN

3. RANCANGAN PENGALAMATAN JARINGAN

Divisi	Network	Prefix/CIDR	Range	Broadcast
Internet	172.16.247.0	/24	172.16.247.1 - 172.16.247.254	172.16.247.255
Hotspot Wikrama (60 host)	192.168.4.0	/26	192.168.4.1 – 192.168.4.62	192.168.4.63
HRD Room (10 host)	192.168.4.64	/27	192.168.4.65 – 192.168.4.94	192.168.4.95
Marketing Room (15 host)	192.168.4.96	/28	192.168.4.97 – 192.168.4.110	192.168.4.111
NOC Room (6 host)	192.168.4.112	/29	192.168.4.113 – 192.168.4.118	192.168.4.119

INSTALASI DAN KONFIGURASI

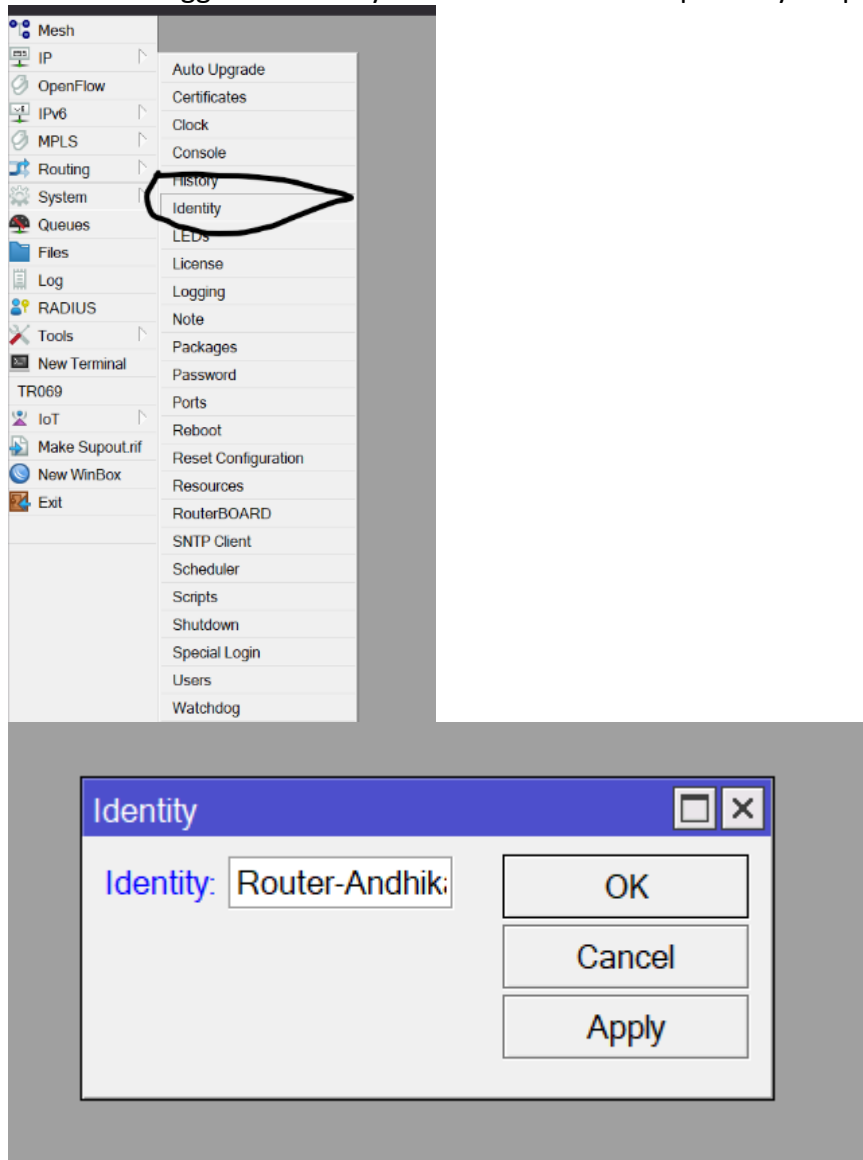
1. MIKROTIK ROUTER



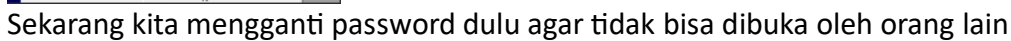
Pertama- tama kita buka winbox, dikarenakan mikrotik nya sudah di reset, kita langsung login menggunakan username admin dengan tanpa password.



Lalu kita mengubah identity terlebih dahulu untuk posisi nya seperti di foto bawah ini



Untuk namanya pakai nama sendiri nya setelah itu klik apply dan ok

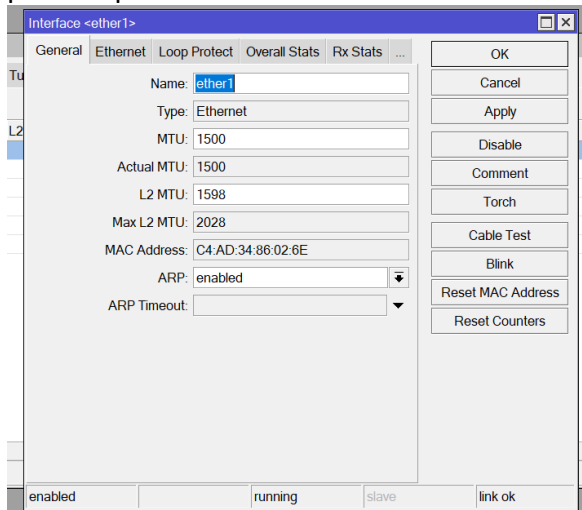


Lalu kita isi password nya terserah kalian yang penting kuat dan jarang di ketahui orang lain, untuk old passwd nya dikosongkan saja soalnya pas login tanpa pakai password

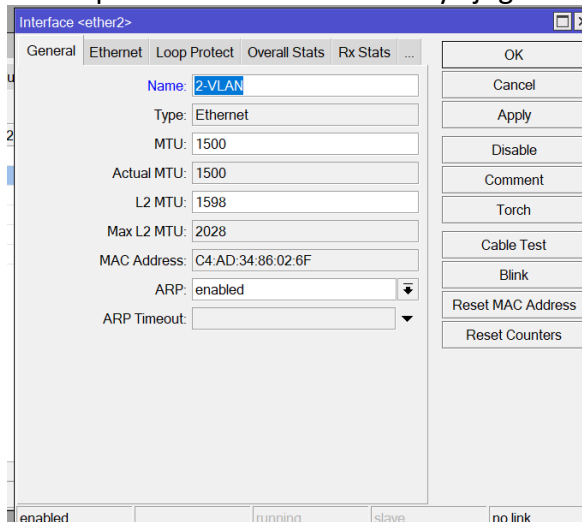


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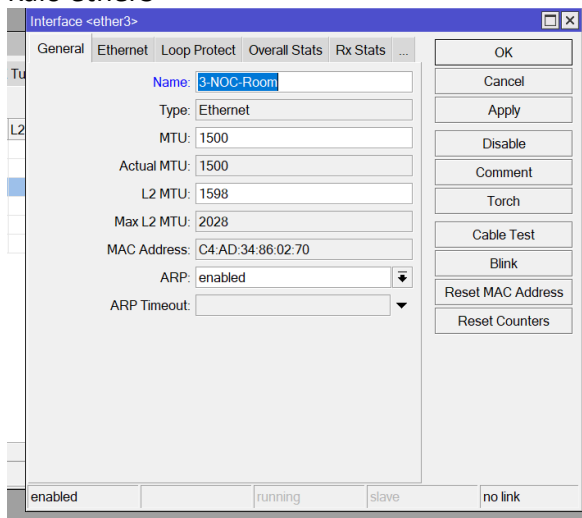
Sekarang kita ubah nama ether1 jadi 1-Internet, untuk caranya klik ether nya nanti muncul panel seperti di foto di bawah



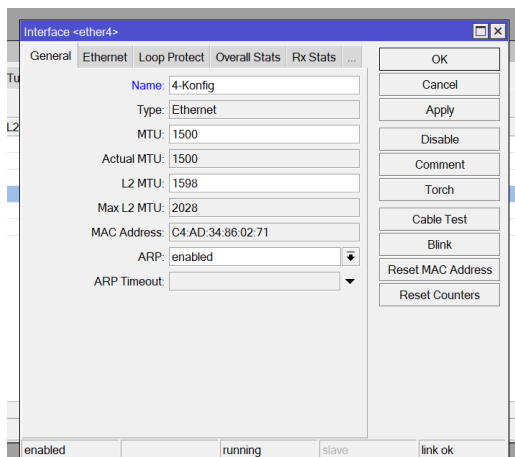
Nah seperti itu untuk sisa ether nya juga sama untuk ether2 itu



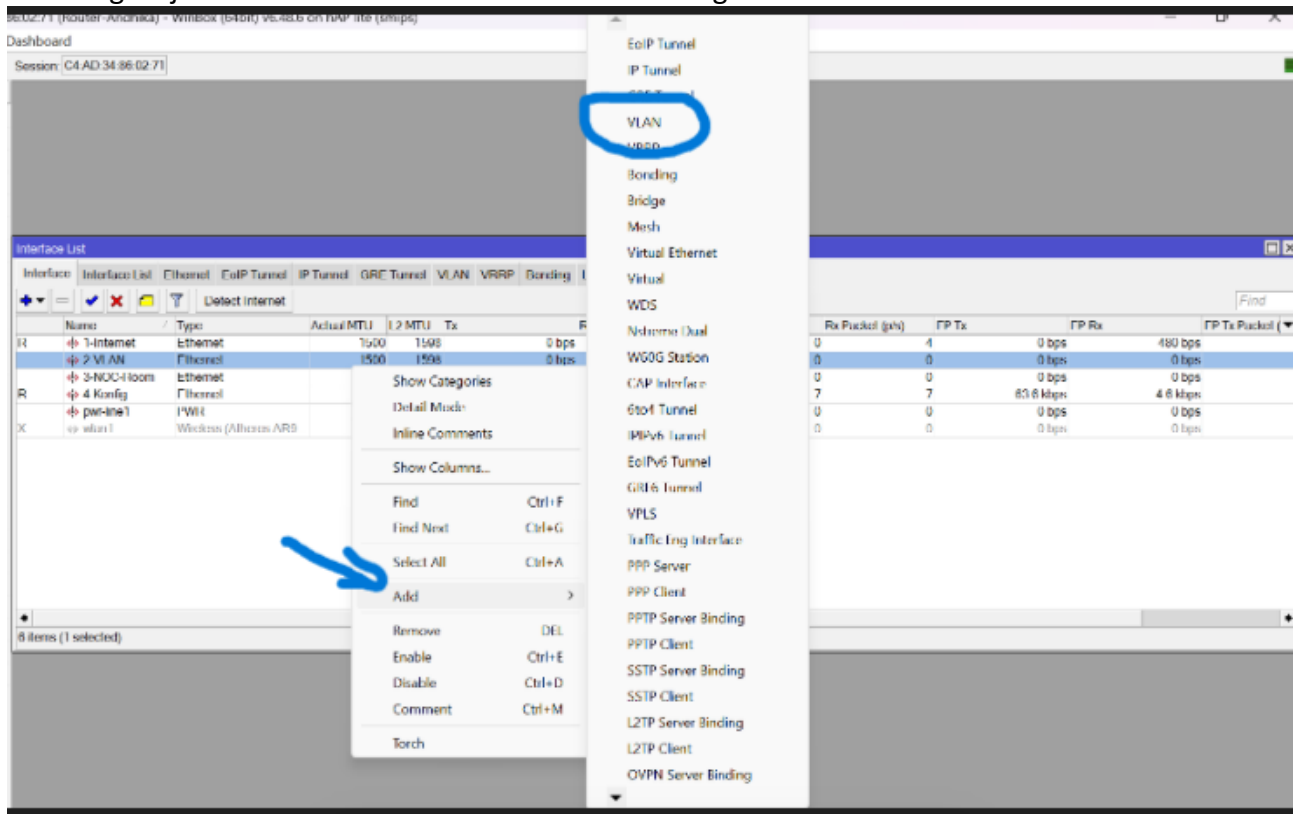
Kalo ether3



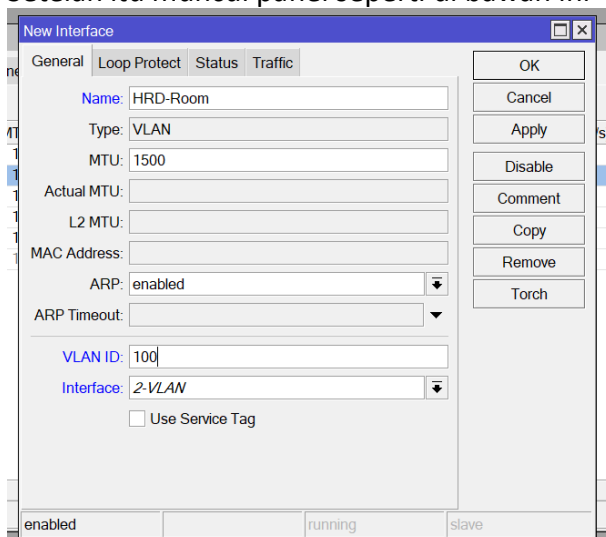
Kalo ether4 itu



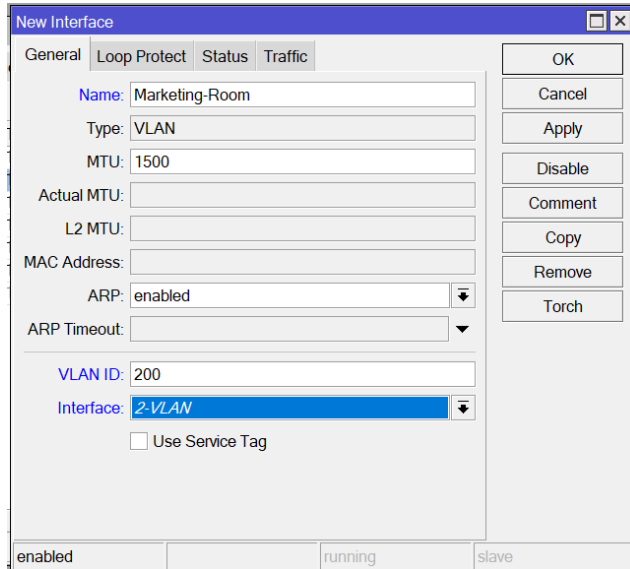
Sekarang lanjut menambahkan vlan untuk 2-VLAN dengan cara di klik kanan ether tsb



Setelah itu muncul panel seperti di bawah ini



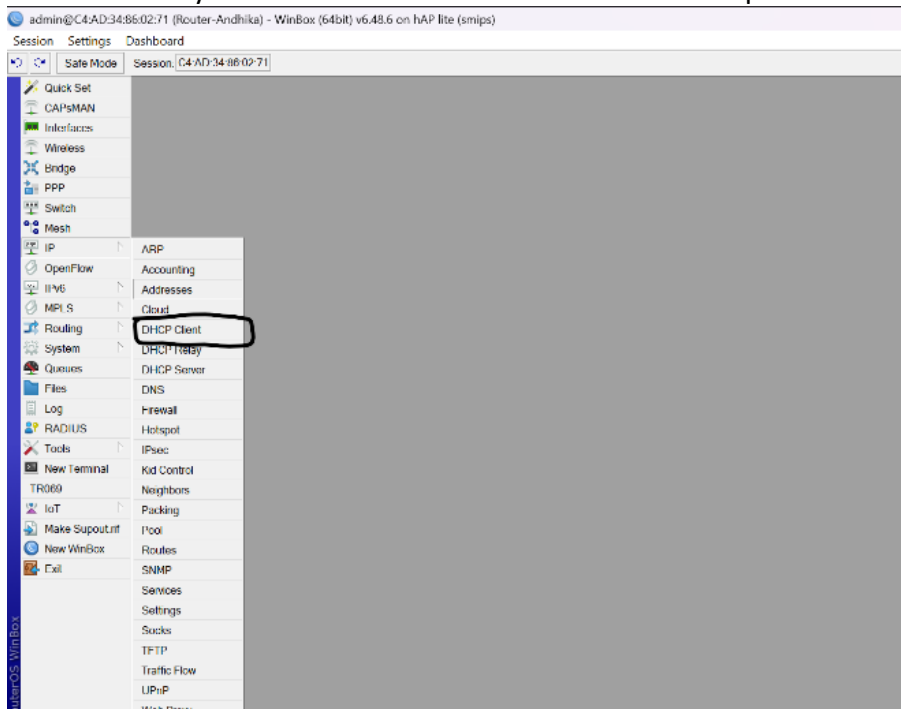
Diganti nama vlan1 jadi HRD-Room, ubah VLAN ID dari 1 ke 100, dan ganti interface nya ke 2-VLAN



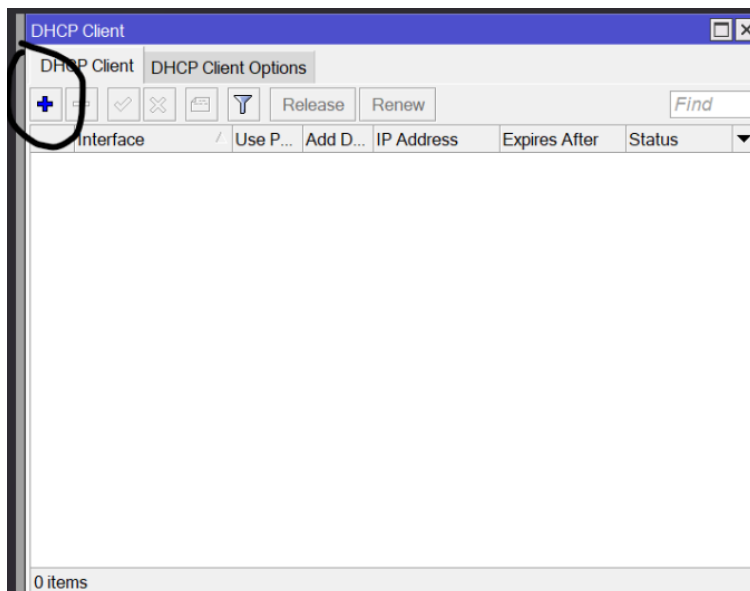
Untuk vlan2 nya juga seperti di foto atas

	Name	Type	Actual MTU	L2 MTU	Tx	Rx	Tx Packet (p/s)
R	1-Internet	Ethernet	1500	1598	0 bps	9.8 kbps	
	2-VLAN	Ethernet	1500	1598	0 bps	0 bps	
	HRD-Room	VLAN	1500	1594	0 bps	0 bps	
	Marketing-...	VLAN	1500	1594	0 bps	0 bps	
	3-NOC-Room	Ethernet	1500	1598	0 bps	0 bps	
R	4-Konfig	Ethernet	1500	1598	73.1 kbps	4.8 kbps	
	pwr-line1	PWR	1500	1598	0 bps	0 bps	
	wlan1	Wireless (Atheros AR9...	1500	1600	0 bps	0 bps	

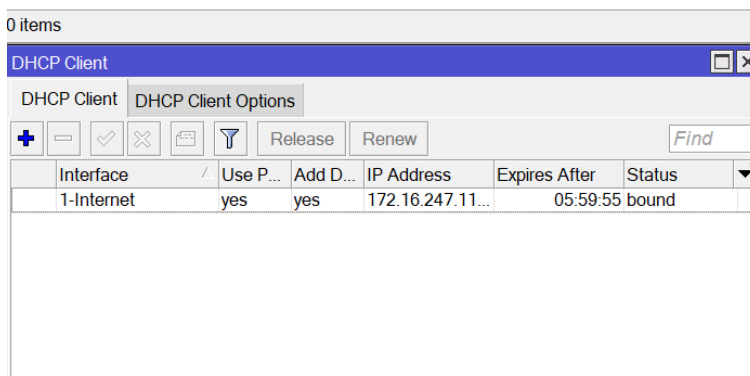
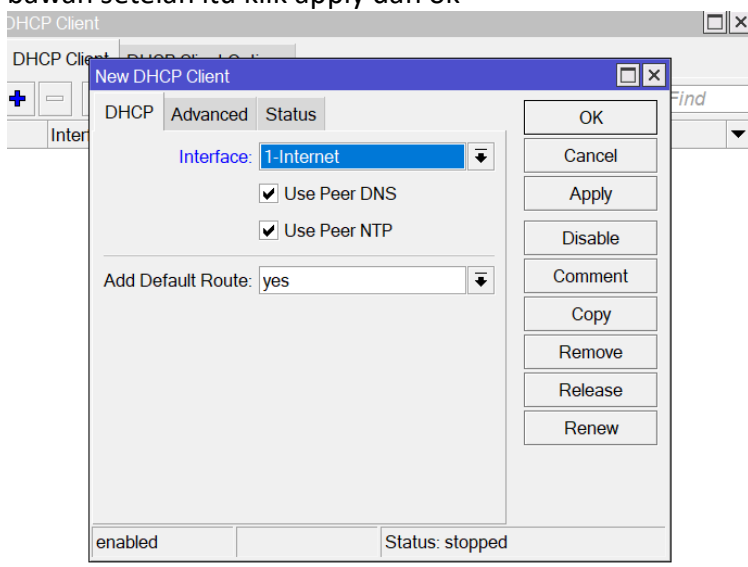
Untuk wlan1 nya di ceklis dulu kalo masih abu abu. Jadi seperti ini tampilan akhir interface nya.

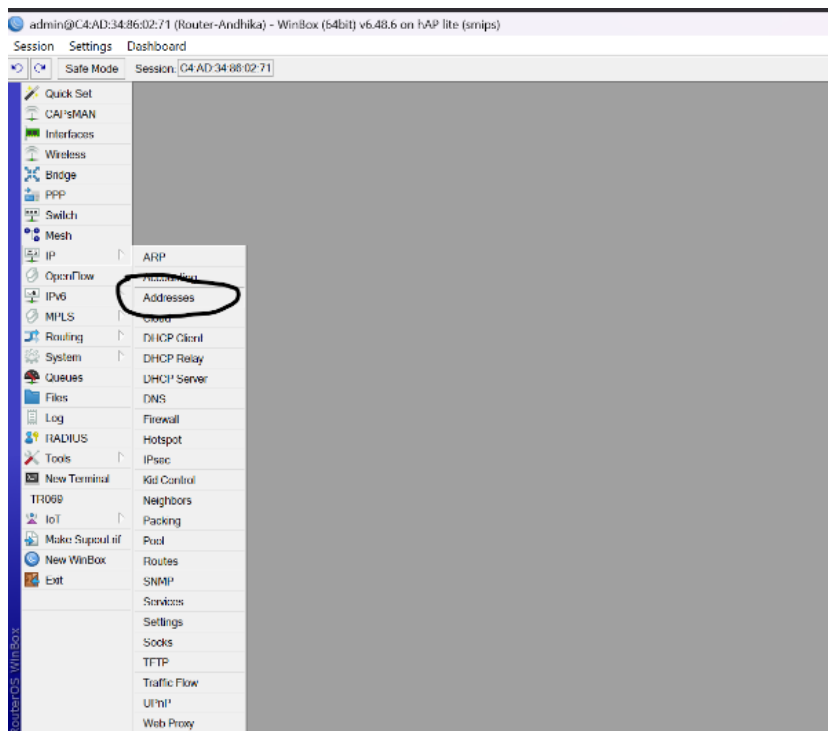


Sekarang lanjut set dhcp client untuk posisi nya liat lingkaran di foto di atas

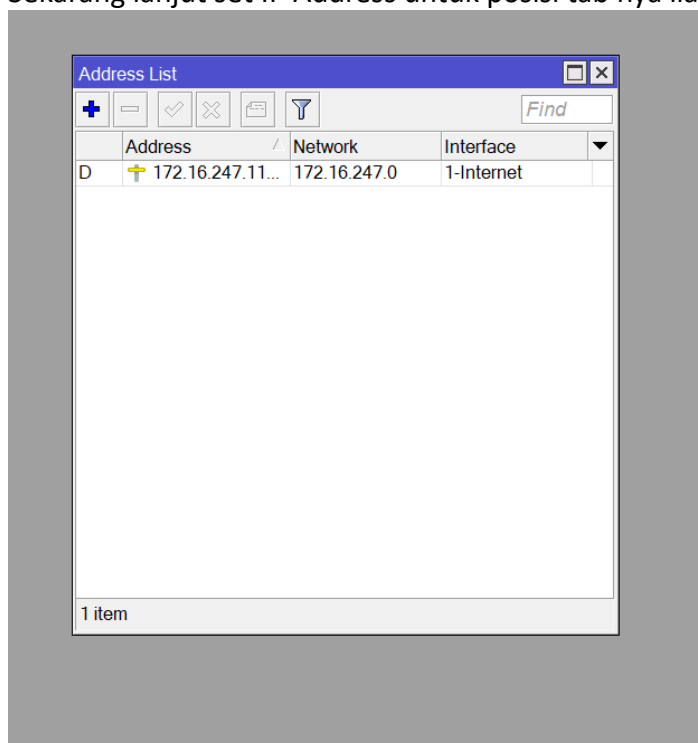


Diklik tanda “+” untuk menambahkan dhcp clientnya dan untuk interface nya di liat di foto di bawah setelah itu klik apply dan ok

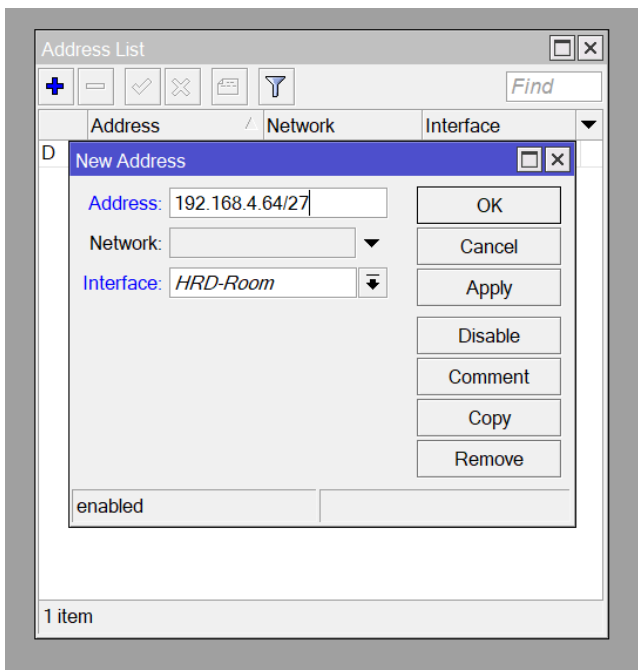




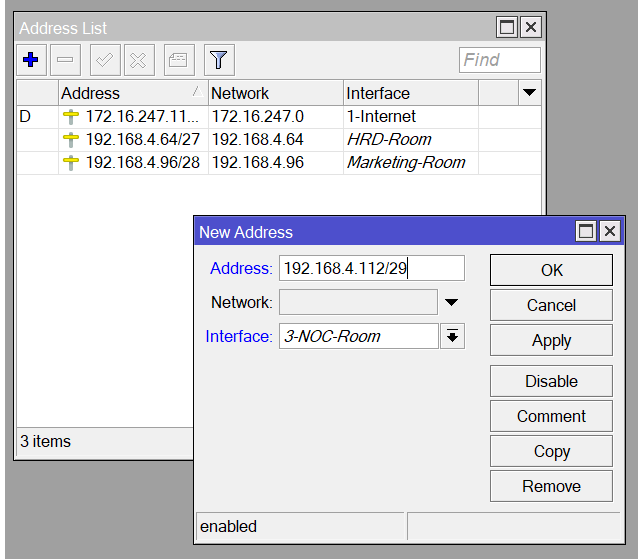
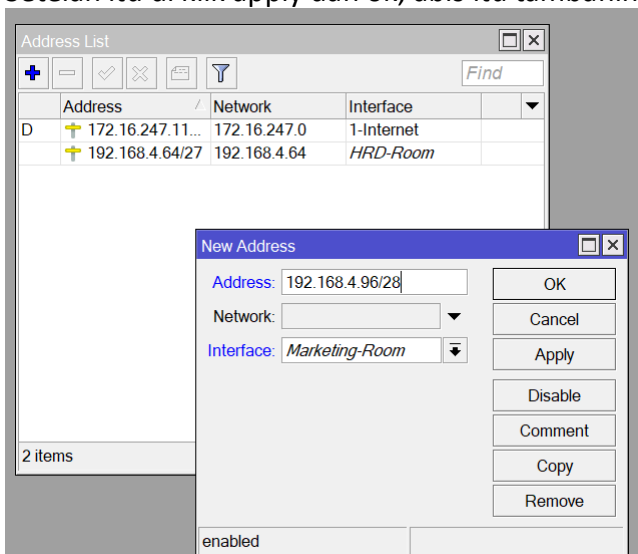
Sekarang lanjut set IP Address untuk posisi tab nya liat lingkaran di foto di atas

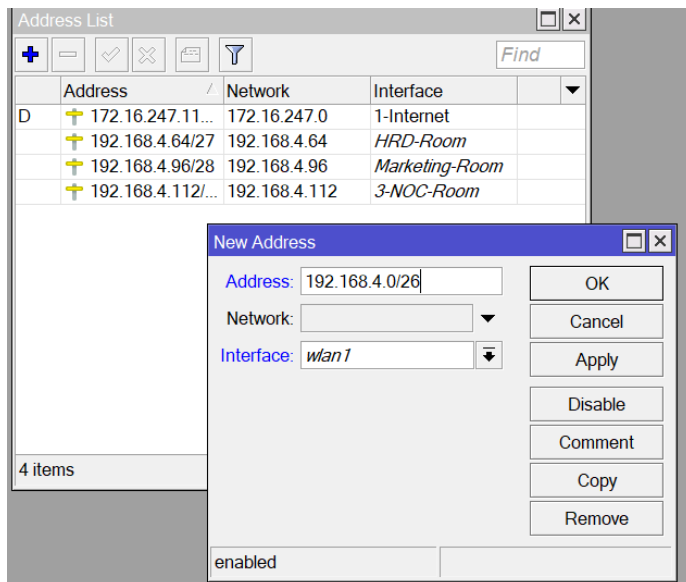


Di klik tanda “+” untuk menambahkan ip address di suatu ether



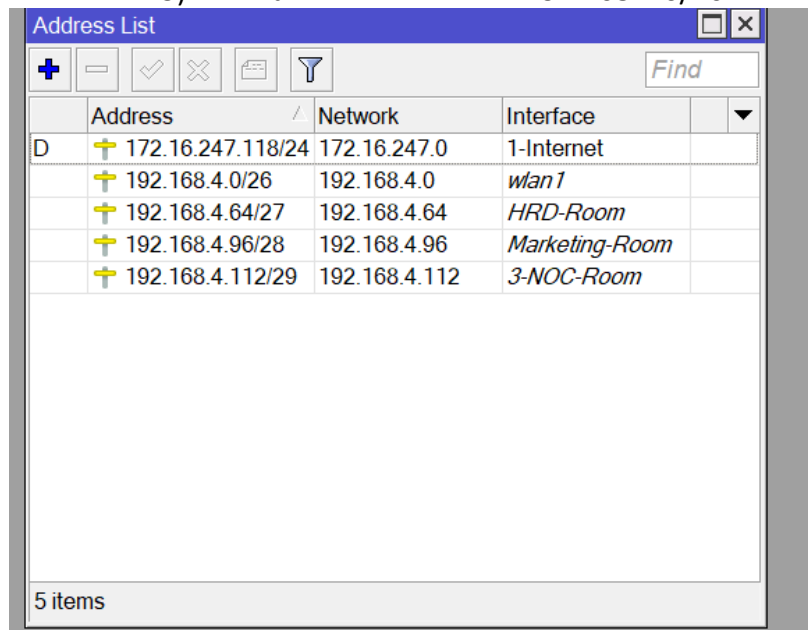
Setelah itu di klik apply dan ok, abis itu tambahin lagi



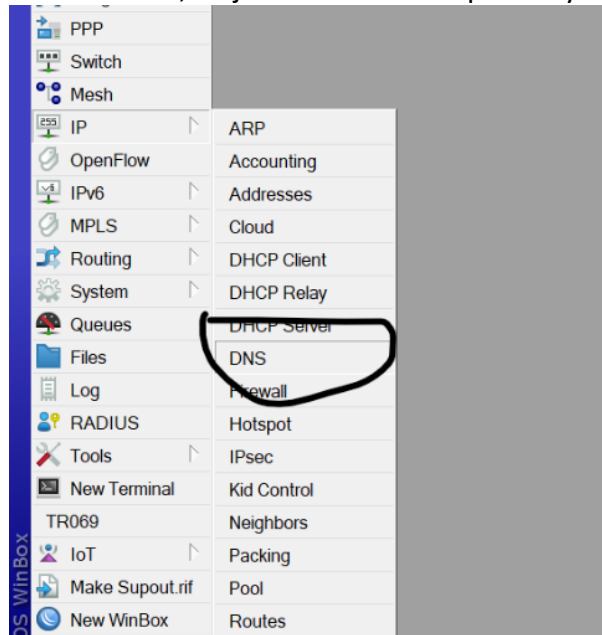


Konfigurasi IP Address

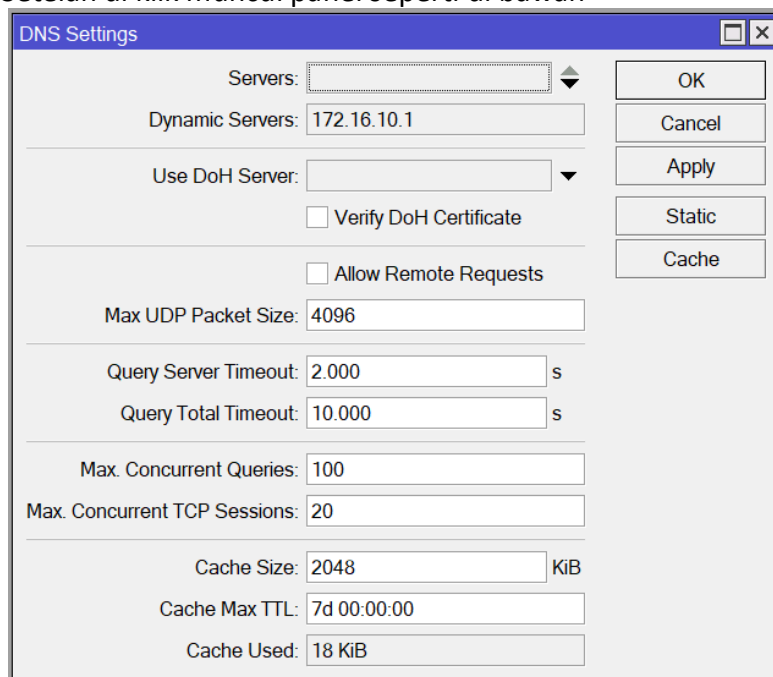
- 1) 1-Internet : Otomatis dari Router
- 2) HRD-Room : 192.168.4.64/27 | VLAN 100
- 3) Marketing-Room : 192.168.4.96/28 | VLAN 102
- 4) NOC-Room : 192.168.4.112/29
- 5) Wlan1 : 192.168.4.0/26



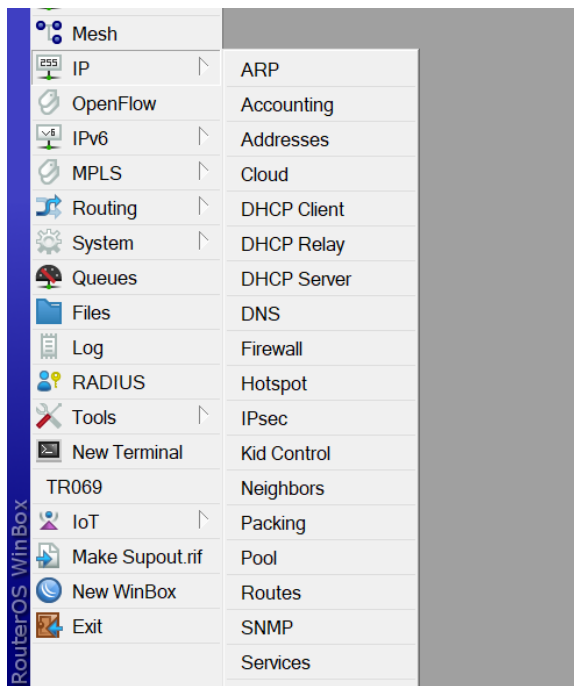
Setelah set IP, lanjut set DNS untuk posisi nya liat lingkaran di bawah ini



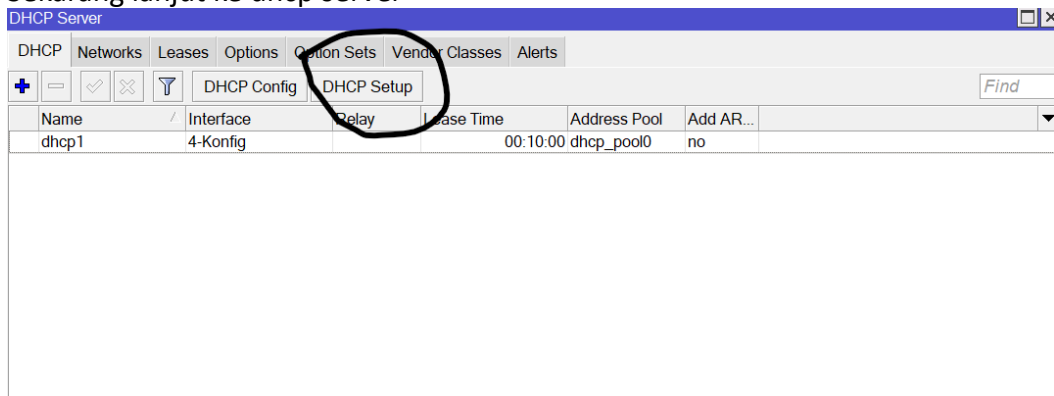
Setelah di klik muncul panel seperti di bawah



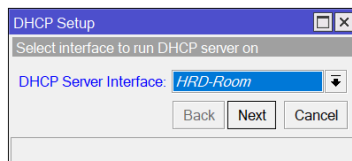
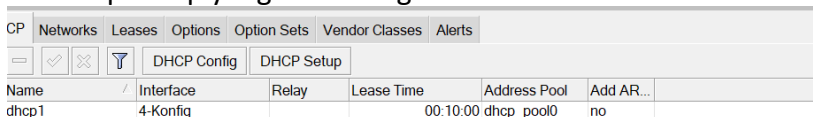
Di centang "Allow Remote Requests" setelah itu klik apply dan ok



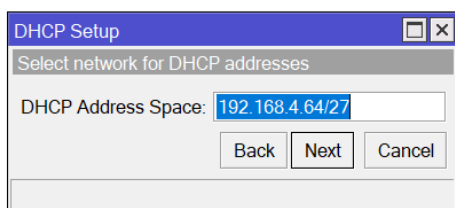
Sekarang lanjut ke dhcp server



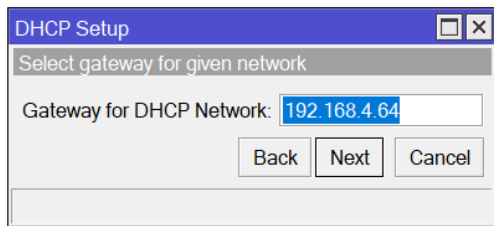
Klik dhcp setup yang ada di lingkaran



Pertama kita pilih *HRD-Room* dulu dan klik next



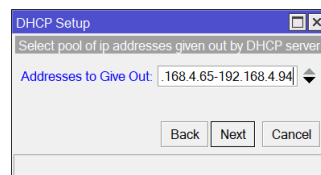
Klik next



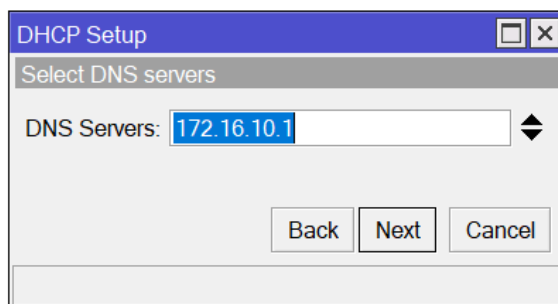
Klik Next

DHCP Server

TCP						
Networks Leases Options Option Sets Vendor Classes Alerts						
DHCP Config DHCP Setup						
Name	Interface	Relay	Lease Time	Address Pool	Add AR...	
dhcp1	4-Konfig		00:10:00	dhcp_pool0	no	



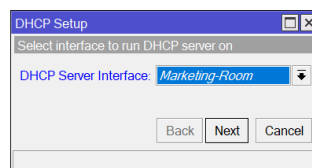
Untuk range nya dari 192.168.4.65 – 192.168.4.94, setelah itu klik next



Klik next dan setelah itu klik next lagi

DHCP Server

TCP						
Networks Leases Options Option Sets Vendor Classes Alerts						
DHCP Config DHCP Setup						
Name	Interface	Relay	Lease Time	Address Pool	Add AR...	
dhcp1	4-Konfig		00:10:00	dhcp_pool0	no	
dhcp2	HRD-Room		00:10:00	dhcp_pool1	no	



Lanjut yang *Marketing-Room* dan klik next

		00:10:00 dhcp_pool0	no
		00:10:00 dhcp_pool1	no

Klik next

		00:10:00 dhcp_pool1	no
--	--	---------------------	----

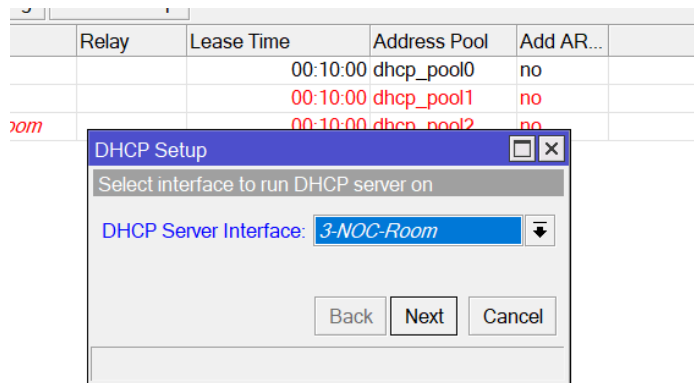
Klik next

	Relay	Lease Time	Address Pool	Add AR...
			00:10:00 dhcp_pool0	no
m			00:10:00 dhcp_pool1	no

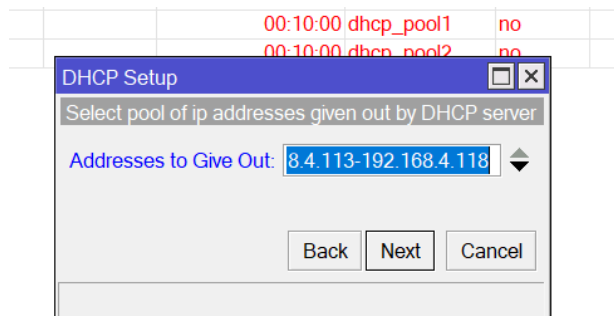
Untuk range nya dari 192.168.4.97 – 192.168.4.110, setelah itu klik next

		00:10:00 dhcp_pool0	no
		00:10:00 dhcp_pool1	no

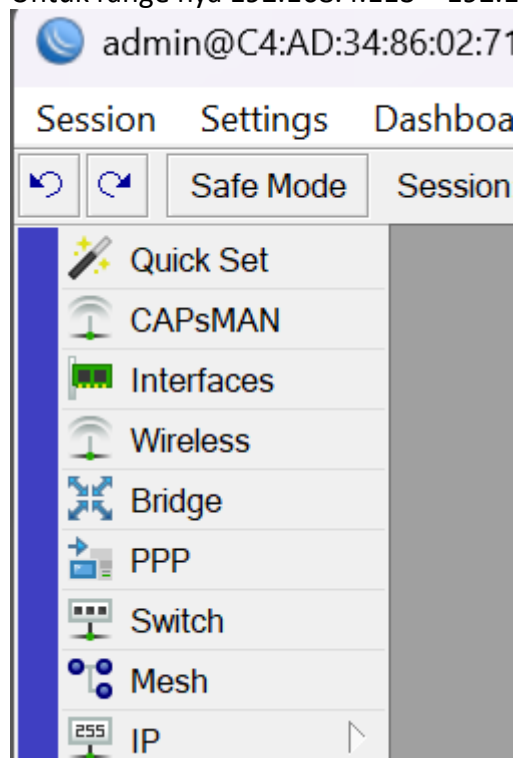
Klik next dan setelah itu klik next lagi



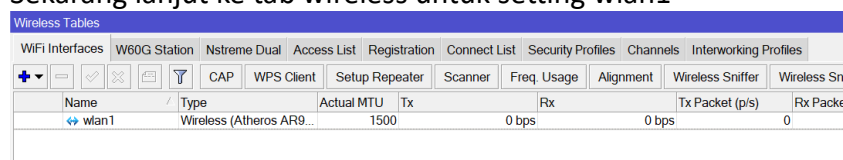
Klik next terus



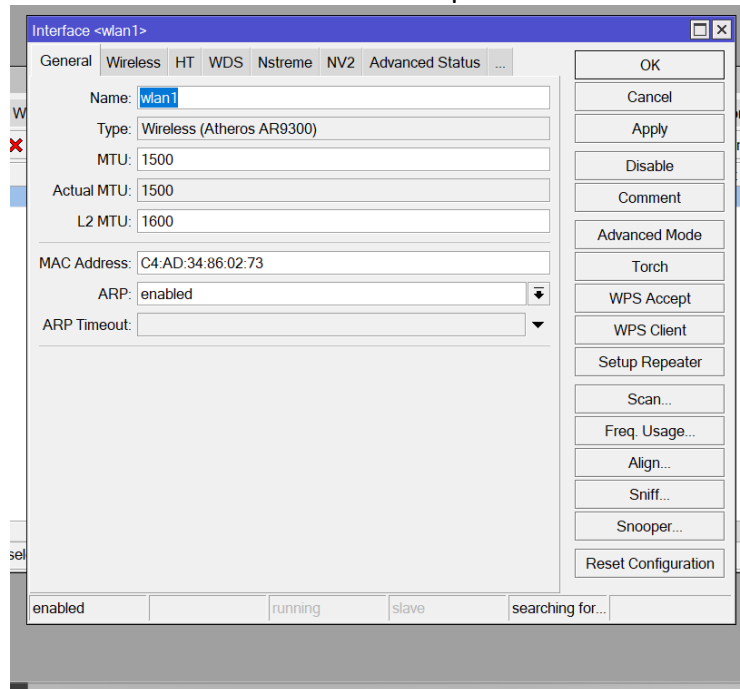
Untuk range nya 192.168.4.113 – 192.168.4.118 setelah itu klik next terus



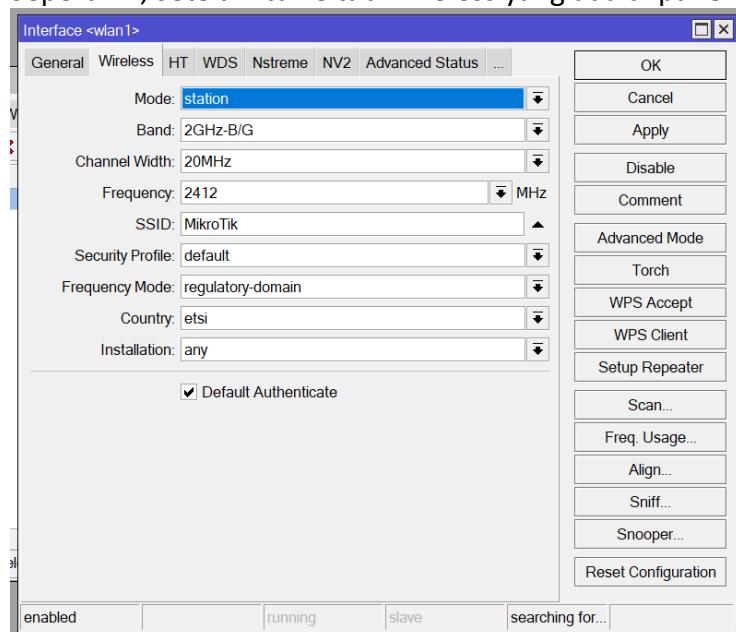
Sekarang lanjut ke tab wireless untuk setting wlan1



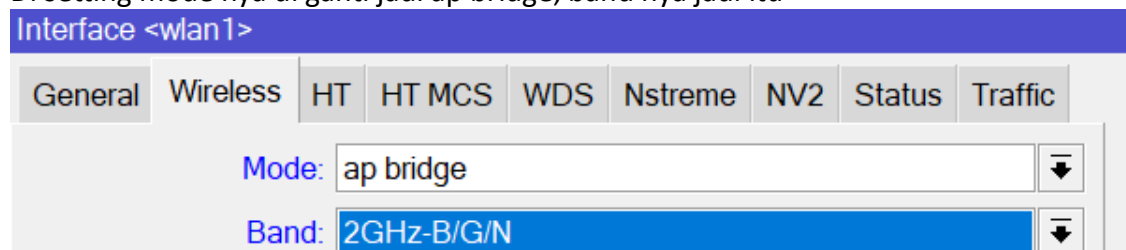
Klik 2x di wlan1 dan nanti muncul panel



Seperti ini, setelah itu ke tab Wireless yang ada di panel



Di setting mode nya di ganti jadi ap bridge, band nya jadi itu



Interface <wlan1>

General Wireless HT HT MCS WDS Nstreme NV2 Status Traffic

Mode: ap bridge

Band: 2GHz-B/G/N

Channel Width: 20MHz

Frequency: 2412 MHz

SSID: Wifi-Andhika

Security Profile: default

WPS Mode: push button

Frequency Mode: regulatory-domain

Country: etsi

Installation: any

Default AP Tx Limit: bps

Default Client Tx Limit: bps

☒ Default Authenticate

☒ Default Forward

☐ Hide SSID

OK

Cancel

Apply

Disable

Comment

Advanced Mode

Torch

WPS Accept

WPS Client

Setup Repeater

Scan...

Freq. Usage...

Align...

Sniff...

Snooper...

Reset Configuration

Untuk SSID nya bebas apa saja, setelah itu di apply dan oke

Wireless Tables

WiFi Interfaces W60G Station Nstreme Dual Access List Registration Connect List Security Profiles Channels Interworking Profiles

+ - ✓ ✗ CAP WPS Client Setup Repeater Scanner Freq. Usage Alignment Wireless Sniffer Wireless Snc

Name	Type	Auth. MTU	To	From	To Packet (s/s)	From Packet

Ke tab Security Profiles untuk mengganti password wlan1

Wireless Tables

WiFi Interfaces W60G Station Nstreme Dual Access List Registration Connect List Security Profiles Channels Interworking Profiles

+ - [icon] [icon]

Name	Mode	Authentication...	Unicast Ciphers	Group Ciphers	WPA Pre-Shared ...	WPA2 Pre-Shared ...
* default	none				*****	*****

Klik 2x di default dan muncul panel

Security Profile <default>

General | RADIUS | EAP | Static Keys

Name: default

Mode: dynamic keys

Authentication Types: ☒ WPA PSK ☒ WPA2 PSK
☒ WPA EAP ☒ WPA2 EAP

Unicast Ciphers: ☒ aes ccm ☐ tkip

Group Ciphers: ☒ aes ccm ☐ tkip

WPA Pre-Shared Key: 12345678

WPA2 Pre-Shared Key: 12345678

Supplicant Identity: MikroTik

Group Key Update: 00:05:00

Management Protection: disabled

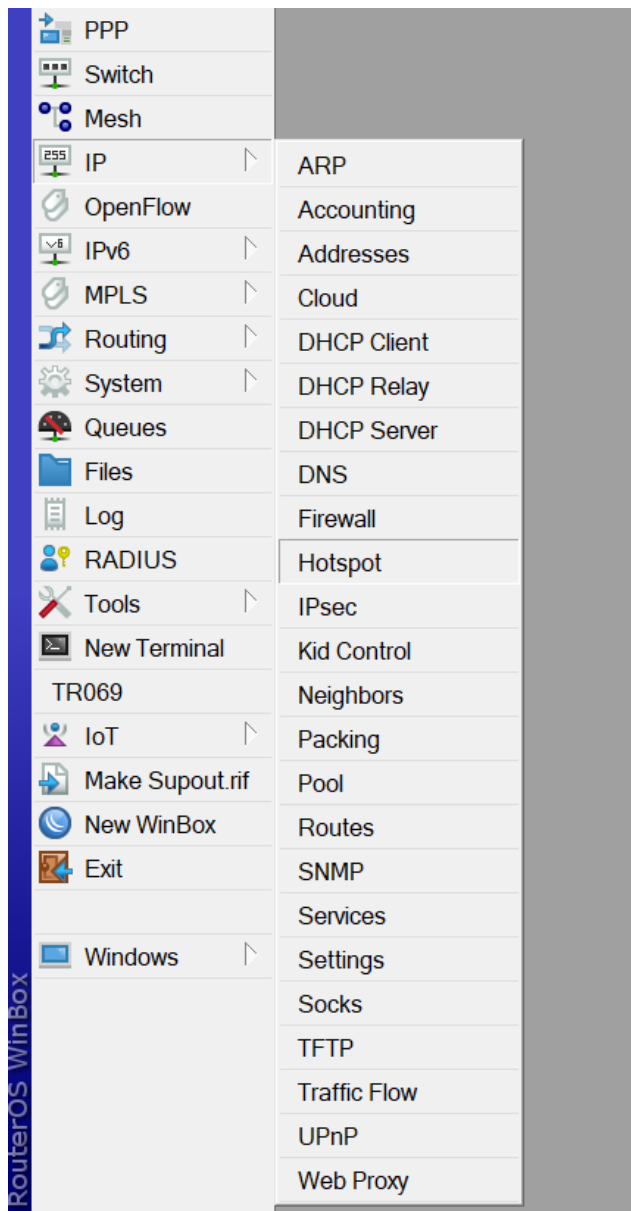
Management Protection Key:

☐ Disable PMKID

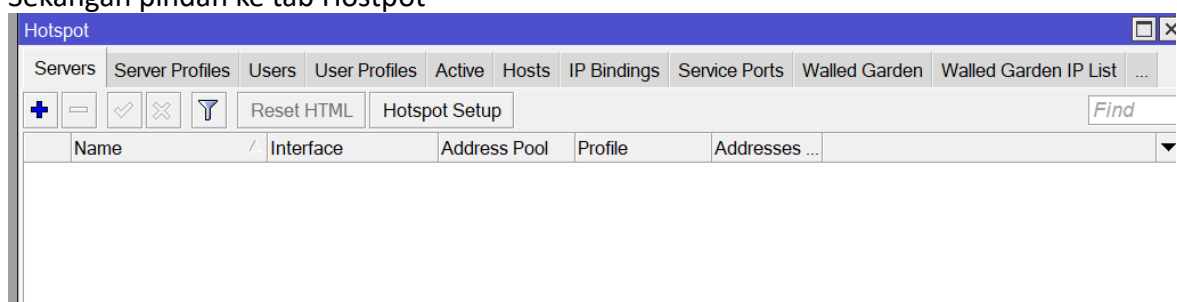
OK
Cancel
Apply
Comment
Copy
Remove

default

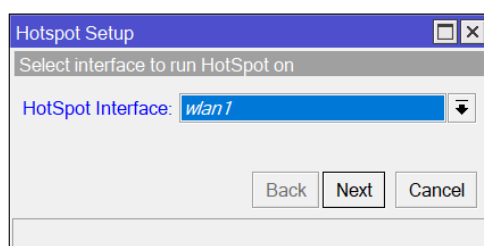
Setting nya seperti itu, tapi untuk password nya mah bebas, btw set password nya yang 1245678 itu. Setelah itu klik apply dan ok

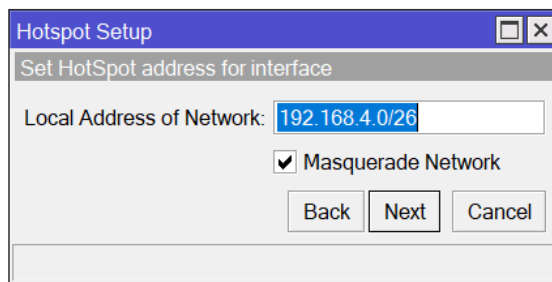


Sekarang pindah ke tab Hotspot

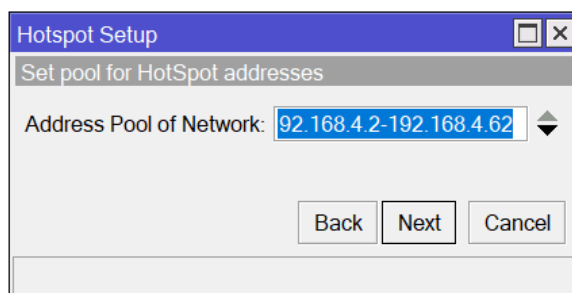


Klik tombol Hotspot Setup

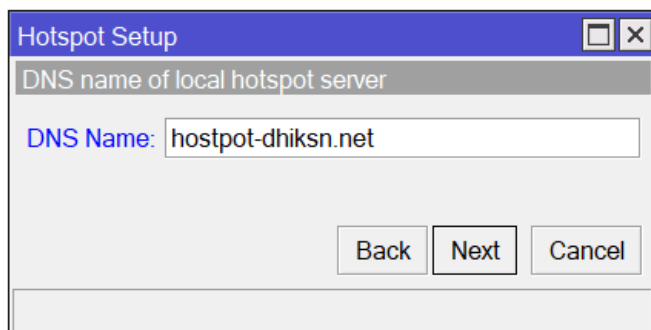




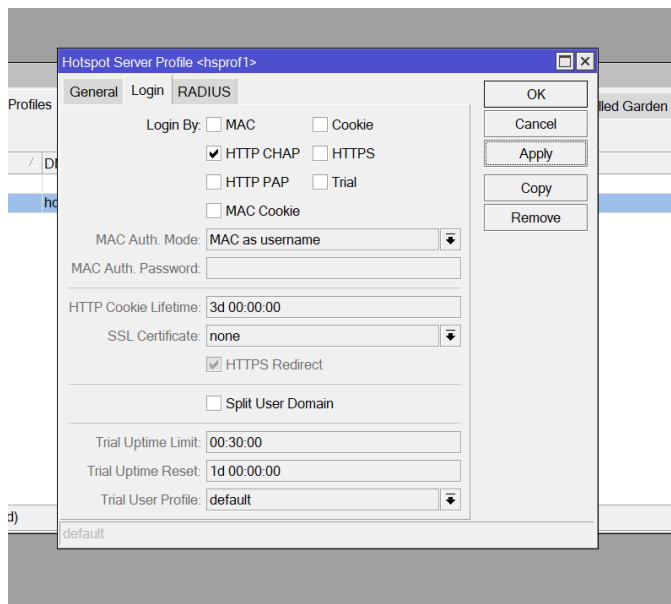
Klik next



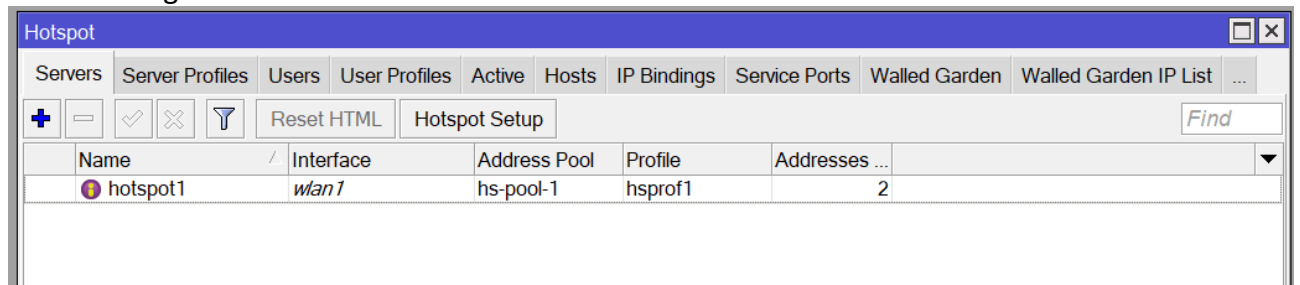
Klik next terus sampai dns name



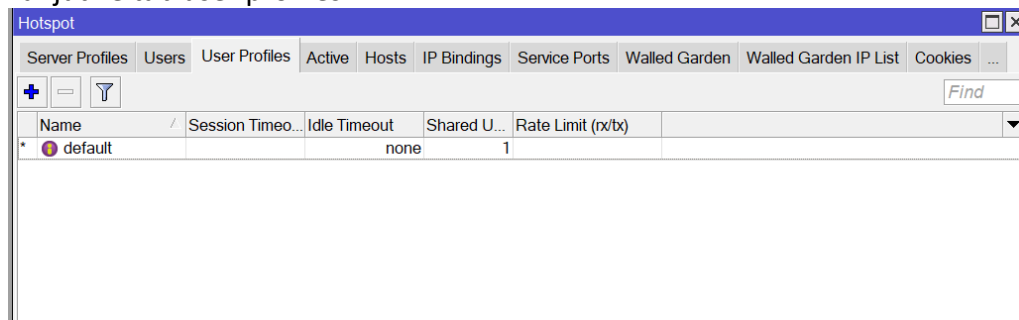
Di DNS name nya di tambahkan domain hostpot-nama.net seperti di foto atas, setelah itu di klik next



Di uncentang cookie



Lanjut ke tab user profiles



Tambahkan user profiles dengan cara klik tombol “+”

Profile	Rate Limit
CEO	4 Mbps
Manager	3 Mbps
Staf	2 Mbps

New Hotspot User Profile

General Queue Scripts

Name: CEO

Address Pool: none

Session Timeout:

Idle Timeout: none

Keepalive Timeout: 00:02:00

Status Autorefresh: 00:01:00

Shared Users: 1

Rate Limit (rx/bx): 4M/4M

☒ Add MAC Cookie

MAC Cookie Timeout: 3d 00:00:00

Address List:

Incoming Filter:

Outgoing Filter:

Incoming Packet Mark:

Outgoing Packet Mark:

Open Status Page: always

☐ Transparent Proxy

OK

Cancel

Apply

Copy

Remove

default

New Hotspot User Profile

General Queue Scripts

Name: Manager

Address Pool: none

Session Timeout:

Idle Timeout: none

Keepalive Timeout: 00:02:00

Status Autorefresh: 00:01:00

Shared Users: 1

Rate Limit (rx/bx): 3M/3M

☒ Add MAC Cookie

MAC Cookie Timeout: 3d 00:00:00

Address List:

Incoming Filter:

Outgoing Filter:

Incoming Packet Mark:

Outgoing Packet Mark:

Open Status Page: always

OK

Cancel

Apply

Copy

Remove

default

Setelah itu klik apply dan ok dan pindah tab ke user

Server	Name	Address	MAC Address	Profile	Uptime
...	counters and limits for trial users				00:00:00
*	all	admin		default	00:00:00

Tambahkan user hotspot dengan ketentuan :

Username	Password	Profile
rudy	rudy	CEO
john	john	Manager
woody	woody	Staf
leanny	leanny	Staf

New Hotspot User

General Limits Statistics

Server: hotspot1

Name: rudy

Password: rudy

Address:

MAC Address:

Profile: CEO

Routes:

Email:

enabled

OK Cancel Apply Disable Comment Copy Remove Reset Counters Reset All Counters

Klik apply dan ok, setelah itu buat baru

New Hotspot User

General Limits Statistics

Server: hotspot1

Name: john

Password: john

Address:

MAC Address:

Profile: Manager

Routes:

Email:

enabled

OK Cancel Apply Disable Comment Copy Remove Reset Counters Reset All Counters

New Hotspot User

General Limits Statistics

Server: hotspot1

Name: woody

Password: woody

Address:

MAC Address:

Profile: default

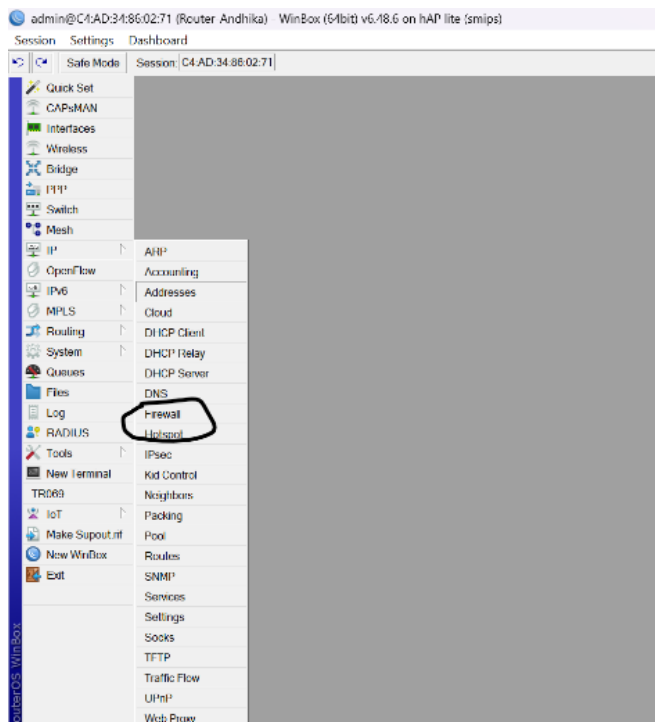
Routes:

Email:

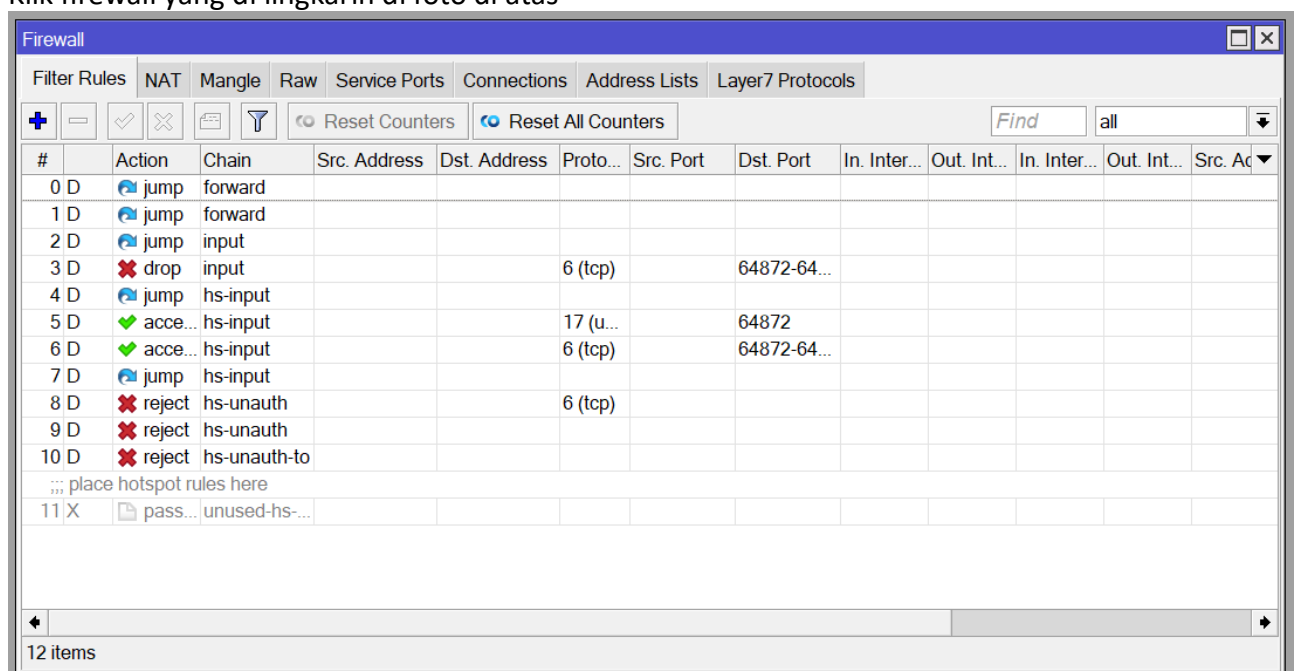
enabled

OK Cancel Apply Disable Comment Copy Remove Reset Counters Reset All Counters

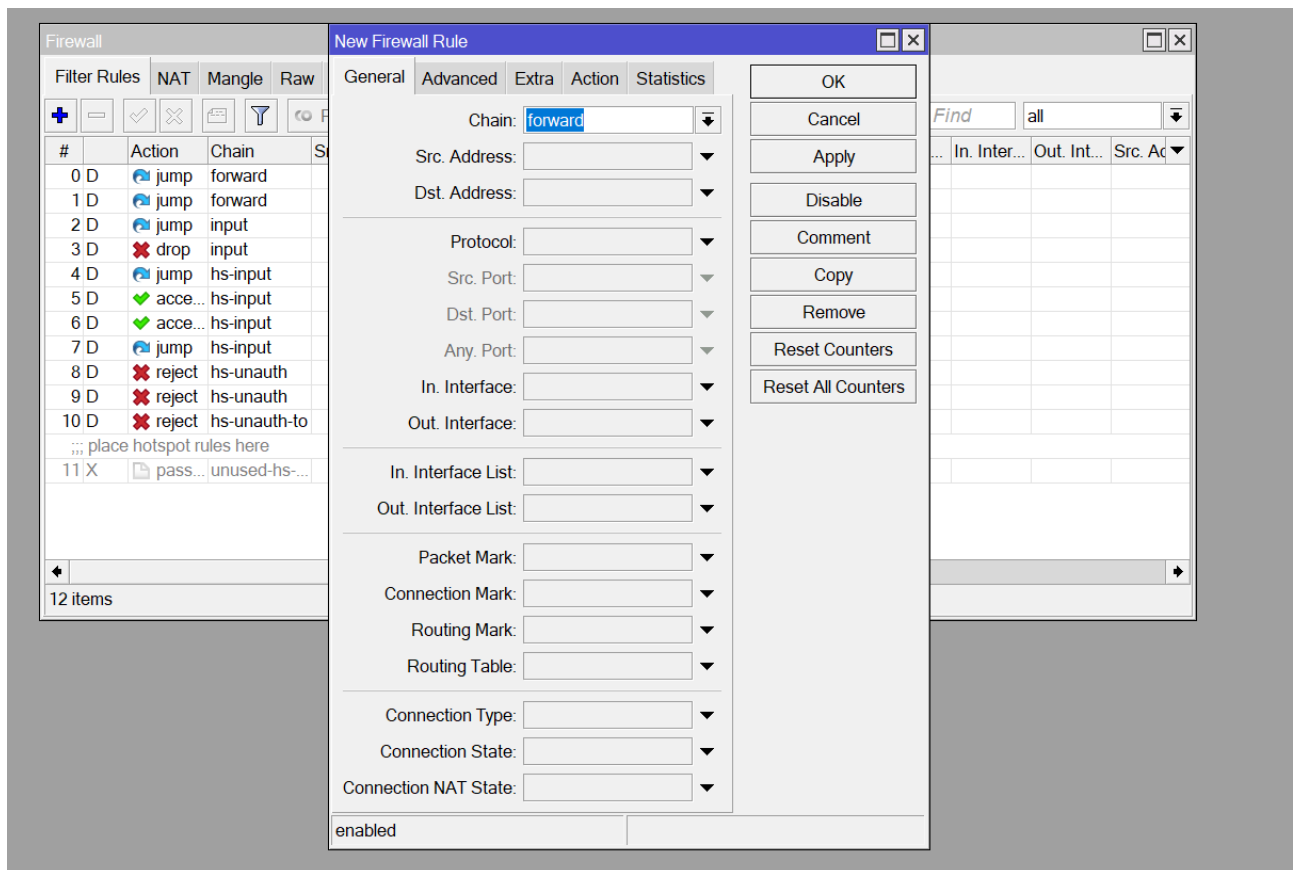
Klik apply dan ok, setelah itu lanjut konfigurasi firewall



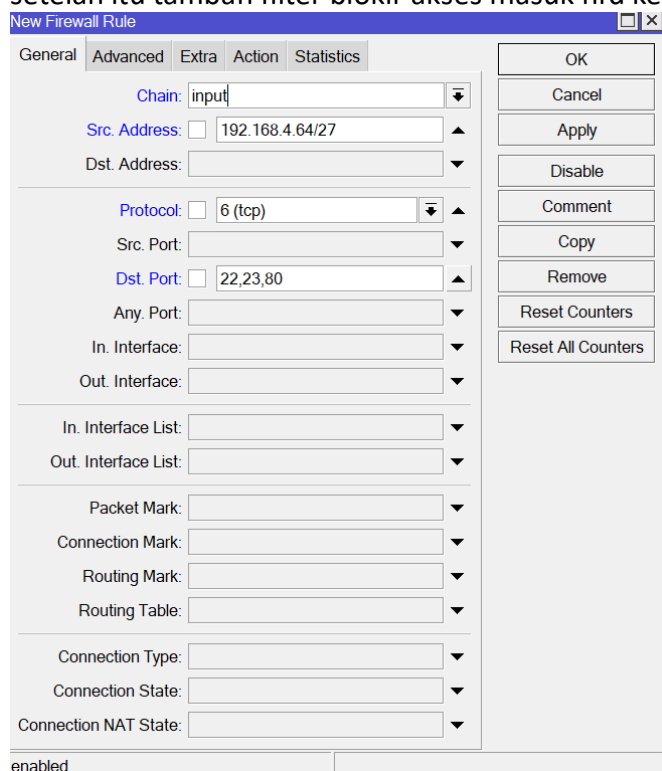
Klik firewall yang di lingkari di foto di atas

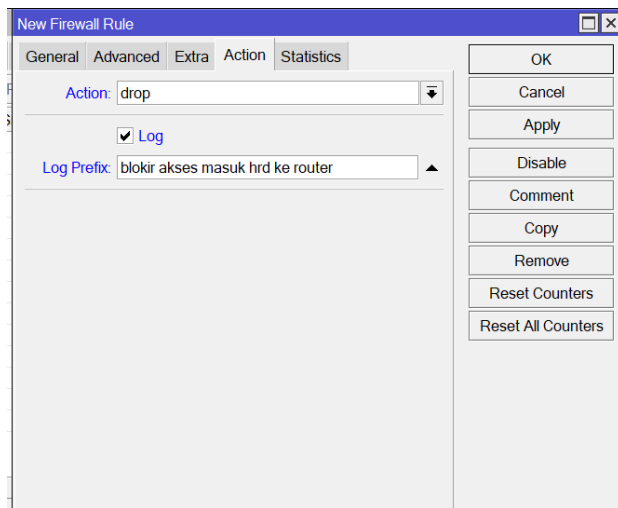


Sekarang tambahkan filter di filters

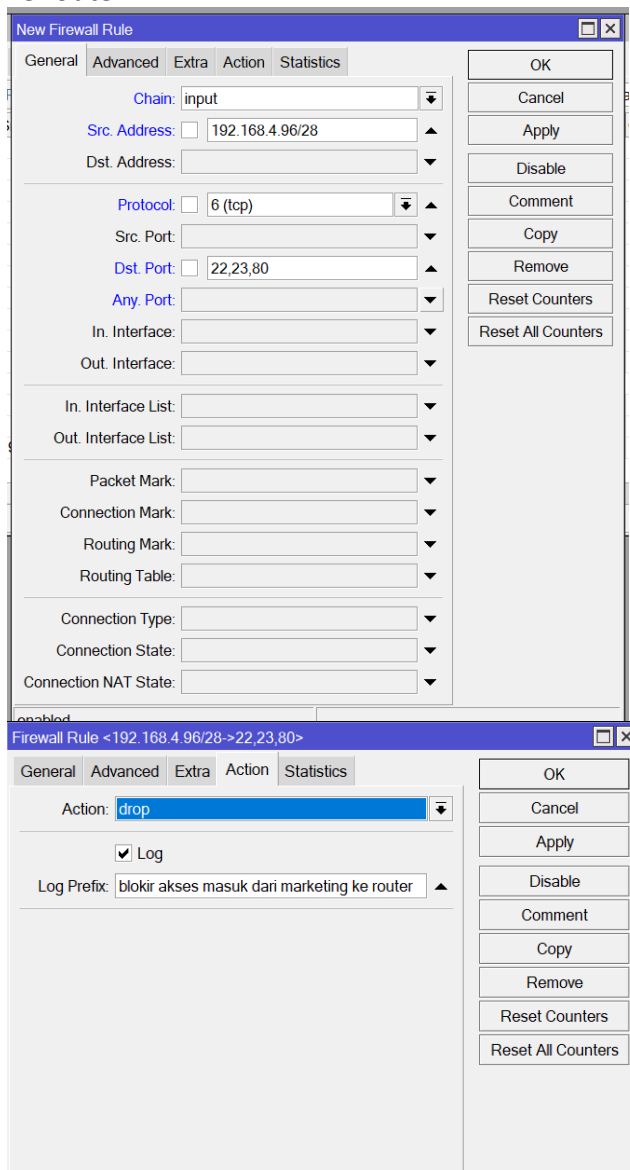


Ini langsung di apply dan ok saja, berfungsi untuk mengizinkan akses forwarding internet, setelah itu tambah filter blokir akses masuk hrd ke router





Setelah itu klik apply dan ok. Sekarang tambahkan filter lagi yang blokir akses masuk hrd ke router



Klik apply dan ok, setelah itu tambahkan filter lagi yang blokir dari public wifi ke router

New Firewall Rule

General Advanced Extra Action Statistics

Chain: input

Src. Address: 192.168.4.1/26

Dst. Address:

Protocol: 6 (tcp)

Src. Port:

Dst. Port: 22,23,80

Any. Port:

In. Interface:

Out. Interface:

In. Interface List:

Out. Interface List:

Packet Mark:

Connection Mark:

Routing Mark:

Routing Table:

Connection Type:

Connection State:

Connection NAT State:

enabled

OK

Cancel

Apply

Disable

Comment

Copy

Remove

Reset Counters

Reset All Counters

Firewall Rule <192.168.4.0/26->22,23,80>

General Advanced Extra Action Statistics

Action: drop

☒ Log

Log Prefix: blokir akses masuk dari wifi ke router

OK

Cancel

Apply

Disable

Comment

Copy

Remove

Reset Counters

Reset All Counters

Klik apply dan ok, tambahkan satu lagi filter untuk blokir akses masuk ke router dari Internet melalui SSH, Telnet, API, Winbox, HTTP, HTTPS

New NAT Rule

General | Advanced | Extra | Action | Statistics

Chain: srcnat

Src. Address:

Dst. Address:

Protocol:

Src. Port:

Dst. Port:

Any. Port:

In. Interface:

Out. Interface:

In. Interface List:

Out. Interface List:

Packet Mark:

Connection Mark:

Routing Mark:

Routing Table:

Connection Type:

enabled

OK

Cancel

Apply

Disable

Comment

Copy

Remove

Reset Counters

Reset All Counters

New NAT Rule

General | Advanced | Extra | Action | Statistics

Action: masquerade

☐ Log

Log Prefix:

To Ports:

OK

Cancel

Apply

Disable

Comment

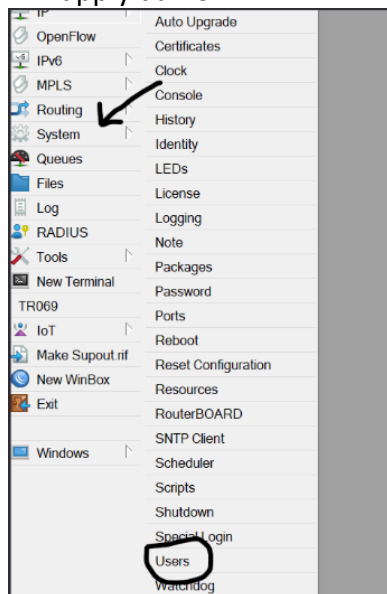
Copy

Remove

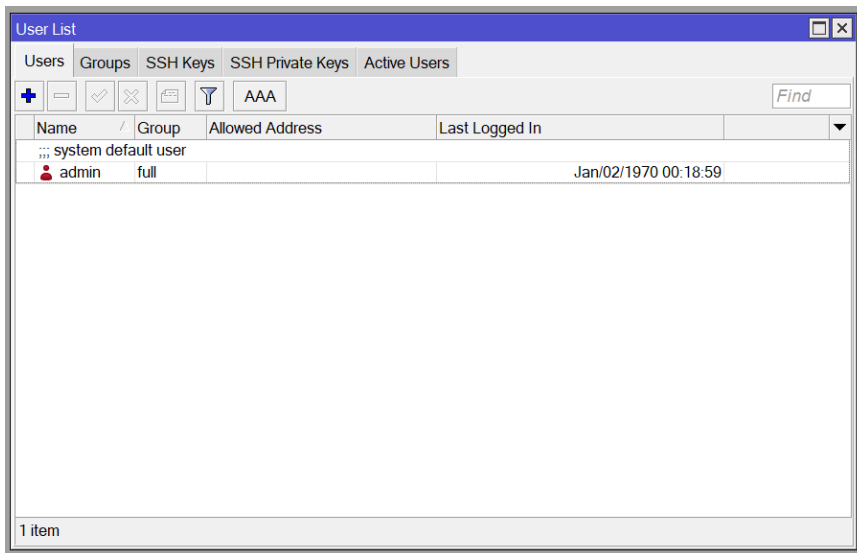
Reset Counters

Reset All Counters

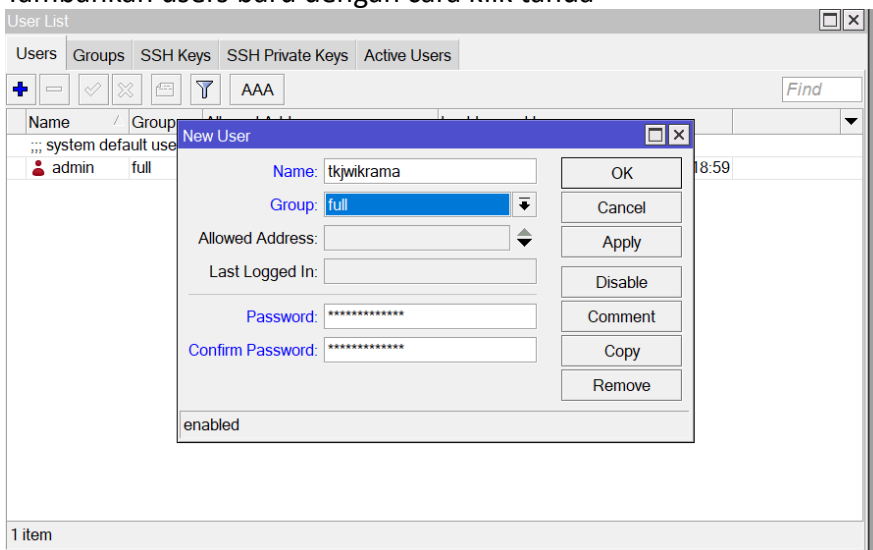
Klik apply dan ok



Sekarang pindah ke tab users untuk posisi nya ada di foto di atas



Tambahkan users baru dengan cara klik tanda “+”

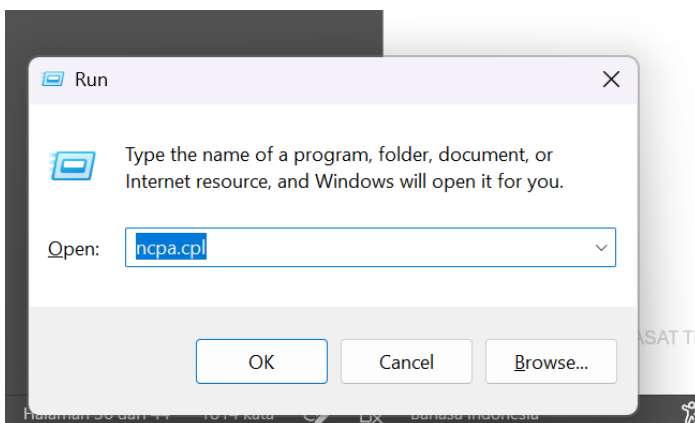


Buat user baru

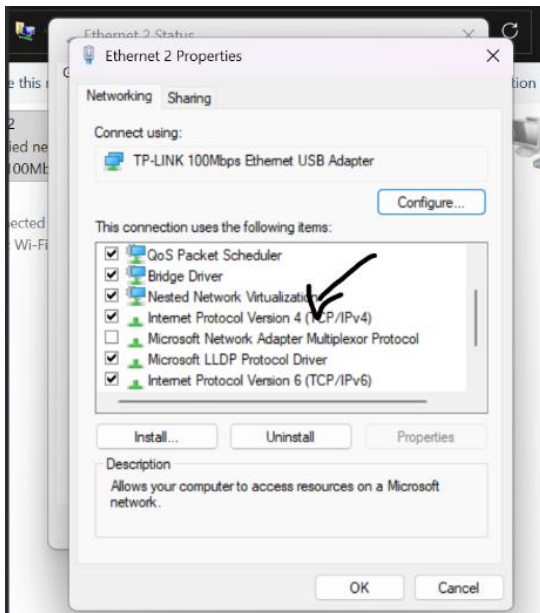
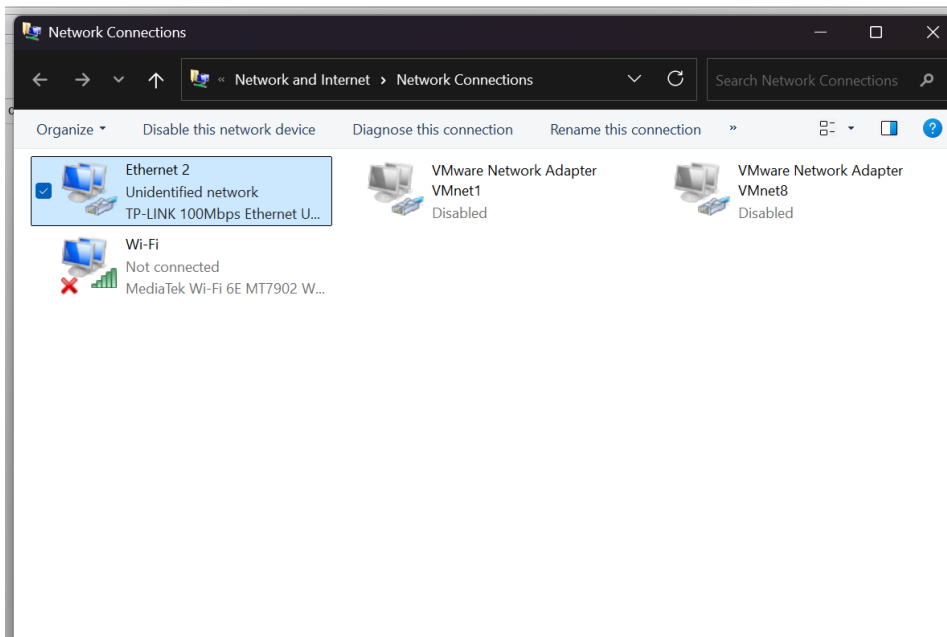
- 6) Username : tkjwikrama
- 7) Password : tkjwikrama123
- 8) Group : full

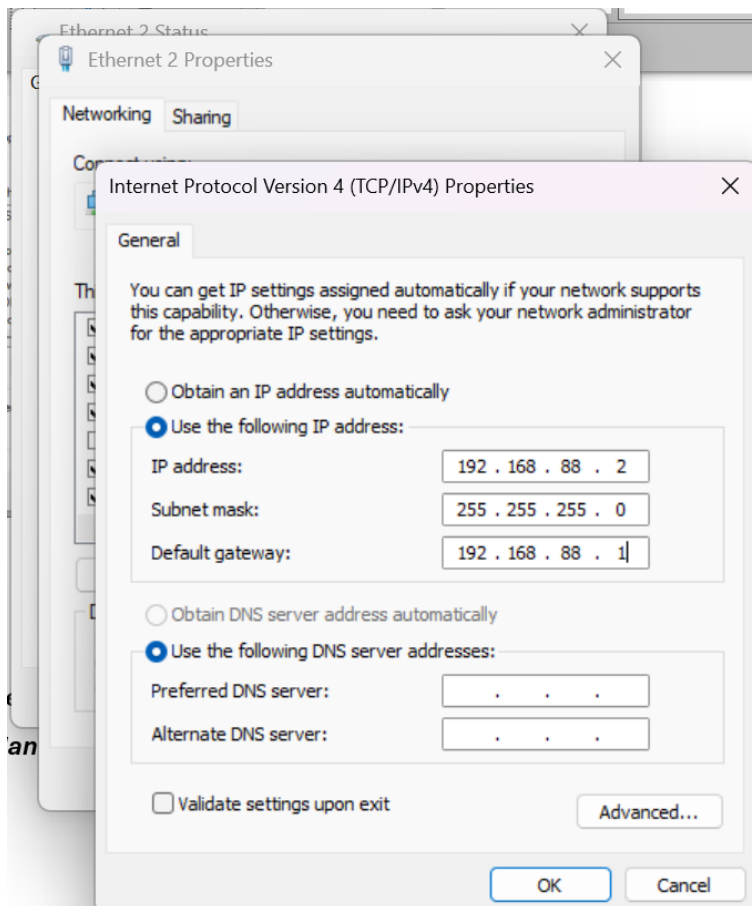
Klik apply dan ok. Sudah selesai konfigurasi mikrotik.

2. MIKROTIK SWITCH

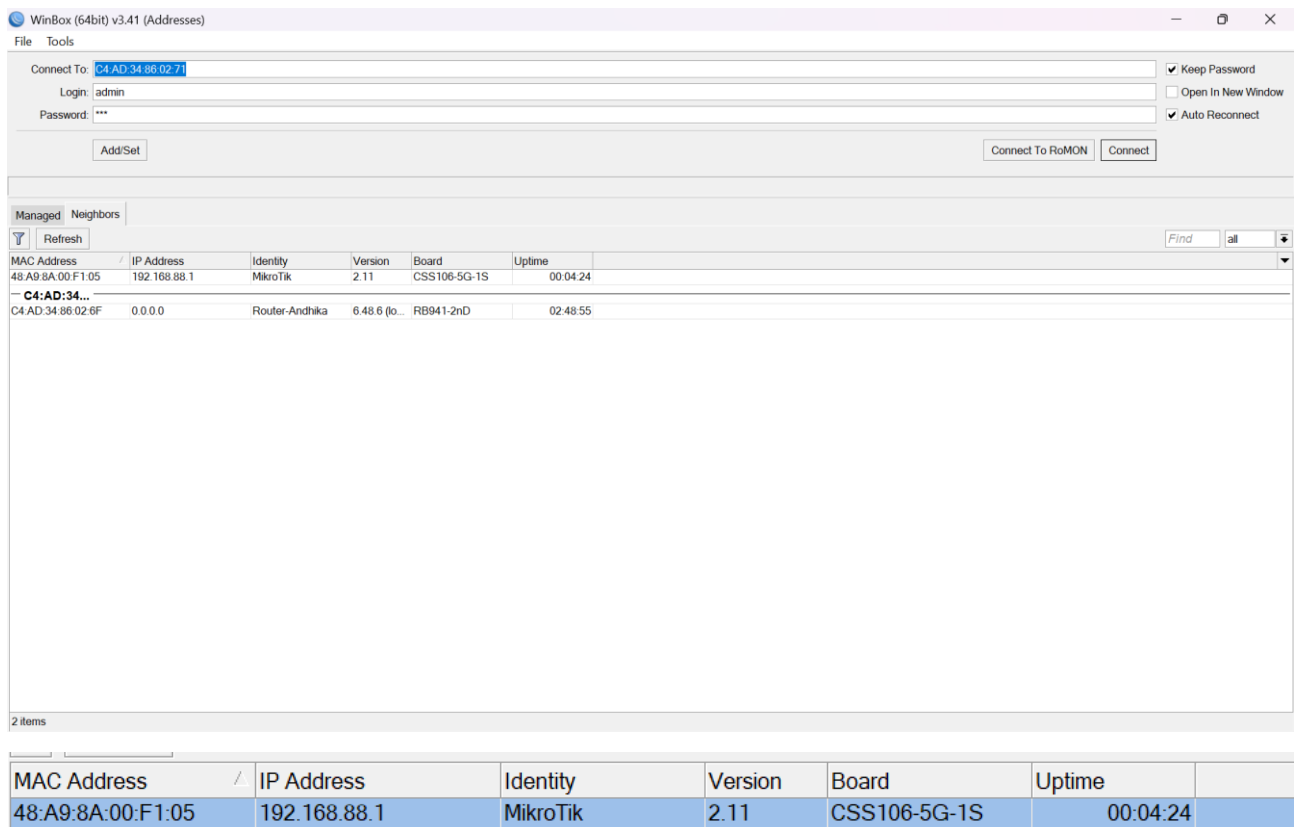


Shortcut untuk ke control panel yang network connection

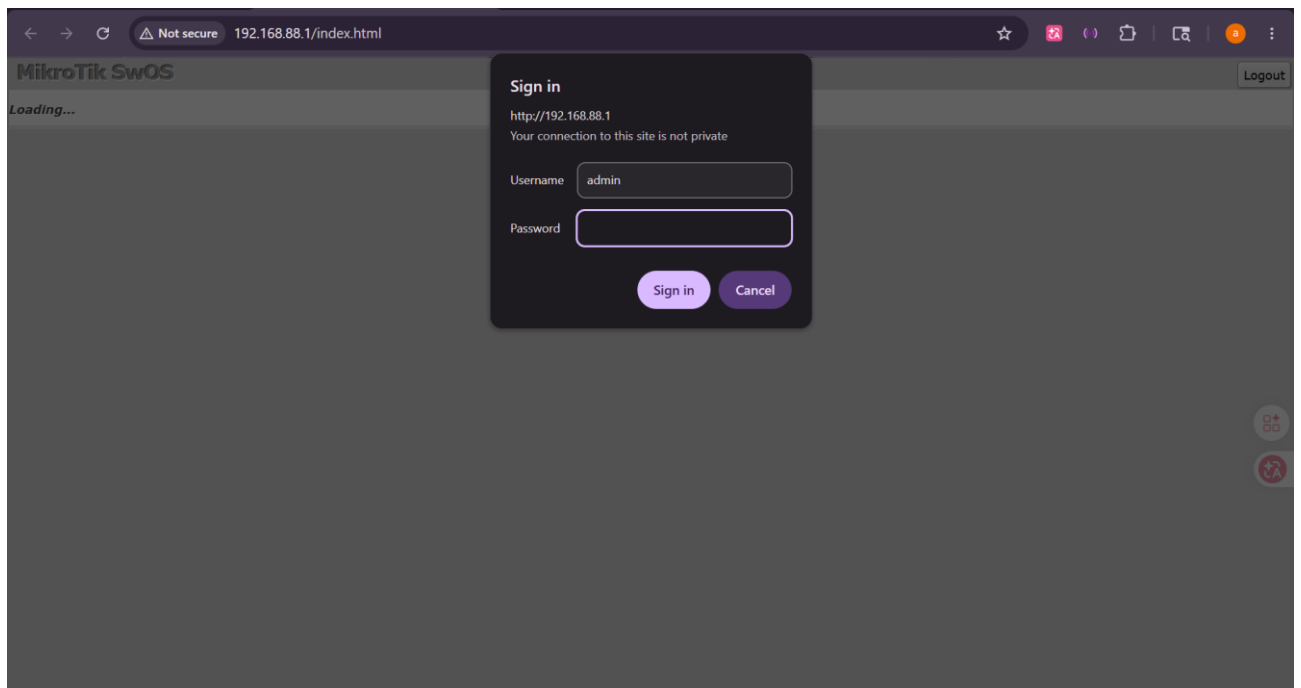




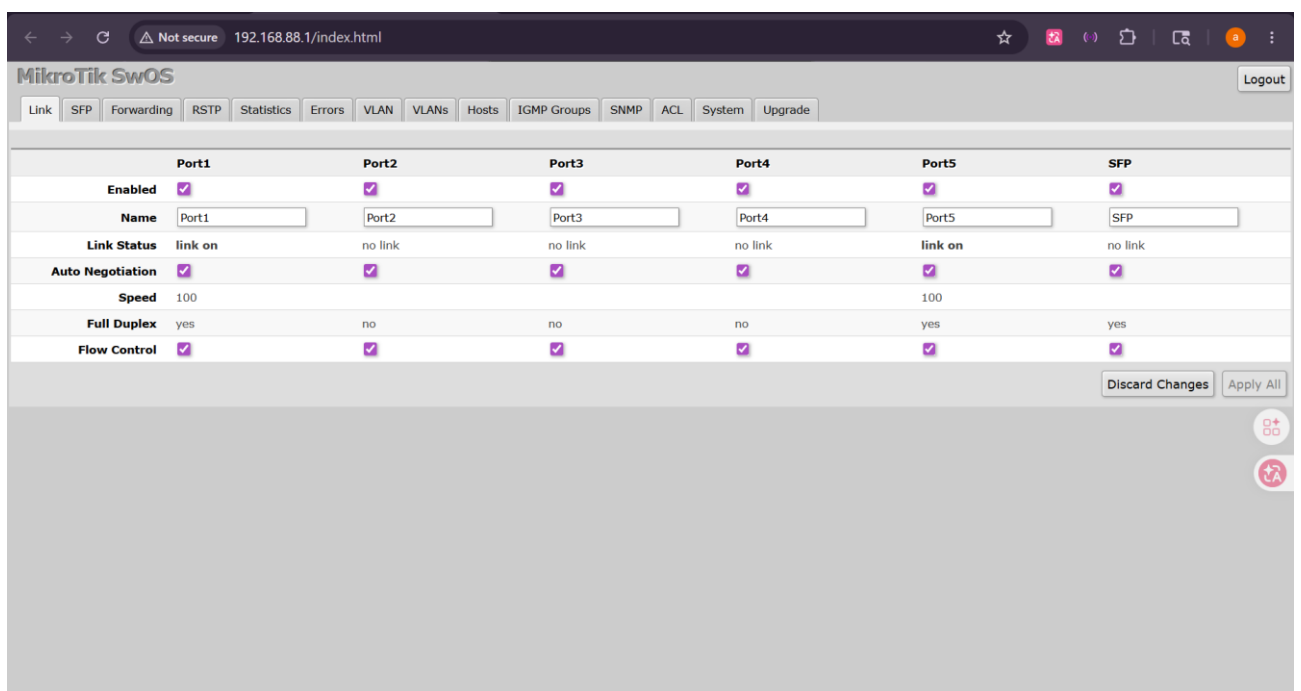
KLIK OK dan balik lagi ke winbox nya



Klik ip address 192.168.88.1 dan klik connect



Untuk username nya menggunakan admin dan tanpa password setelah itu langsung login



Pindah ke tab System untuk mengganti identity

MikroTik SwOS

Link SFP Forwarding RSTP Statistics Errors VLAN VLANs Hosts IGMP Groups SNMP ACL System Upgrade

General

Address Acquisition: DHCP with fallback

Static IP Address: 192.168.88.1

Identity: Switch-Andhika

Allow From:

Allow From Ports: ☒ 1 ☒ 2 ☒ 3 ☒ 4 ☒ 5 SFP

Allow From VLAN:

Watchdog: ☒

Independent VLAN Lookup: ☐

IGMP Snooping: ☐

Mikrotik Discovery Protocol: ☒

MAC Address: 48:a9:8a:00:f1:05

Serial Number: HDF08T387TQ

Board Name: CSS106-5G-1S

Pending changes Discard Changes Apply All

Password Change

Old Password:

New Password:

Mengganti identity jadi Switch-name, setelah itu pindah tab ke VLAN

MikroTik SwOS

Link SFP Forwarding RSTP Statistics Errors VLAN VLANs Hosts IGMP Groups SNMP ACL System Upgrade

Port1 Port2 Port3 Port4 Port5 SFP

Ingress

Port1	Port2	Port3	Port4	Port5	SFP
VLAN Mode: optional	enabled	enabled	optional	optional	optional
VLAN Receive: only tagged	only untagged	only untagged	any	any	any
Default VLAN ID: 1	100	200	1	1	1
Force VLAN ID: ☐	☐	☐	☐	☐	☐

Egress

Port1	Port2	Port3	Port4	Port5	SFP
VLAN Header: add if missing	always strip	always strip	leave as is	leave as is	leave as is

Pending changes Discard Changes Apply All

Setelah di apply all, pindah tab ke VLANS

MikroTik SwOS

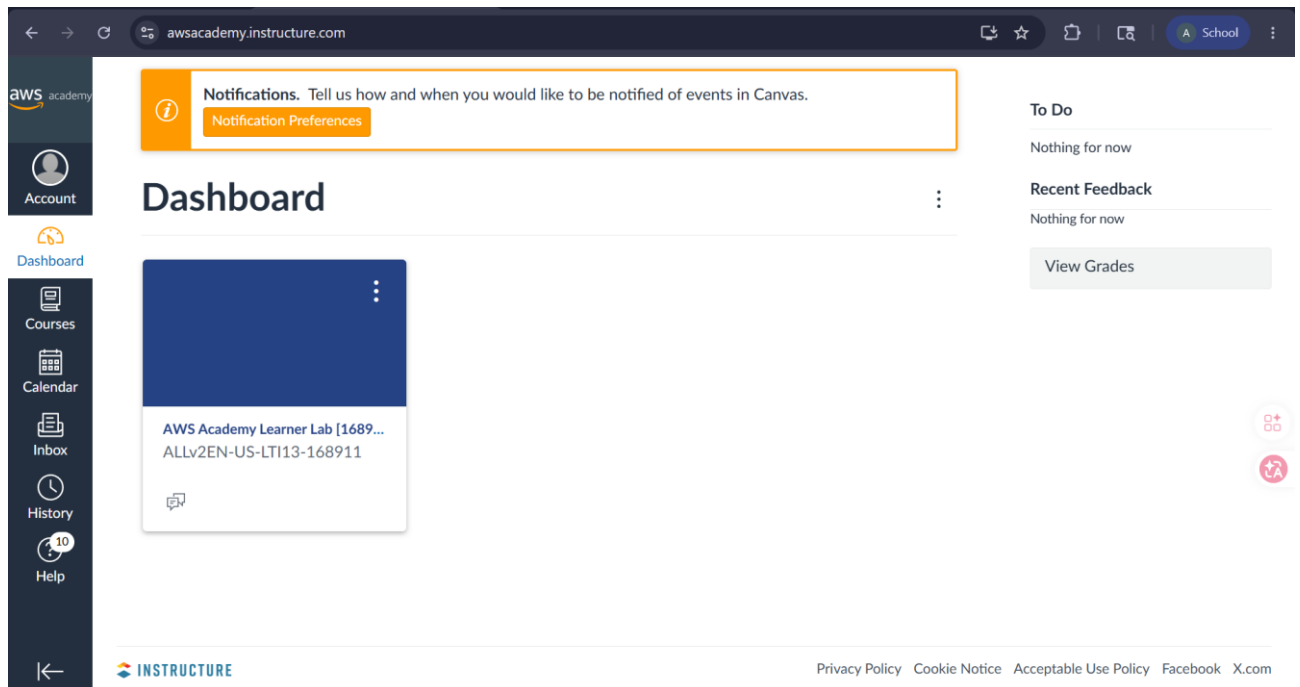
Link SFP Forwarding RSTP Statistics Errors VLAN VLANs Hosts IGMP Groups SNMP ACL System Upgrade

VLAN ID	IVL	IGMP Snooping	Port1	Port2	Port3	Port4	Port5	SFP	
100	☐	☐	add if missing	always strip	leave as is	leave as is	leave as is	leave as is	Cut Insert
200	☐	☐	add if missing	leave as is	always strip	leave as is	leave as is	leave as is	Cut Insert

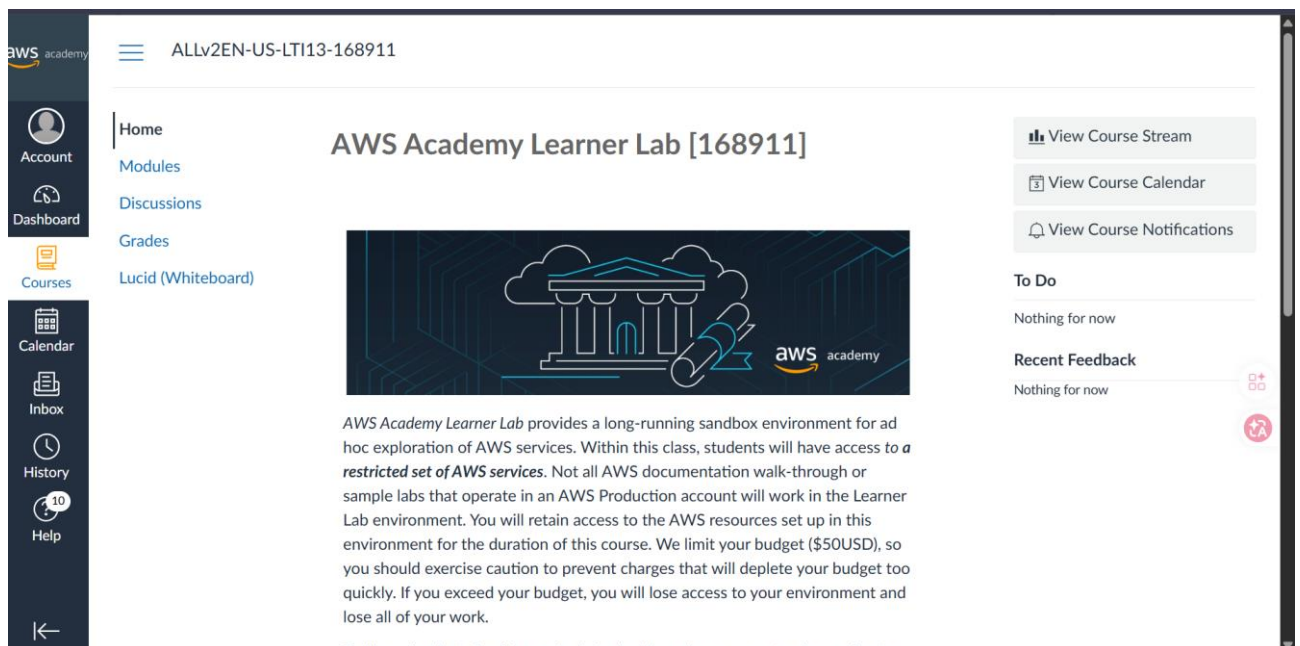
Pending changes Append Sort Discard Changes Apply All

Tambahkan seperti di foto

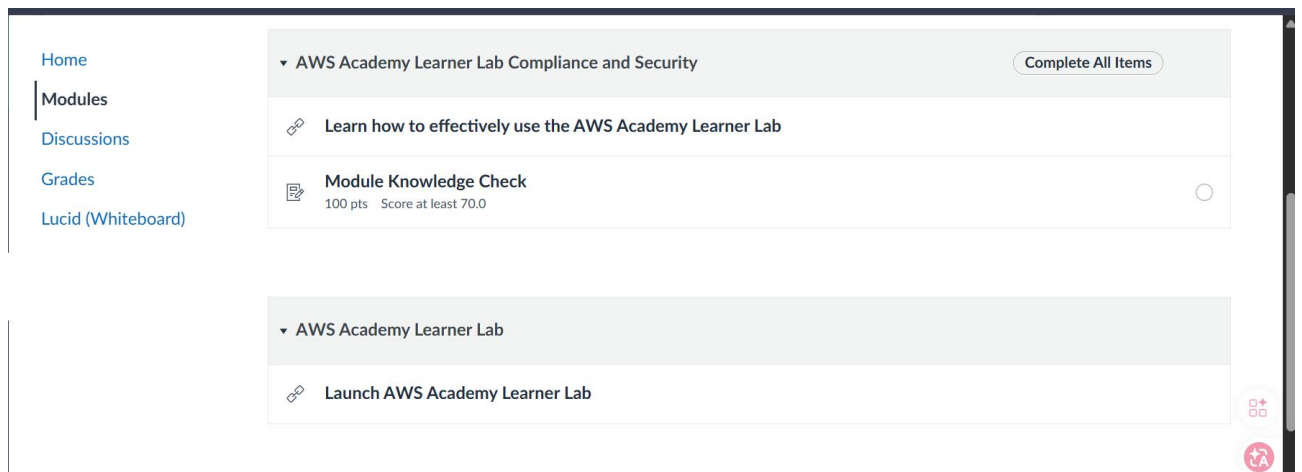
3. AWS



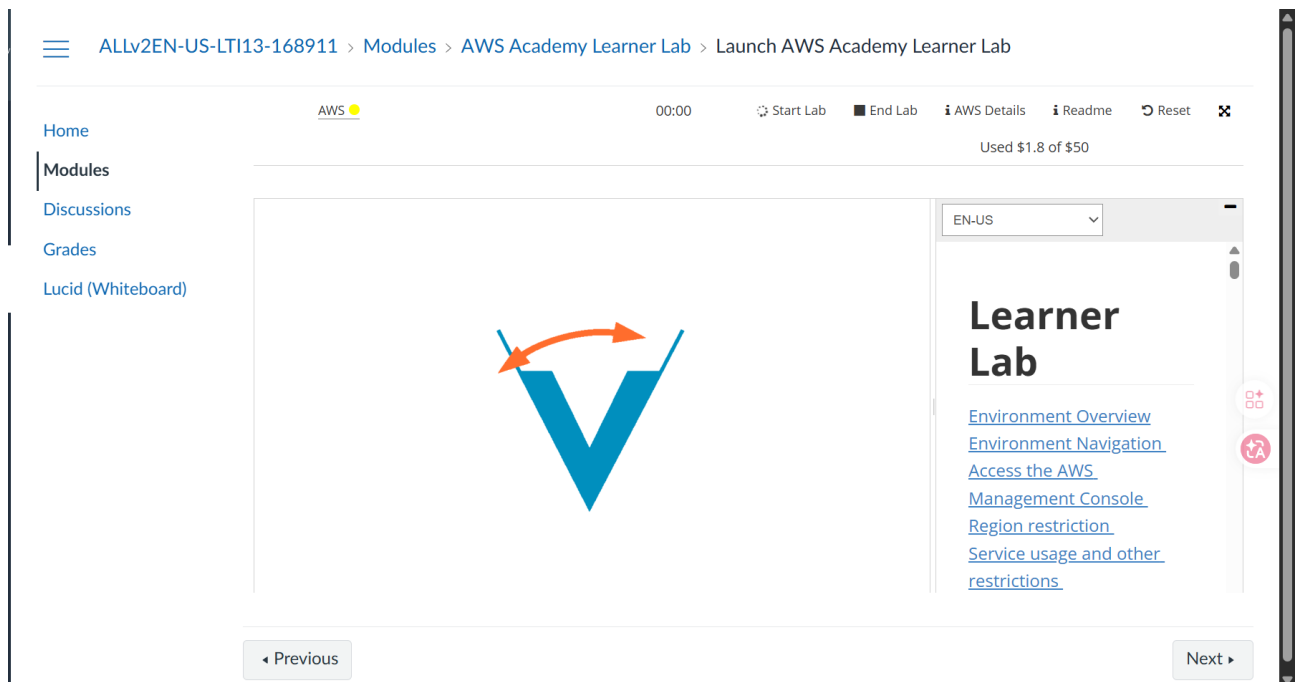
Login [aws](#) nya dulu, di dashboard klik courses nya



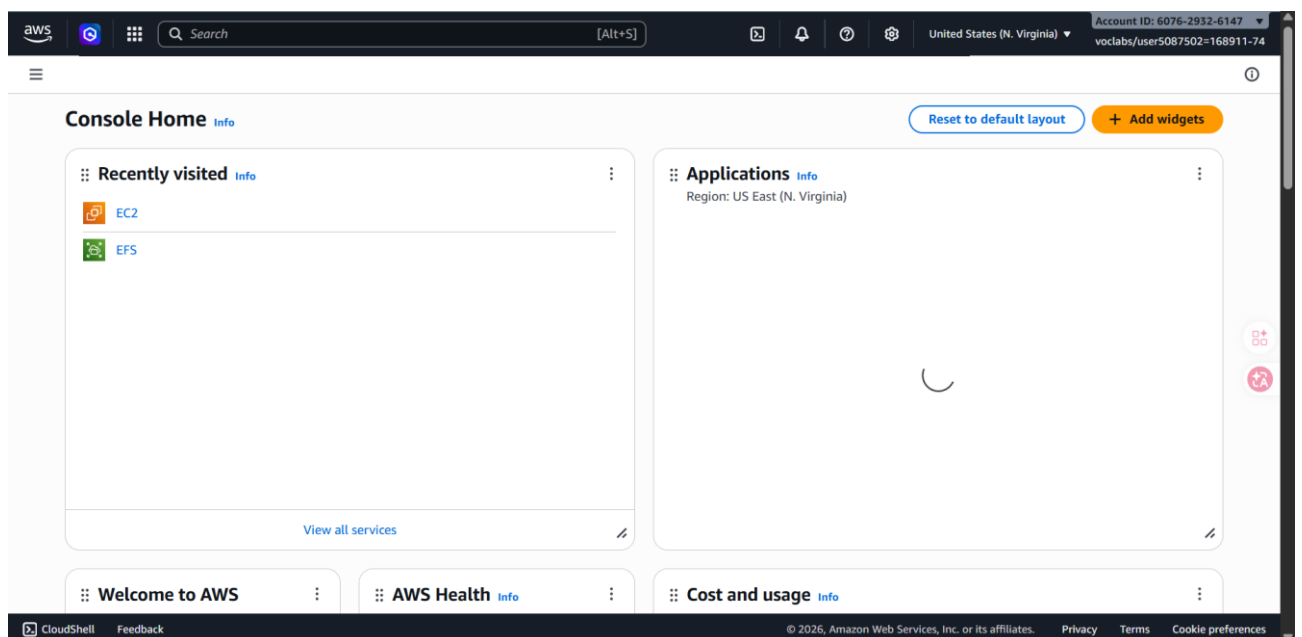
Setelah itu pindah tab ke modules



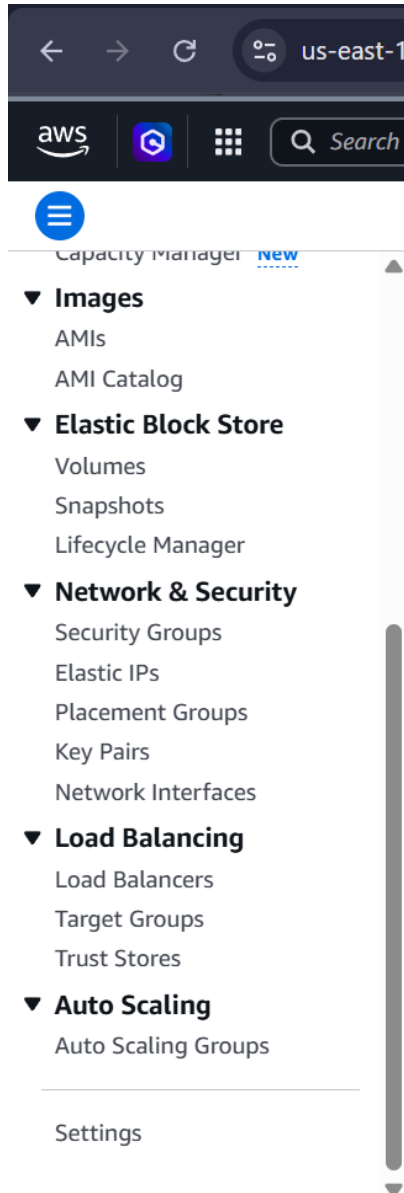
Cara yang launch aws academy learner lab dan klik



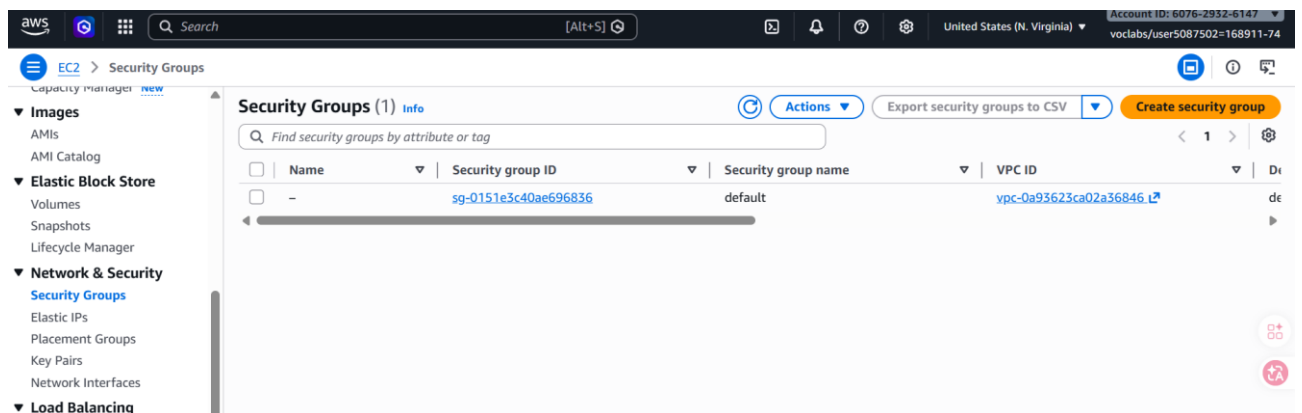
Klik start lab dan masuk ke console nya



Klik EC2



Pindah ke tab Security Groups



Klik create security group

EC2 > Security Groups > Create security group

Basic details

Security group name Info

Name cannot be edited after creation.

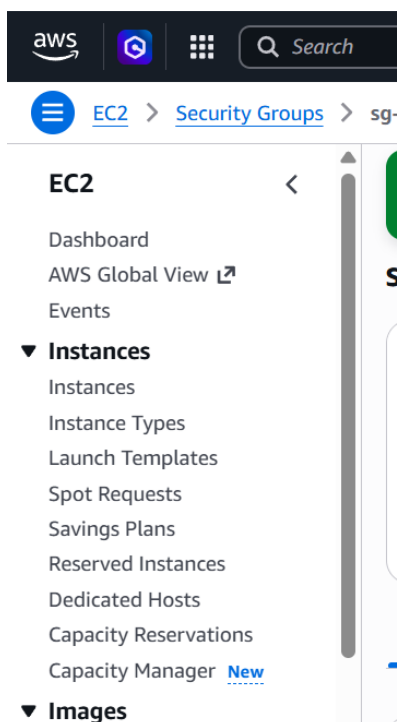
Description Info

VPC Info

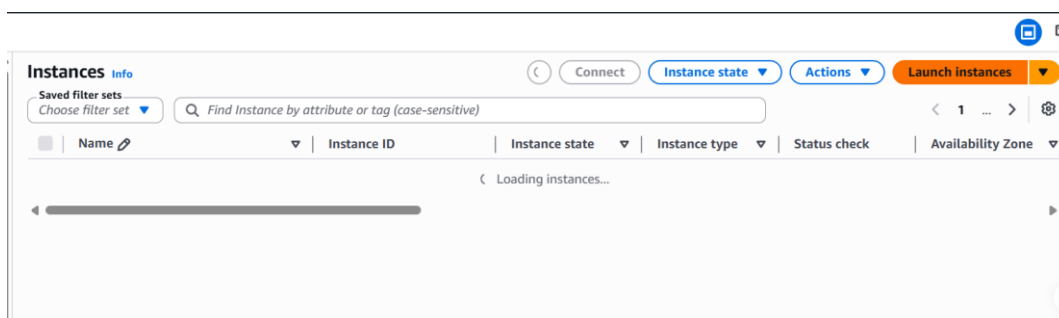
Inbound rules

Type	Protocol	Port range	Source	Description - optional	
Custom TCP	TCP	2202	Any...		Delete
HTTP	TCP	80	Any...	0.0.0.0/0	Delete
HTTPS	TCP	443	Any...	0.0.0.0/0	Delete

Isi nama nya jadi SG-Web-ASAT-Nama dan tambahkan inbound rules ssh, http, dan https. Btw untuk ssh nya berhubung mau pakai port 2202, jadi type nya di ubah jadi custom TCP Setelah itu klik create di paling bawah.



Pindah ke tab instances



Klik launch instances

Launch an instance [Info](#)

Amazon EC2 allows you to create virtual machines, or instances, that run on the AWS Cloud. Quickly get started by following the simple steps below.

▼ Name and tags [Info](#)

Key Info Name	Value Info SRV-ASAT-Andhika	Resource types Info Select resource types	Remove
Add new tag	Use: SRV-ASAT-Andhika	Instances	

You can add up to 49 more tags.

Ganti nama instances

▼ Application and OS Images (Amazon Machine Image) [Info](#)

An AMI contains the operating system, application server, and applications for your instance. If you don't see a suitable AMI below, use the search field or choose [Browse more AMIs](#).

Recents | My AMIs | **Quick Start**

Amazon Linux	macOS	Ubuntu	Windows	Red Hat	SUSE Linux	Debian

[Browse more AMIs](#)
Including AMIs from AWS, Marketplace and the Community

Amazon Machine Image (AMI)

Ubuntu Server 24.04 LTS (HVM), SSD Volume Type
 ami-05cf1e9f73fbad2e2 (64-bit (x86)) / ami-0953e2223326856ce (64-bit (Arm))
 Virtualization: hvm ENA enabled: true Root device type: ebs

Free tier eligible

Description

Ubuntu Server 24.04 LTS (HVM),EBS General Purpose (SSD) Volume Type. Support available from Canonical (<http://www.ubuntu.com/cloud/services>).

Canonical, Ubuntu, 24.04, amd64 noble image

Architecture AMI ID Publish Date Iternama

Pilih ubuntu atau debian bebas yang penting pakai yang free tier eligible

▼ Instance type [Info](#) | [Get advice](#)

Instance type

t2.small
 Family: t2 1 vCPU 2 GiB Memory Current generation: true
 On-Demand Linux base pricing: 0.023 USD per Hour
 On-Demand RHEL base pricing: 0.0376 USD per Hour
 On-Demand Windows base pricing: 0.032 USD per Hour
 On-Demand SUSE base pricing: 0.053 USD per Hour
 On-Demand Ubuntu Pro base pricing: 0.025 USD per Hour

☒ All generations

[Compare instance types](#)

Additional costs apply for AMIs with pre-installed software

▼ Key pair (login) [Info](#)

You can use a key pair to securely connect to your instance. Ensure that you have access to the selected key pair before you launch the instance.

Key pair name - required

Pilih type nya t2.small dan pakai key pair yang vockey

▼ **Network settings** [Info](#) Edit

Network [Info](#)
vpc-0a93623ca02a36846

Subnet [Info](#)
No preference (Default subnet in any availability zone)

Auto-assign public IP [Info](#)
Enable

Firewall (security groups) [Info](#)
A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.

☐ Create security group ☒ Select existing security group

Common security groups [Info](#)
Select security groups ▼ Compare security group rules

SG-Web-ASAT-Andhika sg-00059be165ebd884b ×
VPC: vpc-0a93623ca02a36846

Security groups that you add or remove here will be added to or removed from all your network interfaces.

Klik select existing SG dan pilih yang baru dibuat

▼ **Configure storage** [Info](#) Advanced

1x GiB ▼ Root volume, 3000 IOPS, Not encrypted

Add new volume

Storage nya isi 15GB

▼ **Summary**

Number of instances [Info](#)

Software Image (AMI)
Canonical, Ubuntu, 24.04, amd64...[read more](#)
ami-05cf1e9f73fbad2e2

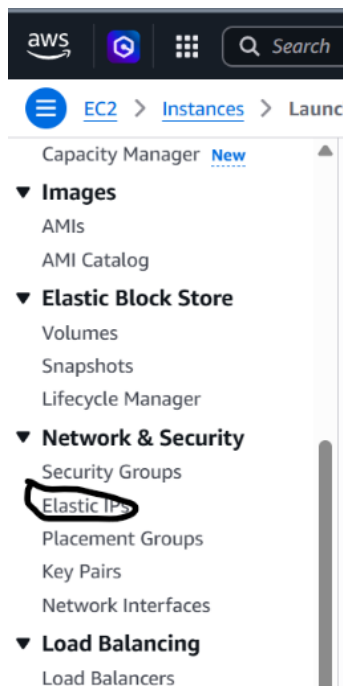
Virtual server type (instance type)
t2.small

Firewall (security group)
SG-Web-ASAT-Andhika

Storage (volumes)
1 volume(s) - 15 GiB

Cancel Launch instance Preview code

Klik launch instance, setelah itu pindah tab ke elastic ip



Elastic IP addresses [Info](#) [Actions](#) [Allocate Elastic IP address](#)

☐	Name	Allocated IPv4 addr...	Type	Allocation ID	Reverse DNS record
No Elastic IP addresses found in this Region					

Elastic IP address settings [Info](#)

Public IPv4 address pool

- ☒ Amazon's pool of IPv4 addresses
- ☐ Public IPv4 address that you bring to your AWS account with BYOIP. (option disabled because no pools found) [Learn more](#)
- ☐ Customer-owned pool of IPv4 addresses created from your on-premises network for use with an Outpost. (option disabled because no customer owned pools found) [Learn more](#)
- ☐ Allocate using an IPv4 IPAM pool (option disabled because no public IPv4 IPAM pools with AWS service as EC2 were found)

Network border group [Info](#)

Global static IP addresses

AWS Global Accelerator can provide global static IP addresses that are announced worldwide using anycast from AWS edge locations. This can help improve the availability and latency for your user traffic by using the Amazon global network. [Learn more](#)

[Create accelerator](#)

Tags - optional

A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.

Key

[Add new tag](#)

Value - optional

[Remove](#)

Use: "IP-ASAT-Andhika"

You can add up to 49 more tag

Elastic IP address allocated successfully.

Elastic IP address 52.204.155.102 / IP-ASAT-Andhika

[Associate this Elastic IP address](#)

Elastic IP addresses (1) [Info](#) [Actions](#) [Allocate Elastic IP address](#)

Klik associate this elastic ip address

EC2 > Elastic IP addresses > Associate Elastic IP address

Associate Elastic IP address [Info](#)

Choose the instance or network interface to associate to this Elastic IP address (52.204.155.102)

Elastic IP address: 52.204.155.102

Resource type
Choose the type of resource with which to associate the Elastic IP address.

☒ Instance
☐ Network interface

Instance
i-0315514b121445340

Private IP address
The private IP address with which to associate the Elastic IP address.
Choose a private IP address

Reassociation
Specify whether the Elastic IP address can be reassociated with a different resource if it already associated with a resource.
☐ Allow this Elastic IP address to be reassociated

[Cancel](#) [Associate](#)

Pilih instance yang di buat tadi dan klik associate

Elastic IP address associated successfully.
Elastic IP address 52.204.155.102 has been associated with instance i-0315514b121445340

Elastic IP addresses (1) [Info](#)

Find elastic IP addresses by attribute or tag

Public IPv4 address : 52.204.155.102 [Clear filters](#)

☐	Name	Allocated IPv4 address	Type	Allocation ID	Reverse DNS record
☐	IP-ASAT-Andhika	52.204.155.102	Public IP	eipalloc-062ebd70c8a6fb420	-

Setelah itu buka winbox lagi

WinBox (64bit) v3.41 (Addresses)

File Tools

Connect To: C4-AD-34-86-02-71 ☒ Keep Password
Login: admin ☐ Open In New Window
Password: *** ☒ Auto Reconnect

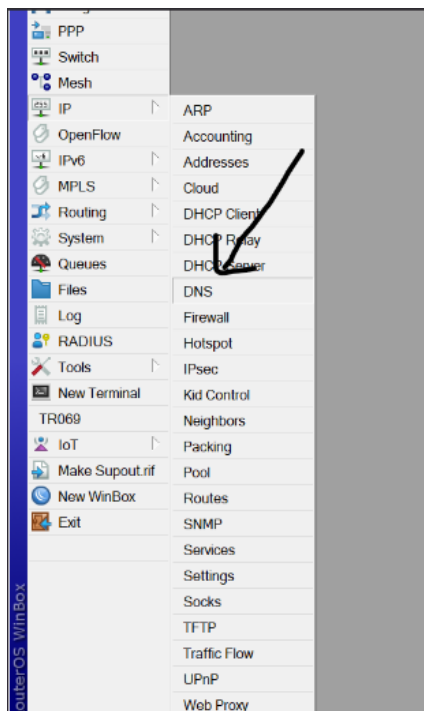
[Add/Set](#) [Connect To RoMON](#) [Connect](#)

Managed Neighbors

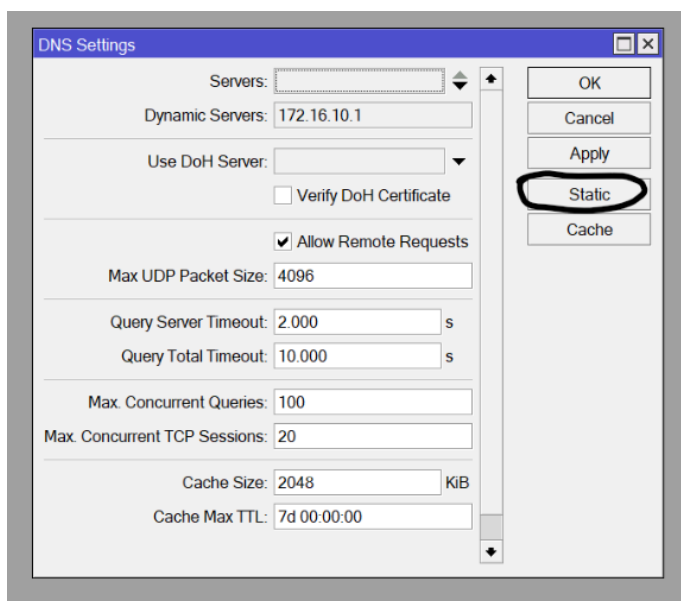
[Refresh](#) [Find](#) all

MAC Address	IP Address	Identity	Version	Board	Uptime
C4-AD-34-86-02-71	10.3.4.1	Router-Andhika	6.48.6 (to...	RB941-2nD	03:30:28

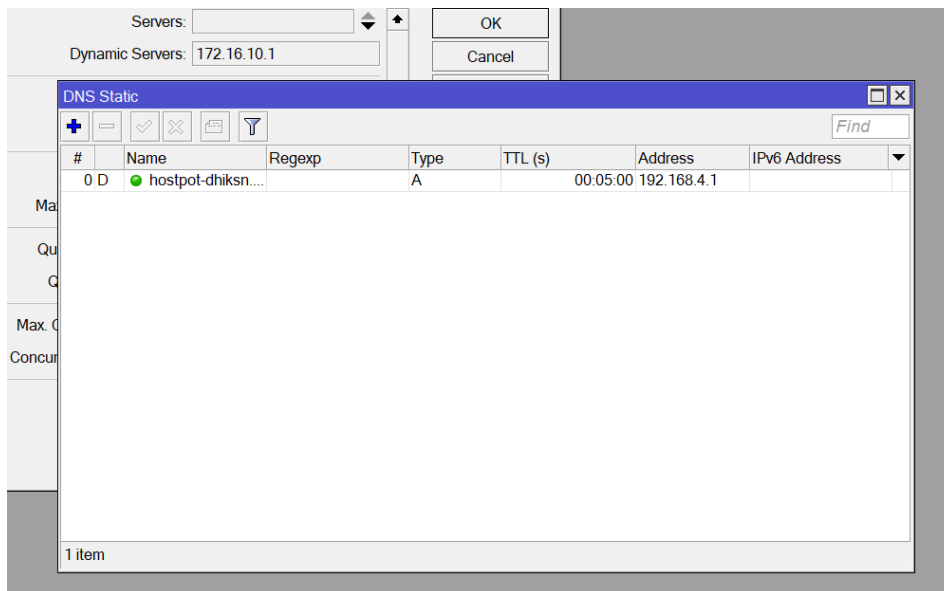
Pilih MAC address itu dan klik connect



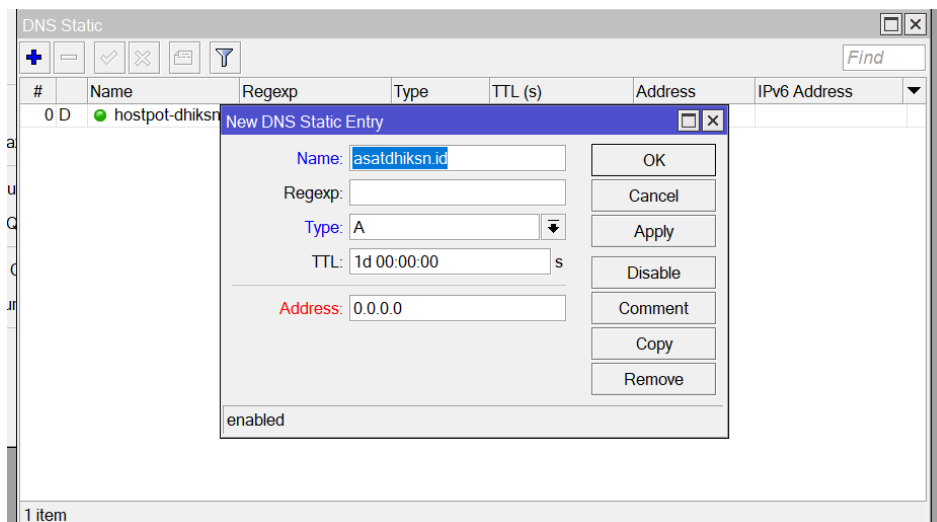
Ke tab DNS dan muncul panel dns settings



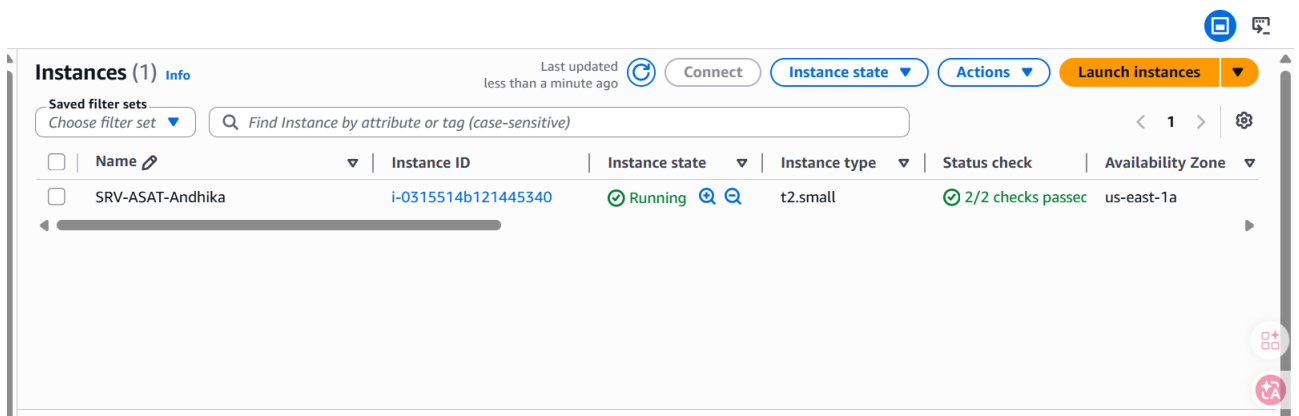
Klik Static yang di lingkaran



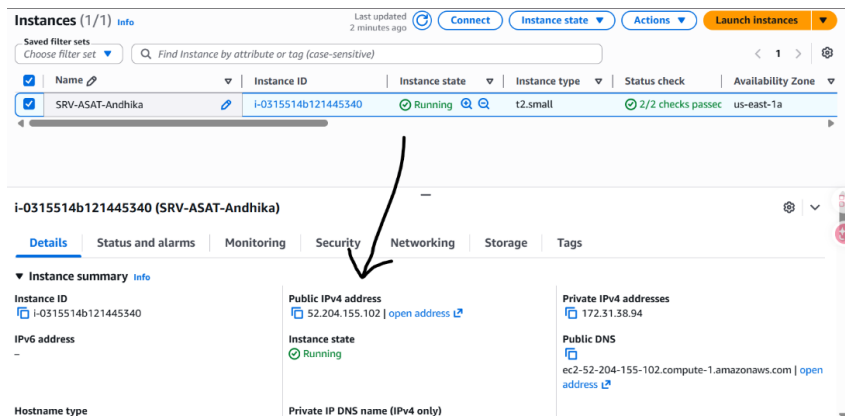
Tambahkan dns static baru dengan cara klik “+”



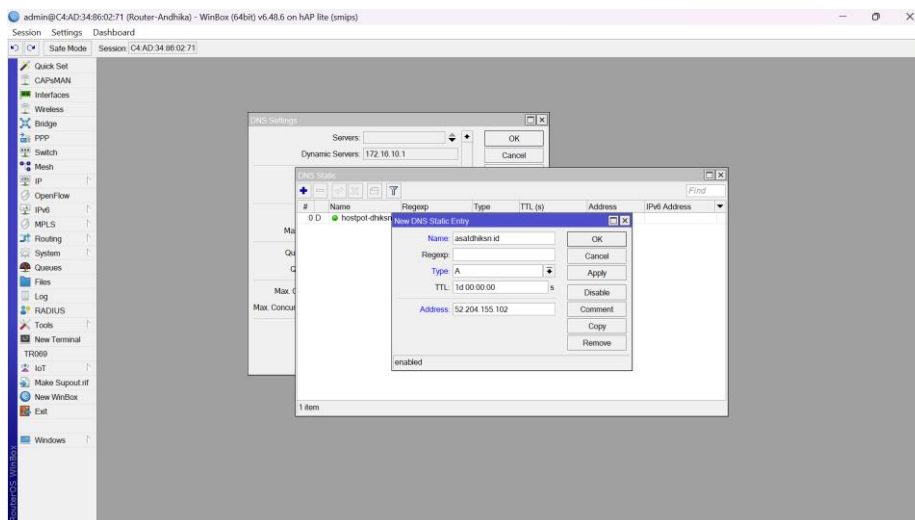
Pakai domain asatname.id untuk address nya pakai ip public EC2, jadi balik lagi ke aws nya



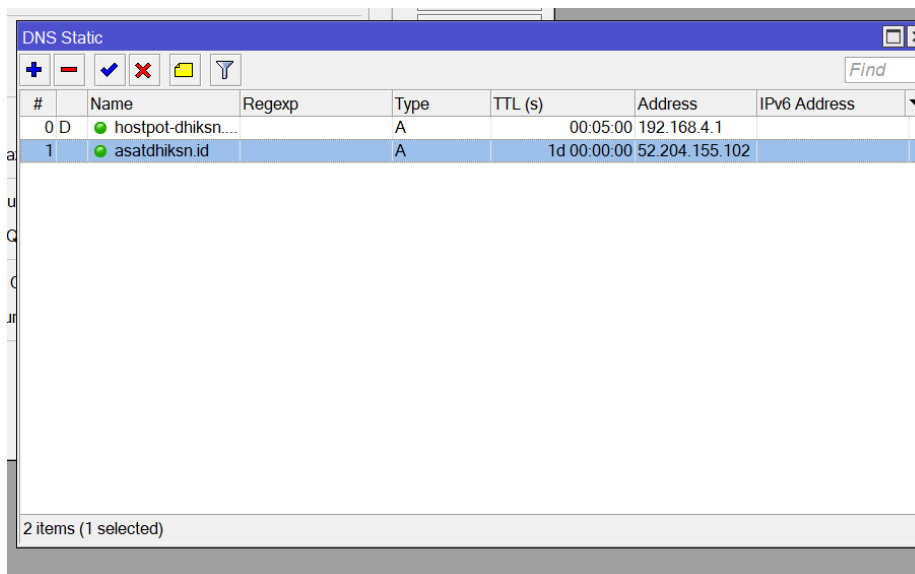
Klik instance nya



Copy ip public EC2 dan balik lagi ke mikrotik nya



Taruh IP Public EC2 nya di Address setelah itu klik apply dan ok



Setelah itu balik ke aws nya dan copy ip public ec2 nya untuk di remote di putty

The screenshot shows the AWS Management Console 'Instances' page. A table lists one instance: 'SRV-ASAT-Andhika' with ID 'i-0315514b121445340', state 'Running', type 't2.small', and '2/2 checks passed'. Below the table, the 'Details' tab for this instance is active, showing its public IP address (52.204.155.102) and public DNS name (ec2-52-204-155-102.compute-1.amazonaws.com).

Tapi sebelum itu ke

The screenshot shows the AWS Academy Learner Lab interface. The main area contains a terminal window with the prompt 'eee_lj_6231188@runweb228350:~\$'. On the right, a 'Cloud Access' sidebar provides session information: 'Remaining session time: 03:27:46(208 minutes)', 'Session started at: 2026-05-19T23:28:59-0700', 'Session to end at: 2026-05-20T03:28:59-0700', and 'Accumulated lab time: 08:00:00 (480 minutes)'. It also states 'No running instance'.

This is a second view of the AWS Academy Learner Lab interface, showing the same session details as the previous screenshot. The 'Cloud Access' sidebar on the right indicates the session is still active with 03:27:46 remaining.

Klik AWS details

ib

AWS Details

Readme

Reset

Used \$1.8 of \$50

minutes)

Session started at: 2026-05-19T23:28:59-0700

Session to end at: 2026-05-20T03:28:59-0700

Accumulated lab time: 08:00:00 (480 minutes)

No running instance

SSH key

Show

Download PEM

Download PPK

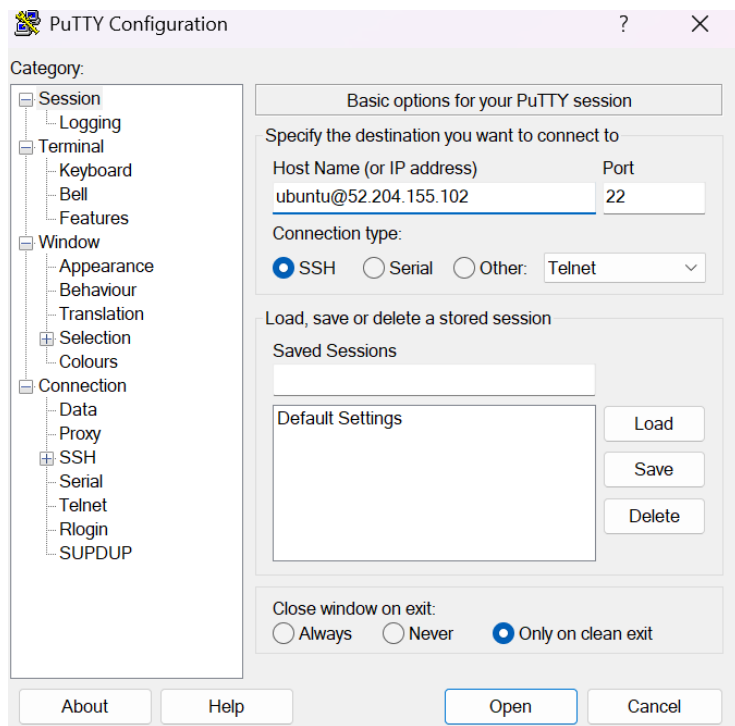
AWS SSO

Download URL

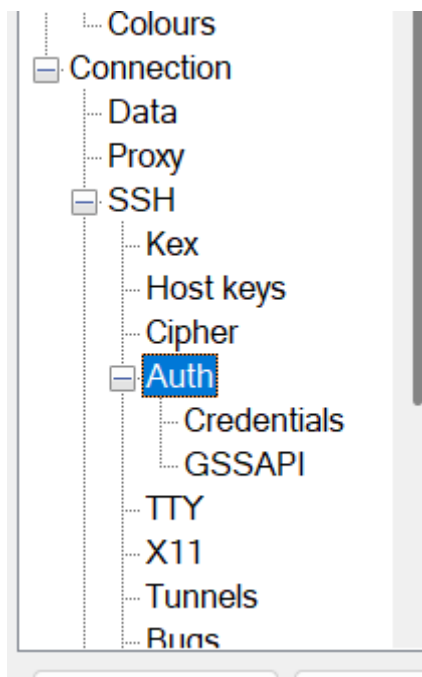
AWSAccountId607629326147

Next ▶

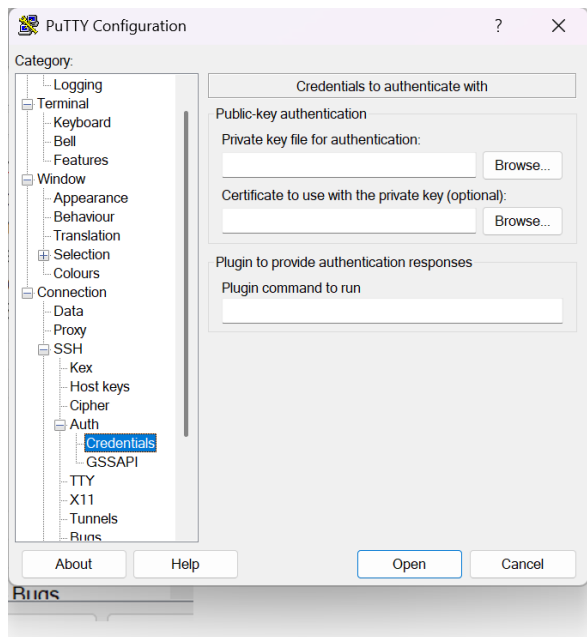
Klik download ppk



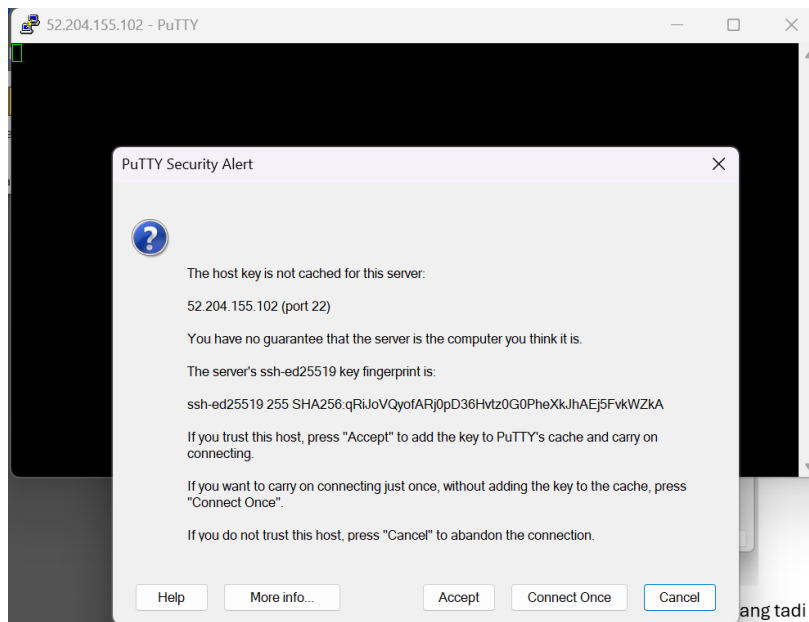
Buka putty dan masukan ubuntu@ip_public_ec2



Masukin ppk nya di ssh – auth – credentials



Pilih browse di private key dan pilih ppk yang tadi di download, setelah itu klik open



Klik accept

```
ubuntu@ip-172-31-38-94: ~  
Expanded Security Maintenance for Applications is not enabled.  
0 updates can be applied immediately.  
Enable ESM Apps to receive additional future security updates.  
See https://ubuntu.com/esm or run: sudo pro status  
The list of available updates is more than a week old.  
To check for new updates run: sudo apt update  
The programs included with the Ubuntu system are free software;  
the exact distribution terms for each program are described in the  
individual files in /usr/share/doc/*/copyright.  
Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by  
applicable law.  
To run a command as administrator (user "root"), use "sudo <command>".  
See "man sudo_root" for details.  
ubuntu@ip-172-31-38-94:~$ sudo su -
```

Setelah masuk langsung type sudo su – dan enter

```
root@ip-172-31-38-94: ~  
root@ip-172-31-38-94:~# apt update  
Hit:1 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble InRelease  
Hit:2 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates InRelease  
Hit:3 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-backports InRelease  
Hit:4 http://security.ubuntu.com/ubuntu noble-security InRelease  
Reading package lists... Done  
Building dependency tree... Done  
Reading state information... Done  
24 packages can be upgraded. Run 'apt list --upgradable' to see them.  
root@ip-172-31-38-94:~#
```

ketik apt update setelah itu Install web server, php, library php, unzip dan git

```
root@ip-172-31-38-94: ~  
root@ip-172-31-38-94:~# sudo apt install git unzip php8.3-fpm php8.3-mysql php8.3-common php8.3  
-cli php8.3-curl php8.3-mbstring php8.3-xml php8.3-zip nginx -y  
Reading package lists... Done  
Building dependency tree... Done  
Reading state information... Done  
git is already the newest version (1:2.43.0-1ubuntu7.3).  
unzip is already the newest version (6.0-28ubuntu4.1).  
php8.3-fpm is already the newest version (8.3.6-0ubuntu0.24.04.8).  
php8.3-mysql is already the newest version (8.3.6-0ubuntu0.24.04.8).  
php8.3-common is already the newest version (8.3.6-0ubuntu0.24.04.8).  
php8.3-cli is already the newest version (8.3.6-0ubuntu0.24.04.8).  
php8.3-curl is already the newest version (8.3.6-0ubuntu0.24.04.8).  
php8.3-mbstring is already the newest version (8.3.6-0ubuntu0.24.04.8).  
php8.3-xml is already the newest version (8.3.6-0ubuntu0.24.04.8).  
php8.3-zip is already the newest version (8.3.6-0ubuntu0.24.04.8).  
nginx is already the newest version (1.24.0-2ubuntu7.8).  
0 upgraded, 0 newly installed, 0 to remove and 24 not upgraded.  
root@ip-172-31-38-94:~#
```

```
root@ip-172-31-38-94: ~
root@ip-172-31-38-94:~# systemctl restart nginx
root@ip-172-31-38-94:~# systemctl status nginx
● nginx.service - A high performance web server and a reverse proxy server
   Loaded: loaded (/usr/lib/systemd/system/nginx.service; enabled; preset: enabled)
   Active: active (running) since Wed 2026-05-20 07:25:02 UTC; 10s ago
     Docs: man:nginx(8)
   Process: 12291 ExecStartPre=/usr/sbin/nginx -t -q -g daemon on; master_process on; (code=exited, status=0/SUCCESS)
   Process: 12295 ExecStart=/usr/sbin/nginx -g daemon on; master_process on; (code=exited, status=0/SUCCESS)
   Main PID: 12296 (nginx)
    Tasks: 2 (limit: 2325)
   Memory: 1.7M (peak: 1.8M)
      CPU: 13ms
   CGroup: /system.slice/nginx.service
           └─12296 "nginx: master process /usr/sbin/nginx -g daemon on; master_process on;"
             └─12297 "nginx: worker process"

May 20 07:25:02 ip-172-31-38-94 systemd[1]: Starting nginx.service - A high performance web server and a reverse proxy server:
May 20 07:25:02 ip-172-31-38-94 systemd[1]: Started nginx.service - A high performance web server and a reverse proxy server:
lines 1-16/16 (END)
```

Restart service web server dan cek status

```
root@ip-172-31-38-94: /var/www/html
root@ip-172-31-38-94:~# cd /var/www/html/
root@ip-172-31-38-94:/var/www/html# ls
index.nginx-debian.html
root@ip-172-31-38-94:/var/www/html# mkdir site1
root@ip-172-31-38-94:/var/www/html# mkdir site2
root@ip-172-31-38-94:/var/www/html# mkdir site3
root@ip-172-31-38-94:/var/www/html# ls
index.nginx-debian.html  site1  site2  site3
root@ip-172-31-38-94:/var/www/html#
```

Pindah ke /var/www/html dan membuat 3 dir baru untuk di git clone

```
root@ip-172-31-38-94: /var/www/html/site1
root@ip-172-31-38-94:/var/www/html# cd site1
root@ip-172-31-38-94:/var/www/html/site1# git clone https://github.com/dhiksn/dhiksn.github.io.git
Cloning into 'dhiksn.github.io'...
remote: Enumerating objects: 12, done.
remote: Counting objects: 100% (12/12), done.
remote: Compressing objects: 100% (8/8), done.
remote: Total 12 (delta 0), reused 12 (delta 0), pack-reused 0 (from 0)
Receiving objects: 100% (12/12), 619.87 KiB | 24.79 MiB/s, done.
root@ip-172-31-38-94:/var/www/html/site1# mv dhiksn.github.io/* /var/www/html/site1/
root@ip-172-31-38-94:/var/www/html/site1#
```

Di git clone di site1 dulu dan pindahkan isi file dhiksn.github.io itu ke /var/www/html/site1

```
root@ip-172-31-38-94: /var/www/html/site2
root@ip-172-31-38-94:/var/www/html/site2# git clone https://github.com/dhiksn/webtjkt1.git
Cloning into 'webtjkt1'...
remote: Enumerating objects: 19, done.
remote: Counting objects: 100% (19/19), done.
remote: Compressing objects: 100% (16/16), done.
remote: Total 19 (delta 5), reused 15 (delta 3), pack-reused 0 (from 0)
Receiving objects: 100% (19/19), 19.15 KiB | 6.38 MiB/s, done.
Resolving deltas: 100% (5/5), done.
root@ip-172-31-38-94:/var/www/html/site2# mv webtjkt1/* /var/www/html/site2
root@ip-172-31-38-94:/var/www/html/site2#
```

Di git clone di site2 dulu dan pindahkan isi file webtjkt itu ke /var/www/html/site2

```
root@ip-172-31-38-94: /var/www/html/site3
root@ip-172-31-38-94:/var/www/html/site3# git clone https://github.com/dhiksn/webtjkt2.git
Cloning into 'webtjkt2'...
remote: Enumerating objects: 18, done.
remote: Counting objects: 100% (18/18), done.
remote: Compressing objects: 100% (15/15), done.
remote: Total 18 (delta 5), reused 14 (delta 3), pack-reused 0 (from 0)
Receiving objects: 100% (18/18), 17.62 KiB | 8.81 MiB/s, done.
Resolving deltas: 100% (5/5), done.
root@ip-172-31-38-94:/var/www/html/site3# mv webtjkt2/* /var/www/html/site3
root@ip-172-31-38-94:/var/www/html/site3#
```

Di git clone di site3 dulu dan pindahkan isi file webtjkt itu ke /var/www/html/site3

```
root@ip-172-31-38-94: /var/www/html
root@ip-172-31-38-94:/var/www/html# systemctl restart nginx && systemctl status nginx
● nginx.service - A high performance web server and a reverse proxy server
   Loaded: loaded (/usr/lib/systemd/system/nginx.service; enabled; preset: enabled)
   Active: active (running) since Wed 2026-05-20 07:33:05 UTC; 13ms ago
     Docs: man:nginx(8)
   Process: 12389 ExecStartPre=/usr/sbin/nginx -t -q -g daemon on; master_process on; (code=ex>
   Process: 12392 ExecStart=/usr/sbin/nginx -g daemon on; master_process on; (code=exited, st>
  Main PID: 12393 (nginx)
    Tasks: 2 (limit: 2325)
   Memory: 1.9M (peak: 1.9M)
      CPU: 13ms
   CGroup: /system.slice/nginx.service
           └─12393 "nginx: master process /usr/sbin/nginx -g daemon on; master_process on;"
             └─12394 "nginx: worker process"

May 20 07:33:05 ip-172-31-38-94 systemd[1]: Starting nginx.service - A high performance web se>
May 20 07:33:05 ip-172-31-38-94 systemd[1]: Started nginx.service - A high performance web ser>
lines 1-16/16 (END)
```

Di restart dan di cek status web server nya, setelah itu install mariadb-server dan konfig

```
root@ip-172-31-38-94: /var/www/html
root@ip-172-31-38-94:/var/www/html# apt install mariadb-server
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
mariadb-server is already the newest version (1:10.11.14-0ubuntu0.24.04.1).
0 upgraded, 0 newly installed, 0 to remove and 24 not upgraded.
root@ip-172-31-38-94:/var/www/html#
```

Setelah install, Lakukan instalasi keamanan mysql (mysql secure installation)

```
root@ip-172-31-38-94: /var/www/html
root@ip-172-31-38-94:/var/www/html# mysql_secure_installation

NOTE: RUNNING ALL PARTS OF THIS SCRIPT IS RECOMMENDED FOR ALL MariaDB
SERVERS IN PRODUCTION USE! PLEASE READ EACH STEP CAREFULLY!

In order to log into MariaDB to secure it, we'll need the current
password for the root user. If you've just installed MariaDB, and
haven't set the root password yet, you should just press enter here.

Enter current password for root (enter for none):
OK, successfully used password, moving on...

Setting the root password or using the unix_socket ensures that nobody
can log into the MariaDB root user without the proper authorisation.

You already have your root account protected, so you can safely answer 'n'.

Switch to unix_socket authentication [Y/n] y
Enabled successfully!
Reloading privilege tables..
... Success!

You already have your root account protected, so you can safely answer 'n'.

Change the root password? [Y/n] y
New password:
Re-enter new password:
Password updated successfully!
Reloading privilege tables..
... Success!

By default, a MariaDB installation has an anonymous user, allowing anyone
to log into MariaDB without having to have a user account created for
them. This is intended only for testing, and to make the installation
a bit smoother. You should remove them before moving into a
production environment.

Remove anonymous users? [Y/n] y
... Success!

Normally, root should only be allowed to connect from 'localhost'. This
ensures that someone cannot guess at the root password from the network.

Disallow root login remotely? [Y/n] y
... Success!

By default, MariaDB comes with a database named 'test' that anyone can
access. This is also intended only for testing, and should be removed
before moving into a production environment.

Remove test database and access to it? [Y/n] y
- Dropping test database...
... Success!
- Removing privileges on test database...
... Success!

Reloading the privilege tables will ensure that all changes made so far
will take effect immediately.

Reload privilege tables now? [Y/n] y
... Success!

Cleaning up...

All done! If you've completed all of the above steps, your MariaDB
installation should now be secure.

Thanks for using MariaDB!
root@ip-172-31-38-94:/var/www/html#
```

Setelah ketik mysql_secure_installation tekan enter, type y, y, y, y, y, y

```
root@ip-172-31-38-94: /var/www/html

root@ip-172-31-38-94:/var/www/html# mysql -u root -p
Enter password:
Welcome to the MariaDB monitor.  Commands end with ; or \g.
Your MariaDB connection id is 41
Server version: 10.11.14-MariaDB-0ubuntu0.24.04.1 Ubuntu 24.04

Copyright (c) 2000, 2018, Oracle, MariaDB Corporation Ab and others.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

MariaDB [(none)]> █
```

Masuk ke mysql -u root -p dan buat database dan user baru

```
Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

MariaDB [(none)]> create database dbwpAndhika;
Query OK, 1 row affected (0.000 sec)

MariaDB [(none)]> create user 'userAndhika'@'localhost' IDENTIFIED BY 'asattkj2026';
Query OK, 0 rows affected (0.001 sec)

MariaDB [(none)]> GRANT ALL PRIVILEGES ON dbwpAndhika.* TO 'userAndhika'@'localhost';
Query OK, 0 rows affected (0.001 sec)

MariaDB [(none)]> flush privileges;
Query OK, 0 rows affected (0.000 sec)

MariaDB [(none)]> exit
Bye
root@ip-172-31-38-94:/var/www/html# █
```

Untuk database nya pakai nama dbwpNama, sedangkan untuk user nya pakai nama userName

```
root@ip-172-31-38-94: /var/www/html

root@ip-172-31-38-94:/var/www/html# sudo systemctl restart mariadb
root@ip-172-31-38-94:/var/www/html# sudo systemctl status mariadb
● mariadb.service - MariaDB 10.11.14 database server
   Loaded: loaded (/usr/lib/systemd/system/mariadb.service; enabled; preset: enabled)
   Active: active (running) since Wed 2026-05-20 07:43:33 UTC; 4s ago
     Docs: man:mariadb(8)
           https://mariadb.com/kb/en/library/systemd/
  Process: 13495 ExecStartPre=/usr/bin/install -m 755 -o mysql -g root -d /var/run/mysqld (code=exited, status=0/SUCCESS)
  Process: 13497 ExecStartPre=/bin/sh -c systemctl unset-environment _WSREP_START_POSITION (code=exited, status=0/SUCCESS)
  Process: 13499 ExecStartPre=/bin/sh -c [ ! -e /usr/bin/galera_recovery ] && VAR= || VAR="/usr/bin/galera_recovery" (code=exited, status=0/SUCCESS)
  Process: 13571 ExecStartPost=/bin/sh -c systemctl unset-environment _WSREP_START_POSITION (code=exited, status=0/SUCCESS)
  Process: 13573 ExecStartPost=/etc/mysql/debian-start (code=exited, status=0/SUCCESS)
 Main PID: 13559 (mariabdd)
   Status: "Taking your SQL requests now..."
    Tasks: 12 (limit: 15347)
  Memory: 78.7M (peak: 81.7M)
     CPU: 419ms
   CGroup: /system.slice/mariadb.service
           └─13559 /usr/sbin/mariabdd

May 20 07:43:33 ip-172-31-38-94 mariabdd[13559]: 2026-05-20 7:43:33 0 [Note] Plugin 'FEEDBACK' is disabled.
May 20 07:43:33 ip-172-31-38-94 mariabdd[13559]: 2026-05-20 7:43:33 0 [Warning] You need to use --log-bin.
May 20 07:43:33 ip-172-31-38-94 mariabdd[13559]: 2026-05-20 7:43:33 0 [Note] Server socket created on IP: 0.0.0.0.
May 20 07:43:33 ip-172-31-38-94 mariabdd[13559]: 2026-05-20 7:43:33 0 [Note] /usr/sbin/mariabdd: ready for connections.
May 20 07:43:33 ip-172-31-38-94 mariabdd[13559]: Version: '10.11.14-MariaDB-0ubuntu0.24.04.1' socket: '/var/run/mysqld/mysqld.sock' port: 3306 MySQL Community Server - GPL.
May 20 07:43:33 ip-172-31-38-94 mariabdd[13559]: 2026-05-20 7:43:33 0 [Note] InnoDB: Buffer pool(s) load completed at 2026-05-20 7:43:33
May 20 07:43:33 ip-172-31-38-94 systemd[1]: Started mariadb.service - MariaDB 10.11.14 database server.
May 20 07:43:33 ip-172-31-38-94 /etc/mysql/debian-start[13576]: Upgrading MariaDB tables if necessary.
May 20 07:43:33 ip-172-31-38-94 /etc/mysql/debian-start[13587]: Checking for insecure root accounts.
May 20 07:43:33 ip-172-31-38-94 /etc/mysql/debian-start[13591]: Triggering myisam-recover for all MyISAM tables.
lines 1-28/28 (END) █
```

Restart dan cek status service mariadb, selanjutnya kita download wordpress

```
root@ip-172-31-38-94: /tmp
root@ip-172-31-38-94:/tmp# wget https://wordpress.org/latest.tar.gz
--2026-05-20 07:46:33-- https://wordpress.org/latest.tar.gz
Resolving wordpress.org (wordpress.org)... 198.143.164.252, 2607:f978:5:8002::c68f:a4fc
Connecting to wordpress.org (wordpress.org)|198.143.164.252|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 27064862 (26M) [application/octet-stream]
Saving to: 'latest.tar.gz'

latest.tar.gz          100%[=====>] 25.81M  58.4MB/s   in 0.4s

2026-05-20 07:46:33 (58.4 MB/s) - 'latest.tar.gz' saved [27064862/27064862]

root@ip-172-31-38-94:/tmp#
```

```
root@ip-172-31-38-94: /tmp
wordpress/wp-includes/update.php
wordpress/wp-includes/user.php
wordpress/wp-includes/utf8.php
wordpress/wp-includes/vars.php
wordpress/wp-includes/version.php
wordpress/wp-includes/widgets/
wordpress/wp-includes/widgets/class-wp-nav-menu-widget.php
wordpress/wp-includes/widgets/class-wp-widget-archives.php
wordpress/wp-includes/widgets/class-wp-widget-block.php
wordpress/wp-includes/widgets/class-wp-widget-calendar.php
wordpress/wp-includes/widgets/class-wp-widget-categories.php
wordpress/wp-includes/widgets/class-wp-widget-custom-html.php
wordpress/wp-includes/widgets/class-wp-widget-links.php
wordpress/wp-includes/widgets/class-wp-widget-media-audio.php
wordpress/wp-includes/widgets/class-wp-widget-media-gallery.php
wordpress/wp-includes/widgets/class-wp-widget-media-image.php
wordpress/wp-includes/widgets/class-wp-widget-media-video.php
wordpress/wp-includes/widgets/class-wp-widget-media.php
wordpress/wp-includes/widgets/class-wp-widget-meta.php
wordpress/wp-includes/widgets/class-wp-widget-pages.php
wordpress/wp-includes/widgets/class-wp-widget-recent-comments.php
wordpress/wp-includes/widgets/class-wp-widget-recent-posts.php
wordpress/wp-includes/widgets/class-wp-widget-rss.php
wordpress/wp-includes/widgets/class-wp-widget-search.php
wordpress/wp-includes/widgets/class-wp-widget-tag-cloud.php
wordpress/wp-includes/widgets/class-wp-widget-text.php
wordpress/wp-includes/widgets.php
wordpress/wp-includes/wp-db.php
wordpress/wp-includes/wp-diff.php
wordpress/wp-links-opml.php
wordpress/wp-load.php
wordpress/wp-login.php
wordpress/wp-mail.php
wordpress/wp-settings.php
wordpress/wp-signup.php
wordpress/wp-trackback.php
```

Di extract pakai command tar -xvf latest.tar.gz

```
root@ip-172-31-38-94: /tmp
root@ip-172-31-38-94:/tmp# sudo mv wordpress /var/www/html/
root@ip-172-31-38-94:/tmp#
```

Pindahkan folder wordpress ke /var/www/html

```
root@ip-172-31-38-94: /var/www/html
root@ip-172-31-38-94:/var/www/html# nano /etc/nginx/sites-available/default
```

Edit default nya jadi seperti di foto bawah ini

```
root@ip-172-31-38-94: /var/www/html
GNU nano 7.2 /etc/nginx/sites-available/default *
# available underneath a path with that package name, such as /drupal8.
#
# Please see /usr/share/doc/nginx-doc/examples/ for more detailed examples.
##
# Default server configuration
#
server {
    listen 80 default_server;
    listen [::]:80 default_server;

    root /var/www/html;

    # Add index.php to the list if you are using PHP
    index index.html index.php index.nginx-debian.html;

    server_name _;

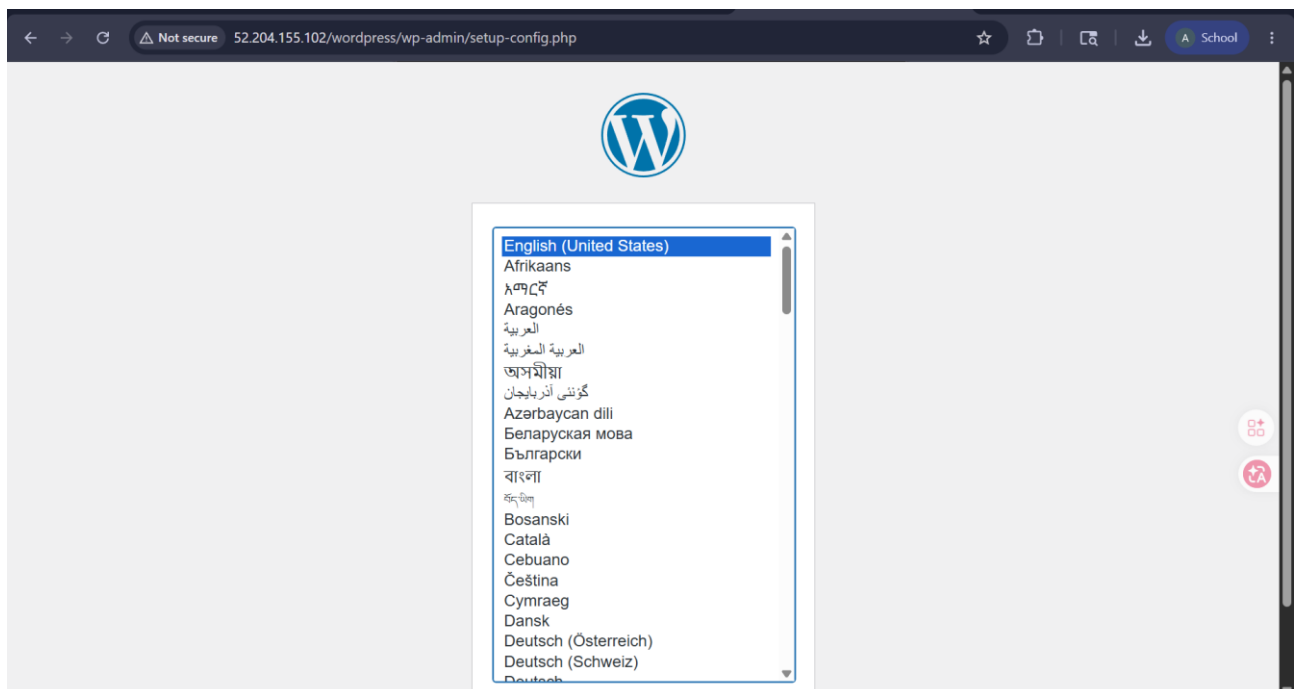
    location / {
        # First attempt to serve request as file, then
        # as directory, then fall back to displaying a 404.
        try_files $uri $uri/ =404;
    }

    # pass PHP scripts to FastCGI server
    #
    location ~ \.php$ {
        include snippets/fastcgi-php.conf;

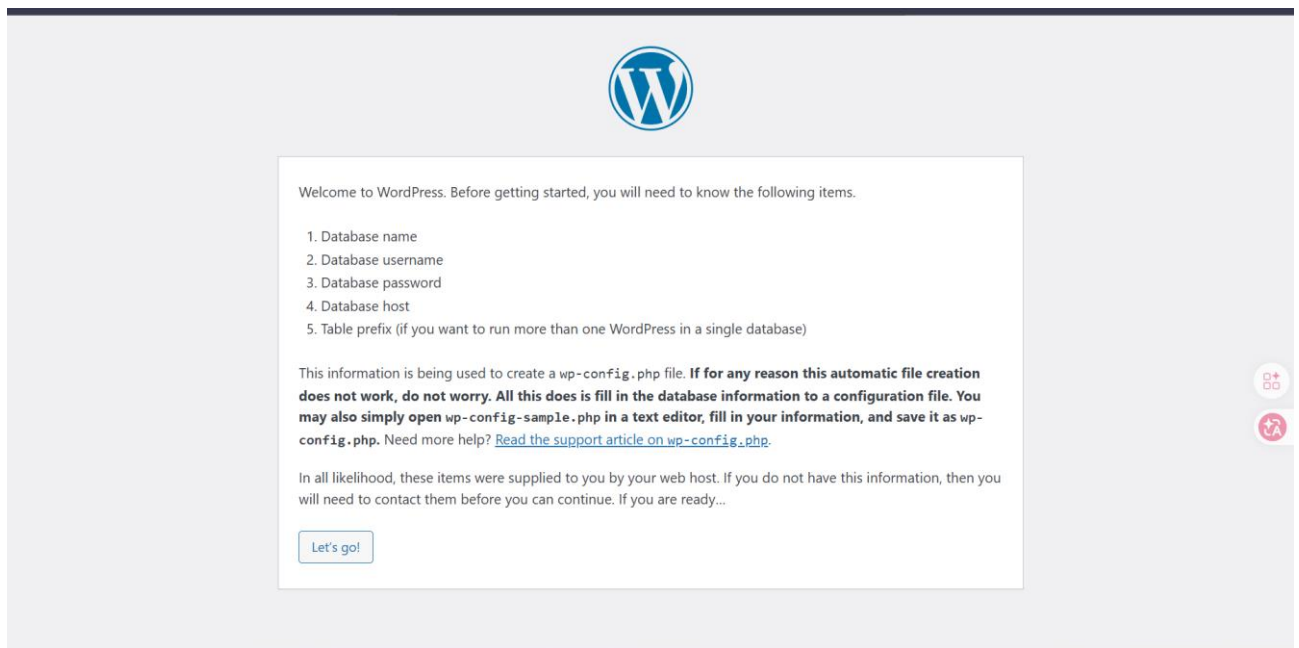
        # With php-fpm (or other unix sockets):
        fastcgi_pass unix:/run/php/php8.3-fpm.sock;
        # With php-cgi (or other tcp sockets):
    }
}
```

Di tambahkan index.php di dan uncomment location ~ \.php\$ {

Setelah itu di restart nginx nya dan dibuka wordpress nya pakai ip_public_ec2/wordpress



Pilih bahasa, default nya English



Klik Let's go

The image shows the WordPress database configuration screen in a web browser. The browser's address bar shows the URL: 52.204.155.102/wordpress/wp-admin/setup-config.php?step=1. The page features the WordPress logo at the top. Below it, a white box contains the text: "Below you should enter your database connection details. If you are not sure about these, contact your host." The form has five fields: "Database Name" with the value "dbwpAndhika", "Username" with the value "userAndhika", "Password" with the value "asattkj2026" and a "Hide" button, "Database Host" with the value "localhost", and "Table Prefix" with the value "wp_". Each field has a descriptive text below it. At the bottom of the form is a "Submit" button.

Isi nama database dan username nya yang di buat tadi, yang password nya juga pakai password database dbwpName, sedangkan database host nya localhost saja

Information needed

Please provide the following information. Do not worry, you can always change these settings later.

Site Title: Wordpress_ASAT-Andhika

Username: dhiksn

Usernames can have only alphanumeric characters, spaces, underscores, hyphens, periods, and the @ symbol.

Password: ... Very weak

Important: You will need this password to log in. Please store it in a secure location.

Confirm Password: ☒ Confirm use of weak password

Your Email: andhikarafi@smkwikrama.sch.id

Double-check your email address before continuing.

Search engine visibility: ☐ Discourage search engines from indexing this site

It is up to search engines to honor this request.

[Install WordPress](#)

Isi site title nya jadi Wordpress_ASAT-Name, untuk username nya bebas soalnya buat login ke wp-admin nya, password nya juga bebas, kalo saya pakai password 123.

Success!

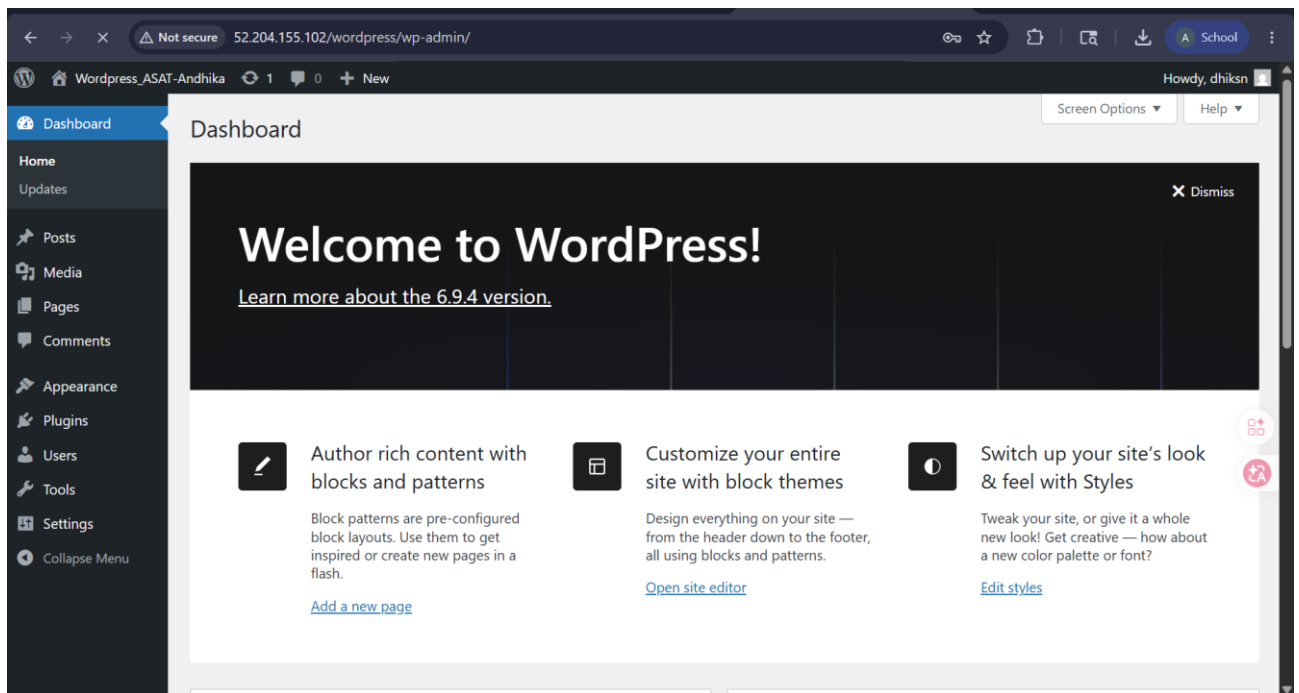
WordPress has been installed. Thank you, and enjoy!

Username: dhiksn

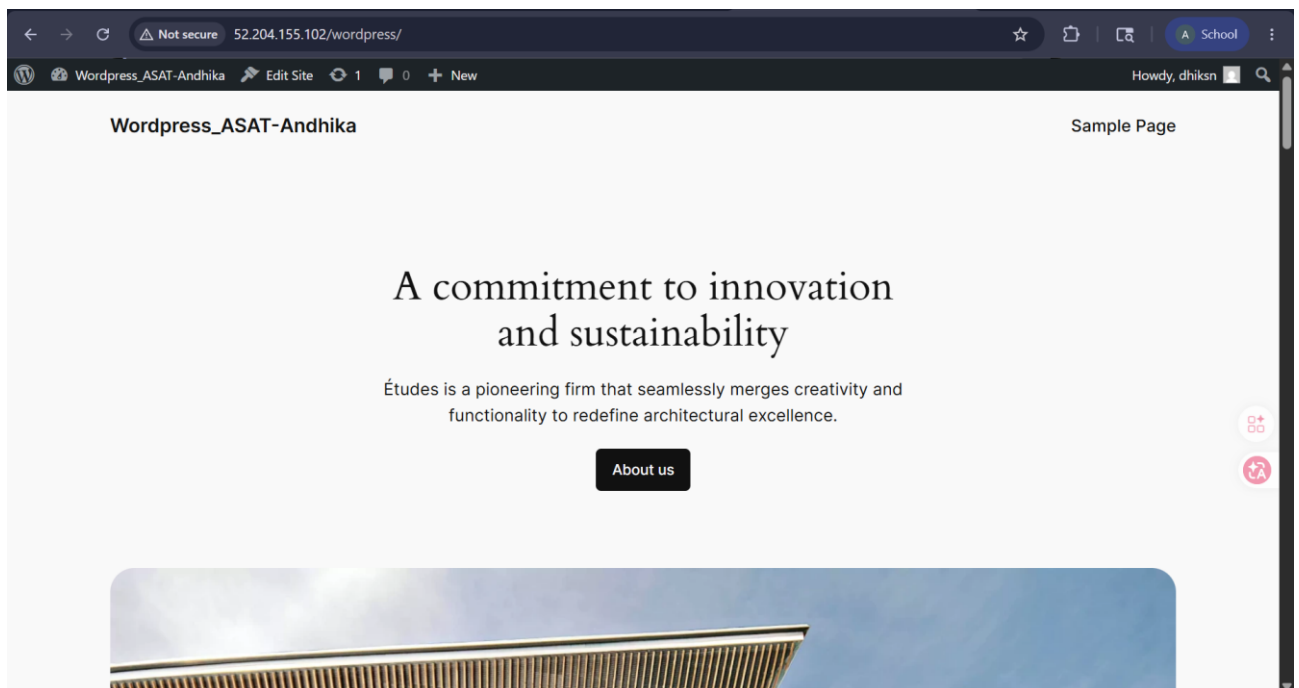
Password: Your chosen password.

[Log In](#)

Klik log in dan nanti masuk ke wp-login



Ini dashboard wordpress



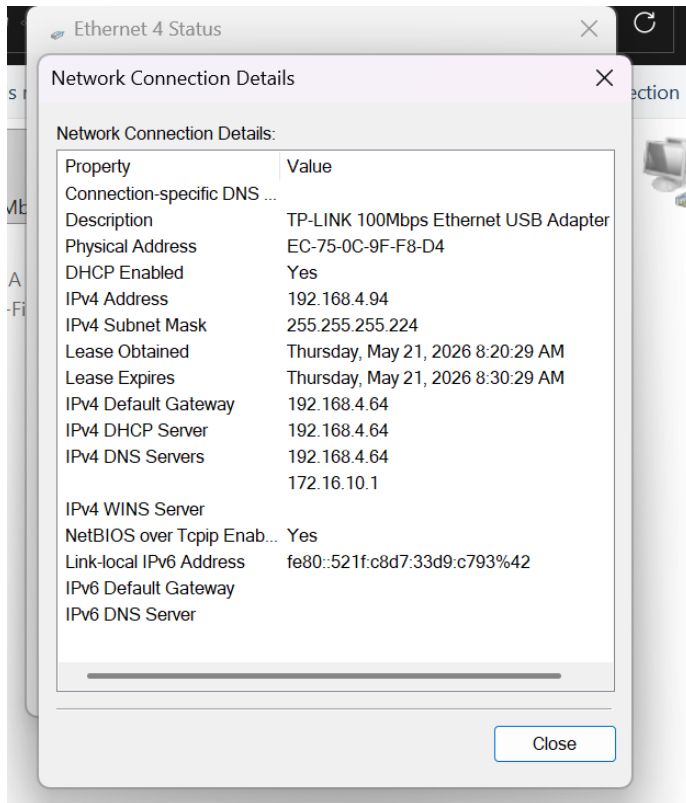
Cek tampilan wordpress

Buatkan alur dokumentasi sesuai dengan apa yang dikerjakan, seluruh rangkaian proses didokumentasikan dan diberikan keterangan/narasi yang jelas.

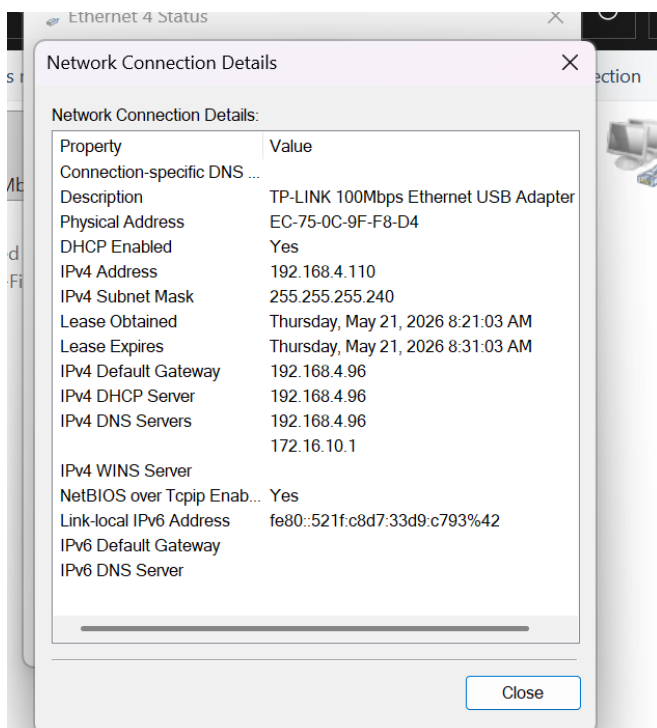
PENGUJIAN HASIL INSTALASI DAN KONFIGURASI

1. PENGUJIAN DHCP SERVER ROUTER

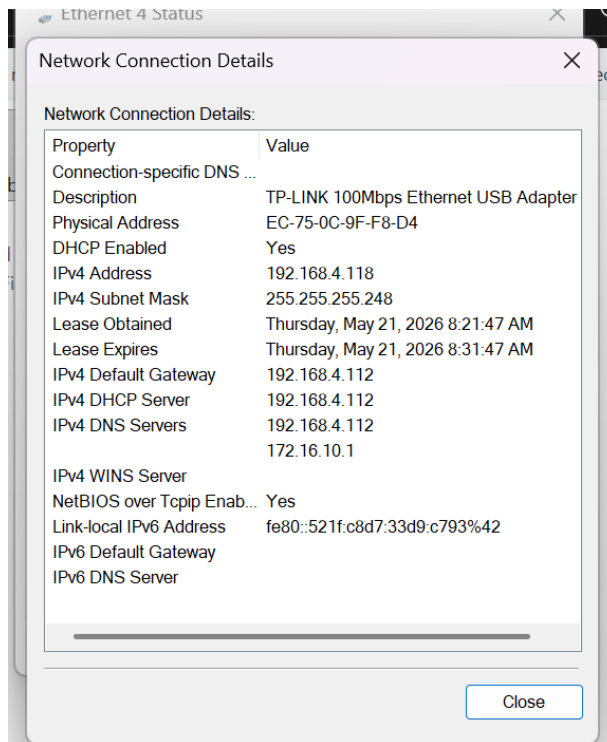
HRD-Room



Marketing-Room

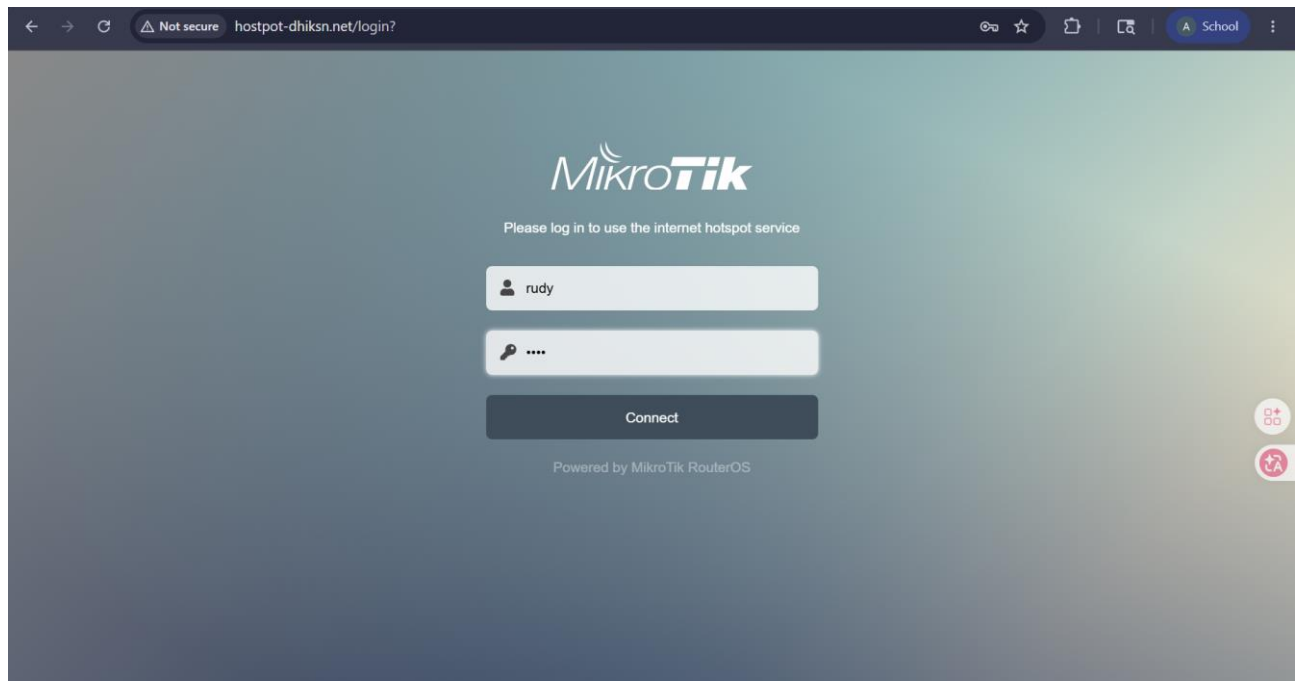


NOC-Room



2. PENGUJIAN HOTSPOT LOGIN

Rudy (CEO)



Login page

← → fast.com

English (US) Privacy

FAST

Your Internet speed is

2.9 Mbps

Show more info

?

f

t

POWERED BY **NETFLIX**

Lakukan speedtest

Network Connection Details

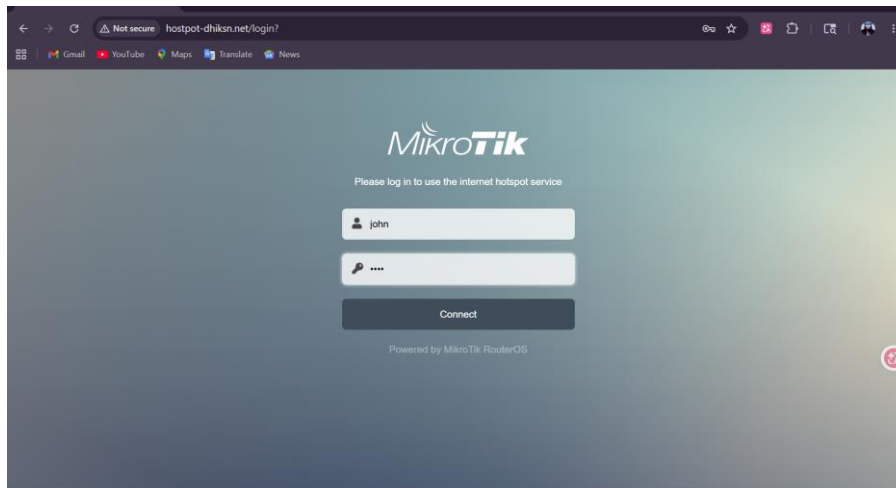
Network Connection Details:

Property	Value
Connection-specific DNS ...	
Description	MediaTek Wi-Fi 6E MT7902 Wireless LAN
Physical Address	A8-41-F4-09-1F-2A
DHCP Enabled	Yes
IPv4 Address	192.168.4.61
IPv4 Subnet Mask	255.255.255.192
Lease Obtained	Wednesday, May 20, 2026 3:46:08 PM
Lease Expires	Wednesday, May 20, 2026 4:46:08 PM
IPv4 Default Gateway	192.168.4.1
IPv4 DHCP Server	192.168.4.1
IPv4 DNS Server	192.168.4.1
IPv4 WINS Server	
NetBIOS over Tcpip Enab...	Yes
Link-local IPv6 Address	fe80::b8bd:b85c:81cf:aedd%17
IPv6 Default Gateway	
IPv6 DNS Server	

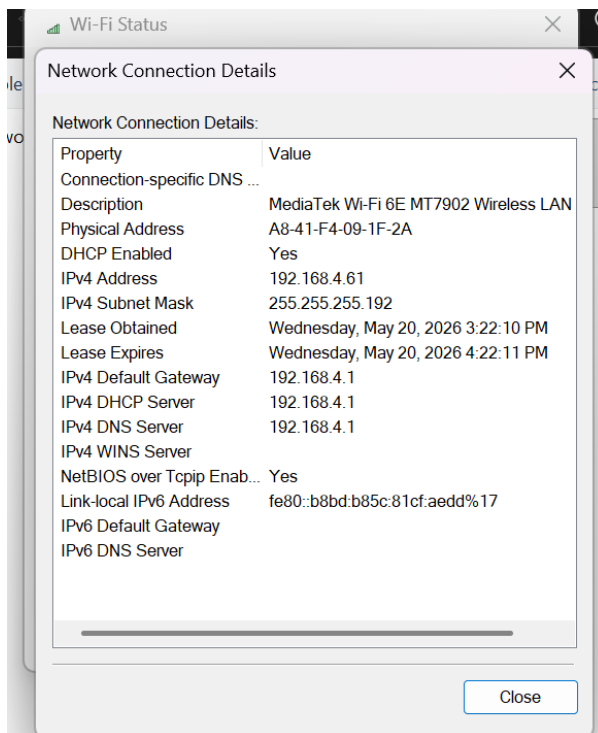
Close

Ip address client

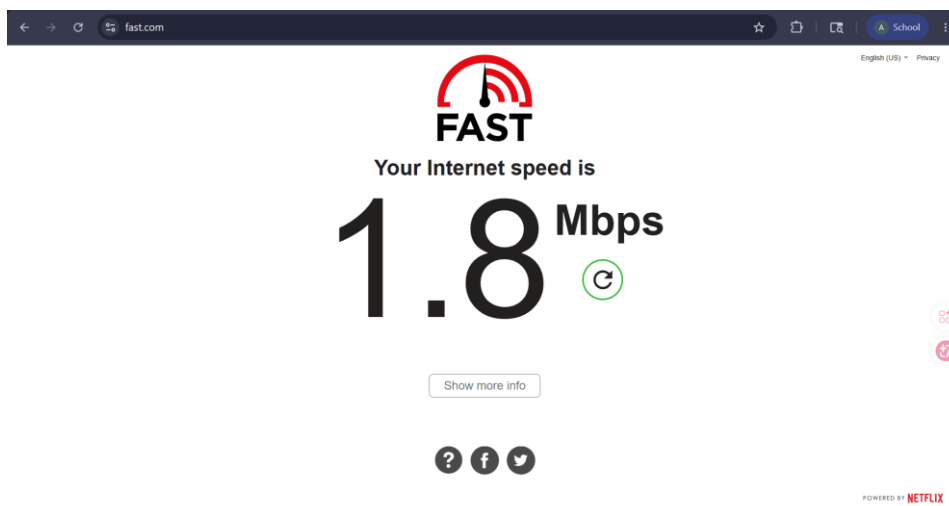
John (Manager)



Login page

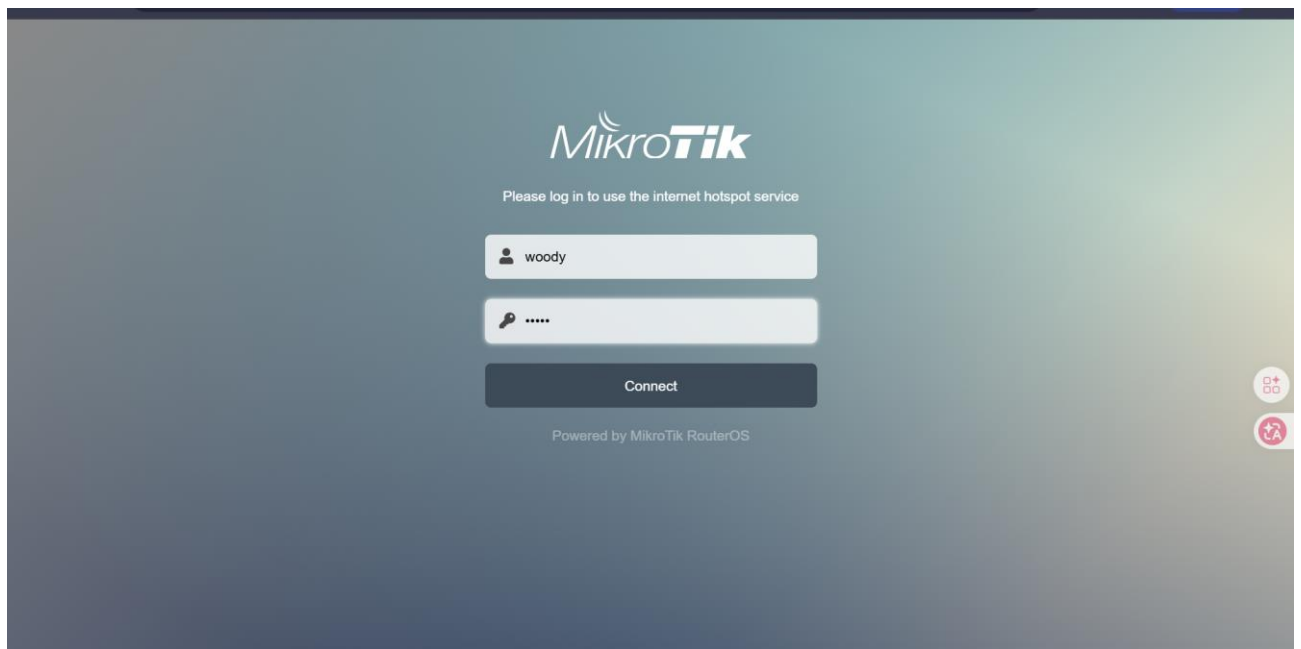


IP Address client

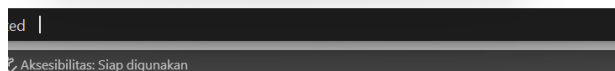
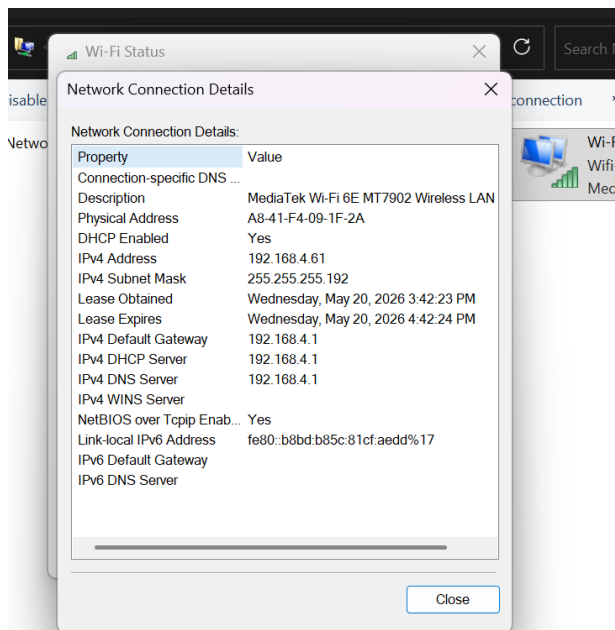


Lakukan speedtest

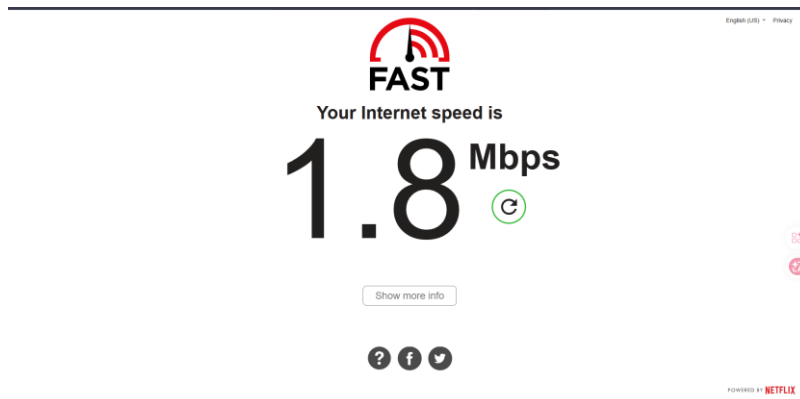
Woody (Staf)



Login page



Ip address client



Speedtest

3. PENGUJIAN KONEKSI INTERNET

```
PS C:\Users\andhika> ping 8.8.8.8

Pinging 8.8.8.8 with 32 bytes of data:
Reply from 8.8.8.8: bytes=32 time=3ms TTL=115
Reply from 8.8.8.8: bytes=32 time=3ms TTL=115
Reply from 8.8.8.8: bytes=32 time=25ms TTL=115
Reply from 8.8.8.8: bytes=32 time=3ms TTL=115

Ping statistics for 8.8.8.8:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 3ms, Maximum = 25ms, Average = 8ms
PS C:\Users\andhika> |
```

4. PENGUJIAN KONEKTIVITAS ANTAR CLIENT

```

Windows PowerShell
PS C:\Users\andhika> ping 192.168.4.64

Pinging 192.168.4.64 with 32 bytes of data:
Reply from 192.168.4.64: bytes=32 time=1ms TTL=64
Reply from 192.168.4.64: bytes=32 time=1ms TTL=64
Reply from 192.168.4.64: bytes=32 time=1ms TTL=64
Reply from 192.168.4.64: bytes=32 time=1ms TTL=64

Ping statistics for 192.168.4.64:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 1ms, Average = 1ms

```

Ping ke IP HRD dari ether4

```

PS C:\Users\andhika> ping 192.168.4.96

Pinging 192.168.4.96 with 32 bytes of data:
Reply from 192.168.4.96: bytes=32 time=1ms TTL=64
Reply from 192.168.4.96: bytes=32 time=1ms TTL=64
Reply from 192.168.4.96: bytes=32 time=1ms TTL=64
Reply from 192.168.4.96: bytes=32 time=1ms TTL=64

Ping statistics for 192.168.4.96:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 1ms, Average = 1ms

```

Ping ke IP Marketing dari ether4

```

PS C:\Users\andhika> ping 192.168.4.112

Pinging 192.168.4.112 with 32 bytes of data:
Reply from 192.168.4.112: bytes=32 time=1ms TTL=64
Reply from 192.168.4.112: bytes=32 time=1ms TTL=64
Reply from 192.168.4.112: bytes=32 time=1ms TTL=64
Reply from 192.168.4.112: bytes=32 time=1ms TTL=64

Ping statistics for 192.168.4.112:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),

```

Ping ke IP NOC dari ether4

```

PS C:\Users\andhika> ping 192.168.4.1

Pinging 192.168.4.1 with 32 bytes of data:
Reply from 192.168.4.1: bytes=32 time=1ms TTL=64
Reply from 192.168.4.1: bytes=32 time=1ms TTL=64
Reply from 192.168.4.1: bytes=32 time=1ms TTL=64
Reply from 192.168.4.1: bytes=32 time=1ms TTL=64

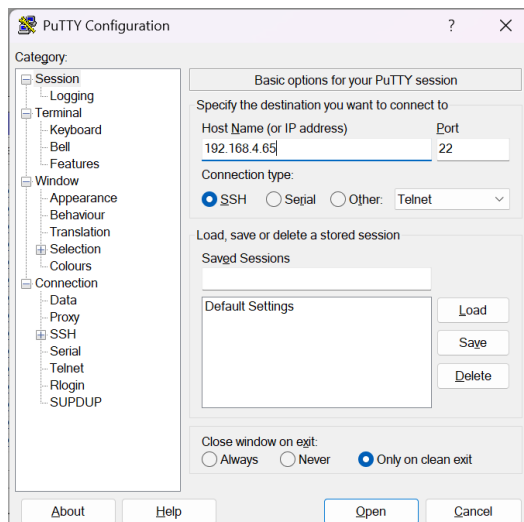
Ping statistics for 192.168.4.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 1ms, Average = 1ms
PS C:\Users\andhika> |

```

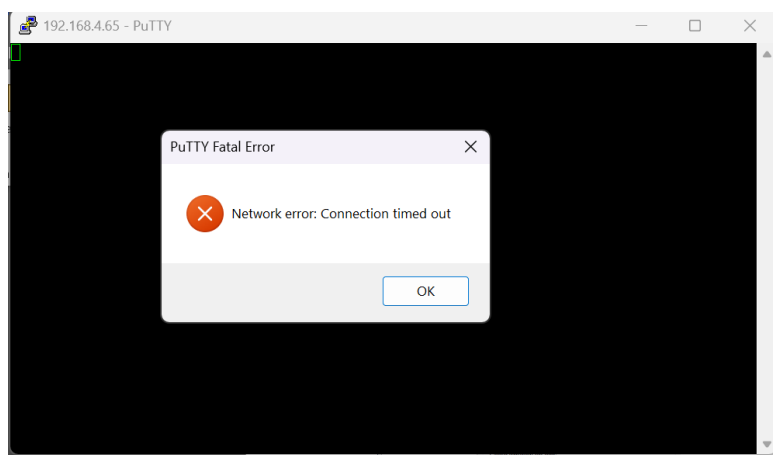
Ping ke IP wlan dari ether4

5. PENGUJIAN BLOK AKSES

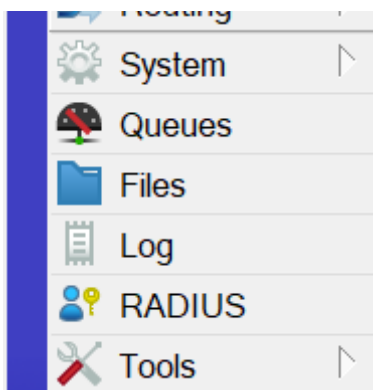
Blokir SSH, HTTP, Telnet dari HRD



Buka putty dan masukan IP HRD-Room, setelah itu klik open



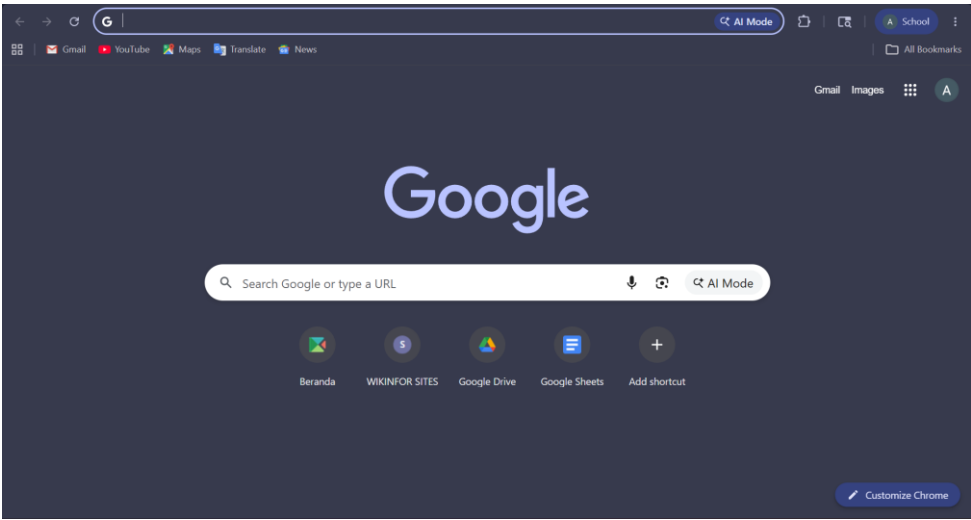
Pasti hasil nya timed out



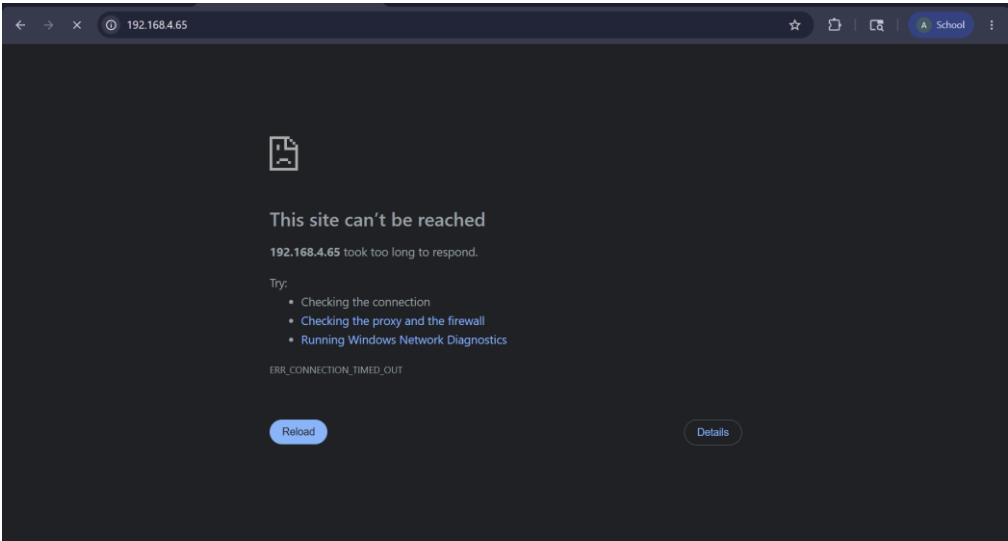
Di winbox pilih tab Log

84	May/21/2026 08:57:15	memory	firewall, info	blokir akses masuk ssh, marketi: in:HRD-Room out:(unknown 0), src-ma...
85	May/21/2026 08:57:16	memory	firewall, info	blokir akses masuk ssh, marketi: in:HRD-Room out:(unknown 0), src-ma...
86	May/21/2026 08:57:18	memory	firewall, info	blokir akses masuk ssh, marketi: in:HRD-Room out:(unknown 0), src-ma...
87	May/21/2026 08:57:22	memory	firewall, info	blokir akses masuk ssh, marketi: in:HRD-Room out:(unknown 0), src-ma...

Ini hasil block ssh, lanjut block HTTP



Ke browser dan ketik IP HRD nya



Lanjut buka log di winbox lagi

#	Time	Buffer	Topics	Message
125	May/21/2026 09:18:20	memory	system, info	filter rule changed by admin
126	May/21/2026 09:18:28	memory	firewall, info	blokir ssh, http, telnet dari h: in:HRD-Room out:(unknown 0), src-mac ec...
127	May/21/2026 09:18:28	memory	firewall, info	blokir ssh, http, telnet dari h: in:HRD-Room out:(unknown 0), src-mac ec...
128	May/21/2026 09:18:28	memory	firewall, info	blokir ssh, http, telnet dari h: in:HRD-Room out:(unknown 0), src-mac ec...
129	May/21/2026 09:18:35	memory	firewall, info	blokir ssh, http, telnet dari h: in:HRD-Room out:(unknown 0), src-mac ec...
130	May/21/2026 09:18:35	memory	firewall, info	blokir ssh, http, telnet dari h: in:HRD-Room out:(unknown 0), src-mac ec...
131	May/21/2026 09:18:36	memory	firewall, info	blokir ssh, http, telnet dari h: in:HRD-Room out:(unknown 0), src-mac ec...
132	May/21/2026 09:18:36	memory	firewall, info	blokir ssh, http, telnet dari h: in:HRD-Room out:(unknown 0), src-mac ec...
133	May/21/2026 09:18:36	memory	firewall, info	blokir ssh, http, telnet dari h: in:HRD-Room out:(unknown 0), src-mac ec...
134	May/21/2026 09:18:37	memory	firewall, info	blokir ssh, http, telnet dari h: in:HRD-Room out:(unknown 0), src-mac ec...
135	May/21/2026 09:18:38	memory	firewall, info	blokir ssh, http, telnet dari h: in:HRD-Room out:(unknown 0), src-mac ec...
136	May/21/2026 09:18:38	memory	firewall, info	blokir ssh, http, telnet dari h: in:HRD-Room out:(unknown 0), src-mac ec...
137	May/21/2026 09:18:39	memory	firewall, info	blokir ssh, http, telnet dari h: in:HRD-Room out:(unknown 0), src-mac ec...
138	May/21/2026 09:18:42	memory	firewall, info	blokir ssh, http, telnet dari h: in:HRD-Room out:(unknown 0), src-mac ec...
139	May/21/2026 09:18:42	memory	firewall, info	blokir ssh, http, telnet dari h: in:HRD-Room out:(unknown 0), src-mac ec...
140	May/21/2026 09:18:43	memory	firewall, info	blokir ssh, http, telnet dari h: in:HRD-Room out:(unknown 0), src-mac ec...

Ini hasil dari block http, lanjut yang block telnet

```
Administrator: Windows PowerShell
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows

PS C:\WINDOWS\system32>
>> Enable-WindowsOptionalFeature -Online -FeatureName TelnetClient
```

Kalo belum ada telnet nya diaktifkan dulu di powershell, pastikan sudah dalam Administrator nya powershell nya, nanti di ketik di powershell nya
“Enable-WindowsOptionalFeature -Online -FeatureName TelnetClient” setelah itu tekan enter

```
PS C:\WINDOWS\system32>
>> Enable-WindowsOptionalFeature -Online -FeatureName TelnetClient

Path      :
Online    : True
RestartNeeded : False

PS C:\WINDOWS\system32> █
```

Pastikan Online nya True nyaa, setelah itu

```
Administrator: Windows PowerShell

PS C:\WINDOWS\system32> telnet 192.168.4.65_
```

Ketik telnet ip_hrd dan tekan enter

```
Administrator: Windows PowerShell

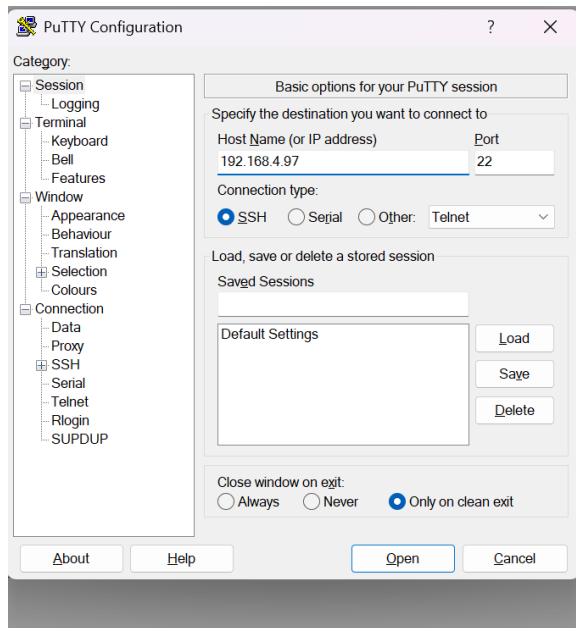
PS C:\WINDOWS\system32> telnet 192.168.4.65
Connecting To 192.168.4.65...Could not open connection to the host, on port 23: Connect failed
PS C:\WINDOWS\system32>
```

Nah lihat connect failed

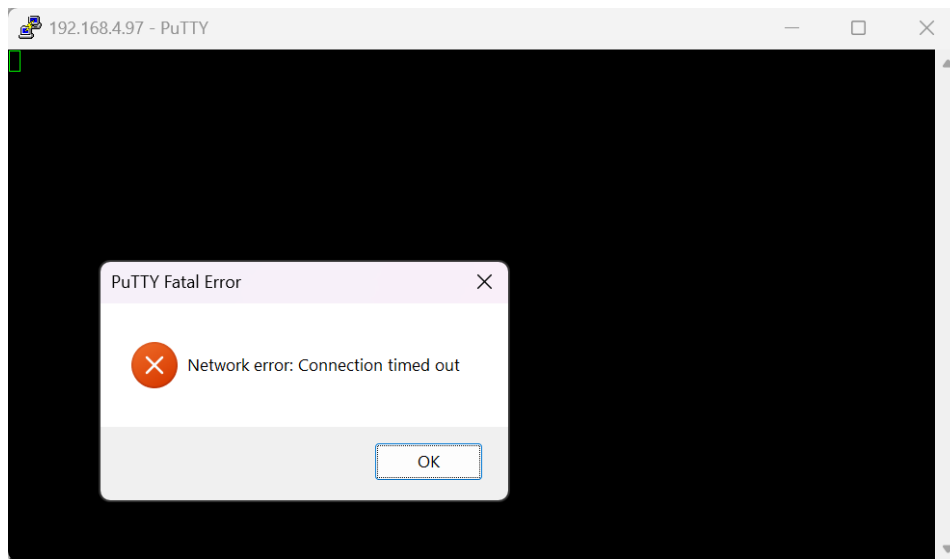
174	May/21/2026 09:28:08	memory	firewall, info	blokir ssh, http, telnet dari h: in:HRD-Room out:(unknown 0), src-mac ec...
175	May/21/2026 09:28:09	memory	firewall, info	blokir ssh, http, telnet dari h: in:HRD-Room out:(unknown 0), src-mac ec...
176	May/21/2026 09:28:11	memory	firewall, info	blokir ssh, http, telnet dari h: in:HRD-Room out:(unknown 0), src-mac ec...
177	May/21/2026 09:28:15	memory	firewall, info	blokir ssh, http, telnet dari h: in:HRD-Room out:(unknown 0), src-mac ec...
178	May/21/2026 09:28:23	memory	firewall, info	blokir ssh, http, telnet dari h: in:HRD-Room out:(unknown 0), src-mac ec...

Di log winbox nya juga muncul info nya

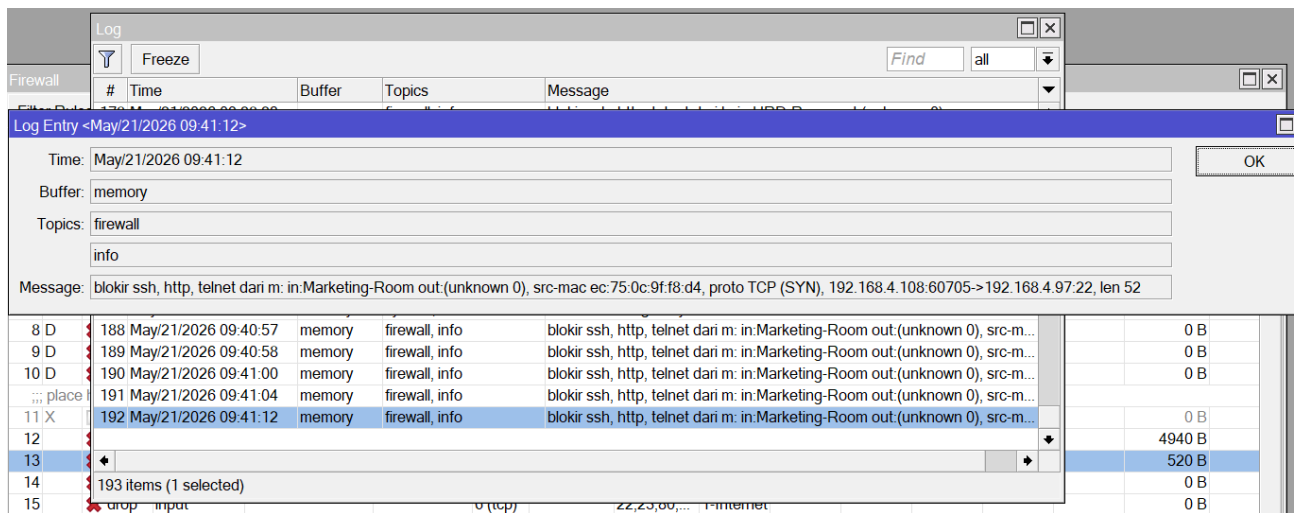
Blokir SSH, HTTP, Telnet dari Marketing



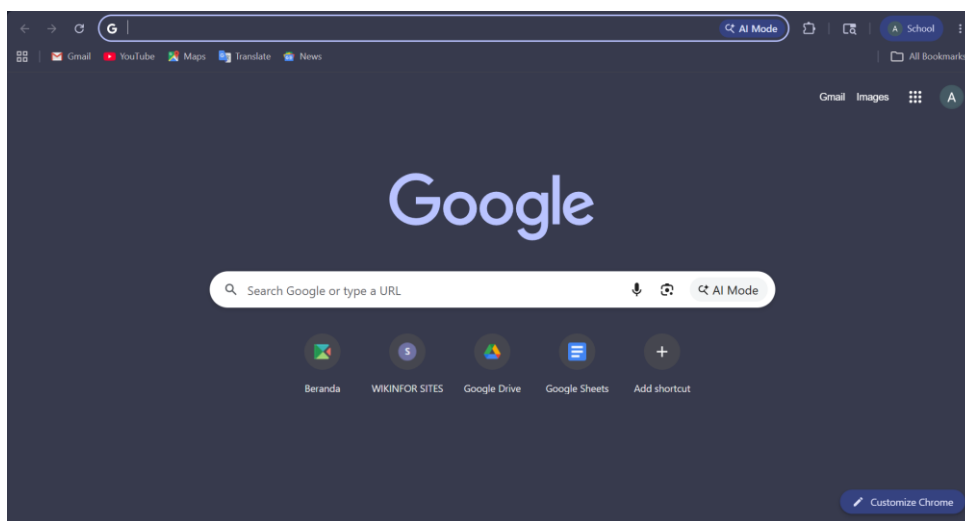
Buka putty dan masukan IP Marketing, setelah itu klik open



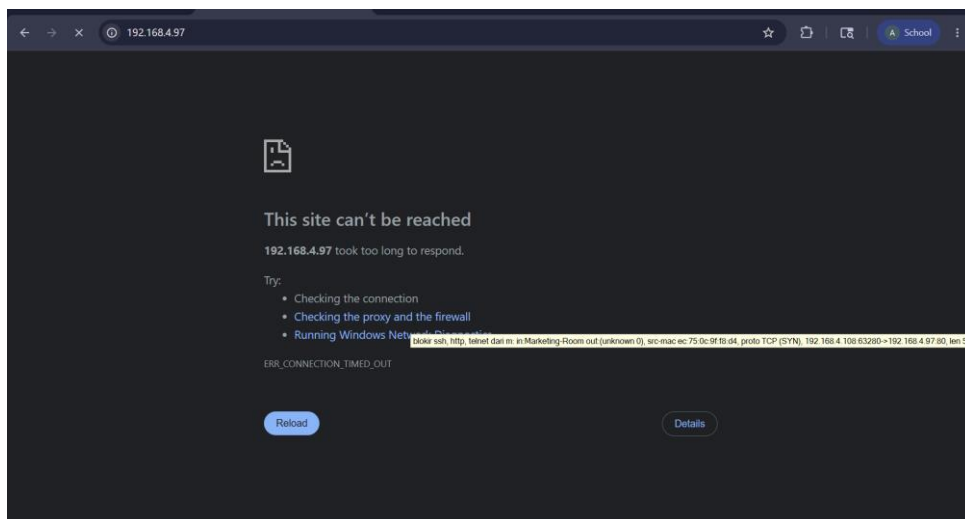
Setelah itu lihat di log winbox nya



Nah masuk info log dari block ssh dari marketing, lanjut yang block http



Buka browser, ketik ip marketing nya, dan tekan enter



Lanjutnya lihat log di winbox

Log Entry <May/21/2026 09:44:59>

Buffer: memory

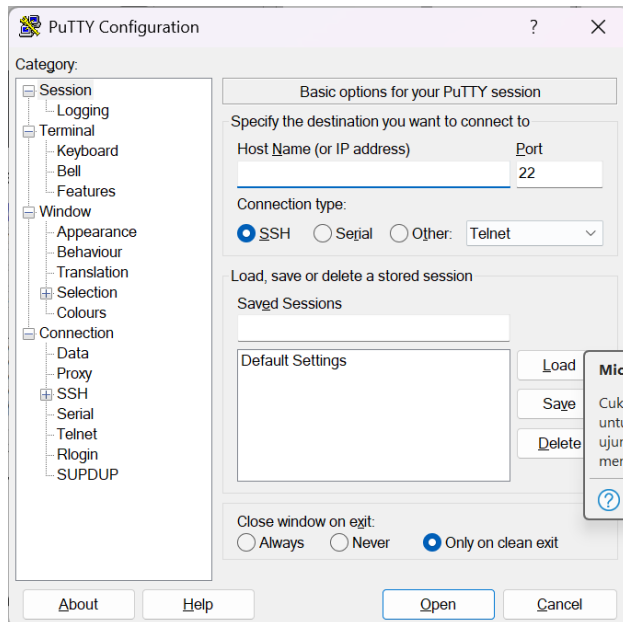
Topics: firewall

info

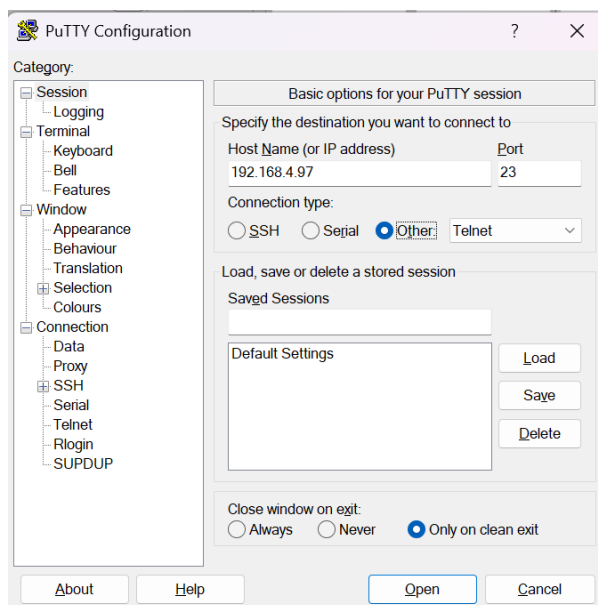
Message: blokir ssh, http, telnet dari m: in:Marketing-Room out:(unknown 0), src-mac ec:75:0c:9f:f8:d4, proto TCP (SYN), 192.168.4.108:62207->192.168.4.97:80, len 52

8 D	233	May/21/2026 09:44:51	memory	firewall, info	blokir ssh, http, telnet dari m: in:Marketing-Room out:(unknown 0), src-m...	0 B	
9 D	234	May/21/2026 09:44:51	memory	firewall, info	blokir ssh, http, telnet dari m: in:Marketing-Room out:(unknown 0), src-m...	0 B	
10 D	235	May/21/2026 09:44:59	memory	firewall, info	blokir ssh, http, telnet dari m: in:Marketing-Room out:(unknown 0), src-m...	0 B	
... place	236	May/21/2026 09:44:59	memory	firewall, info	blokir ssh, http, telnet dari m: in:Marketing-Room out:(unknown 0), src-m...		
11 X	237	May/21/2026 09:44:59	memory	firewall, info	blokir ssh, http, telnet dari m: in:Marketing-Room out:(unknown 0), src-m...	0 B	
12						4940 B	
13						2860 B	
14	238 items (1 selected)						0 B

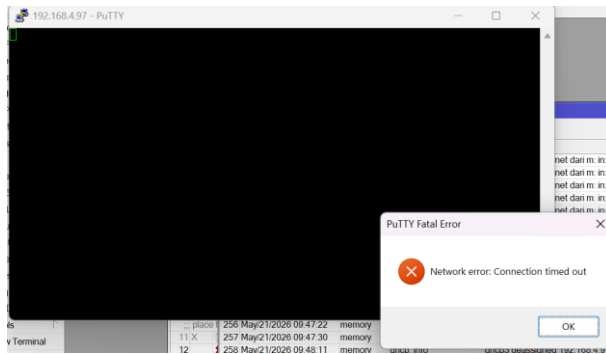
Selanjutnya block telnet pakai ip marketing



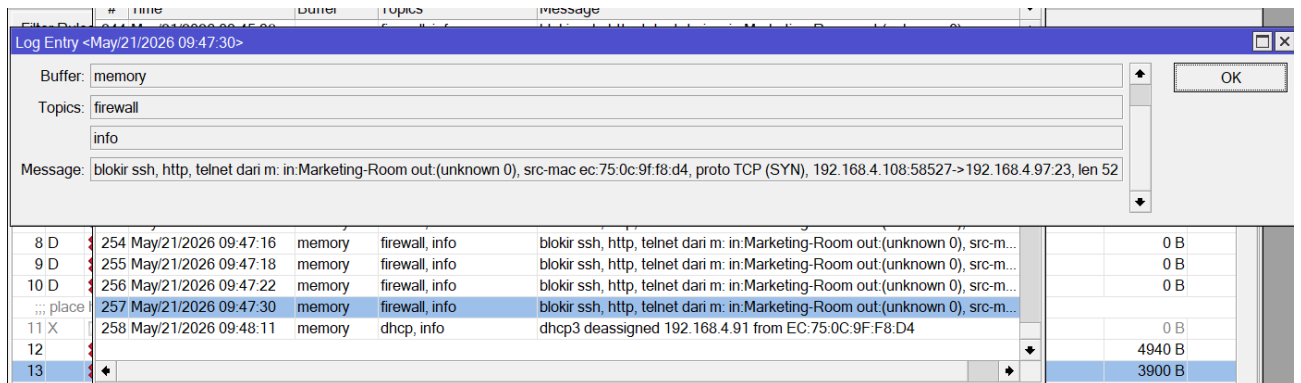
Buka putty, masukan ip marketing, dan ganti type jadi other(telnet)



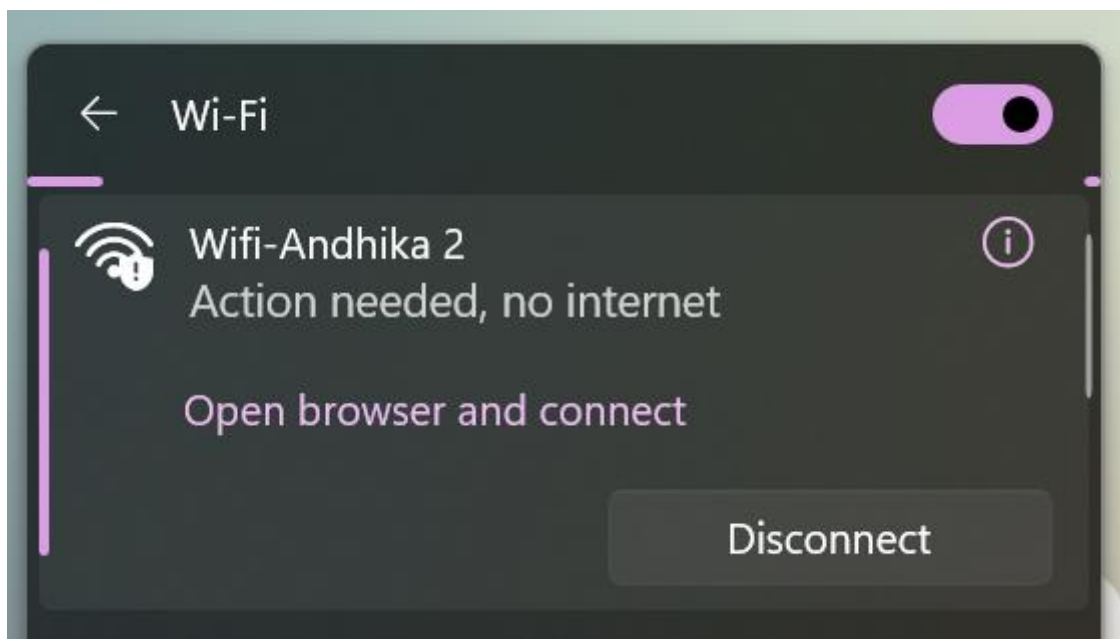
Seperti ini, setelah itu klik open



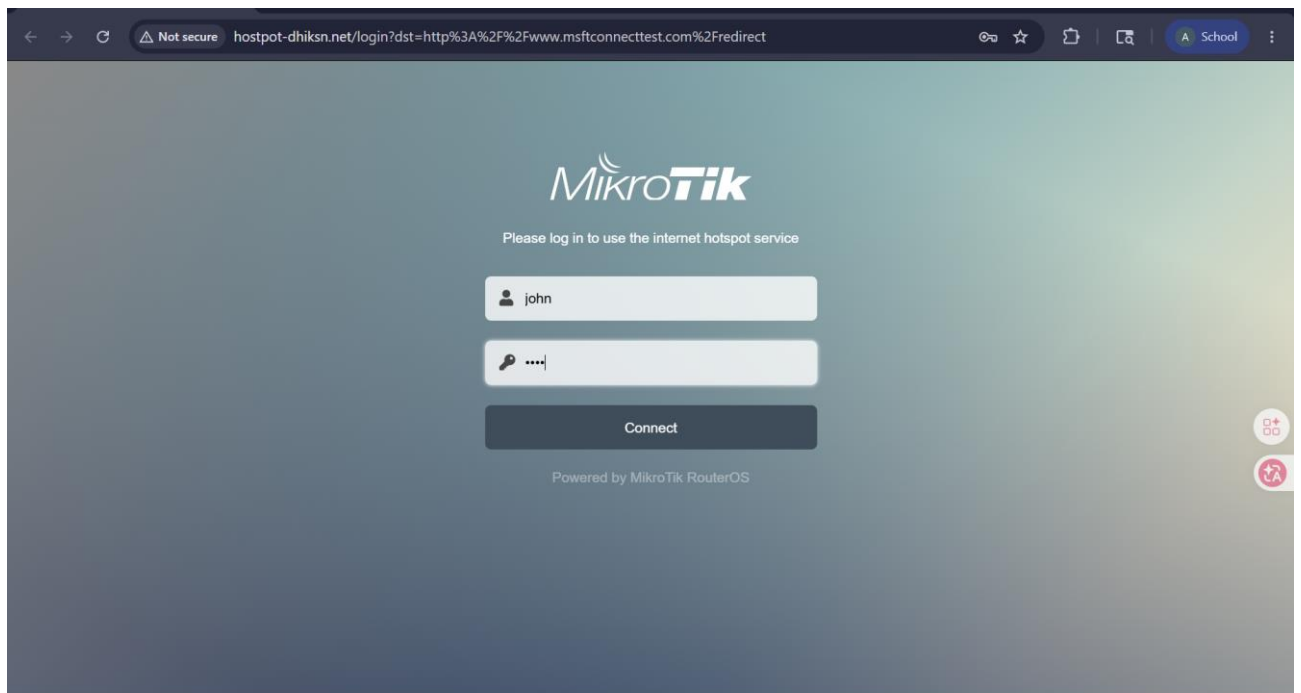
Selanjutnya masuk ke winbox dan pilih tab log



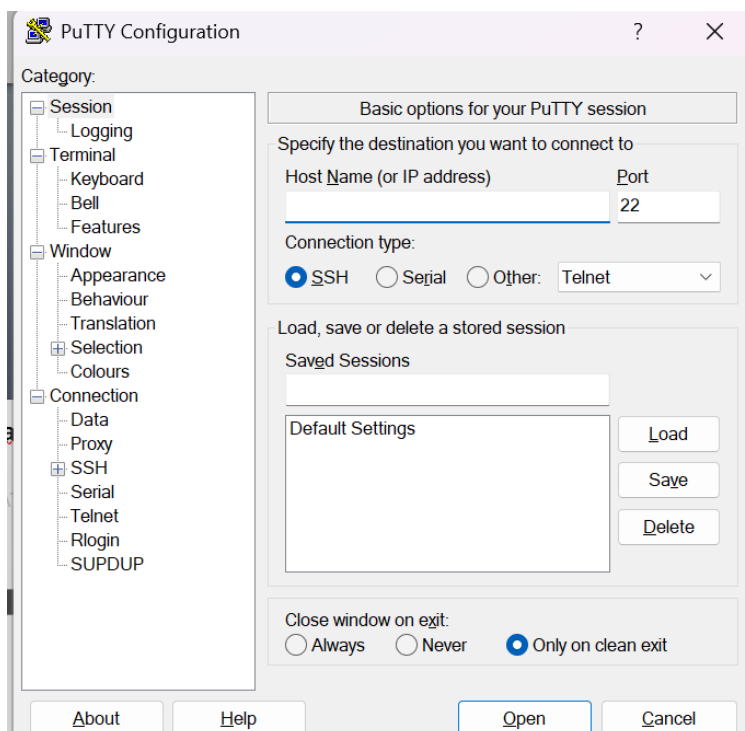
Ini hasil block telnet pakai ip marketing, selanjutnya block ssh,http,telnet pakai wlan



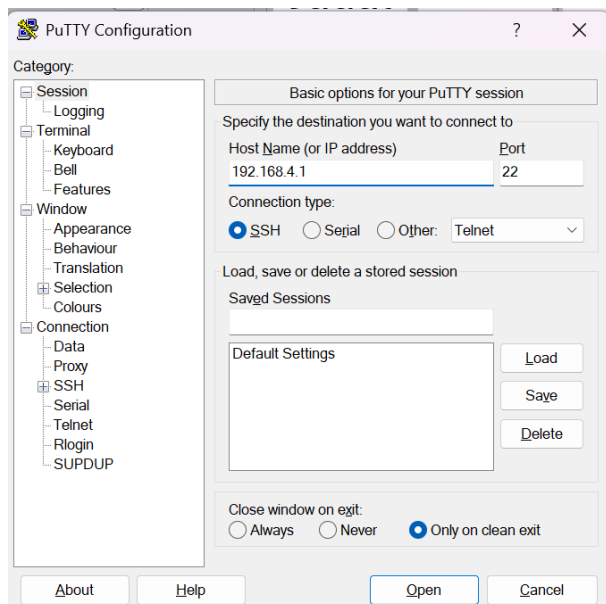
Connect dulu ke wifi nya



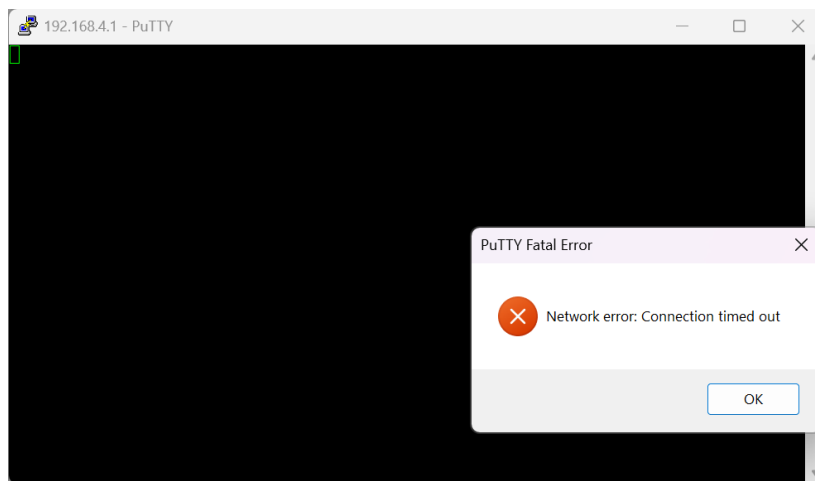
Login nya pakai user john saja, setelah itu klik connect



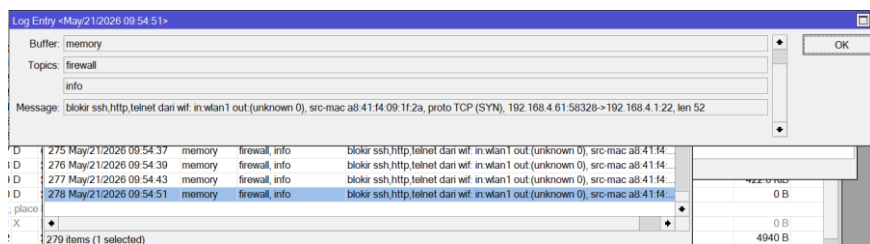
Kita mau block ssh, jadi buka putty terlebih dahulu dan ketik ip wlan nya



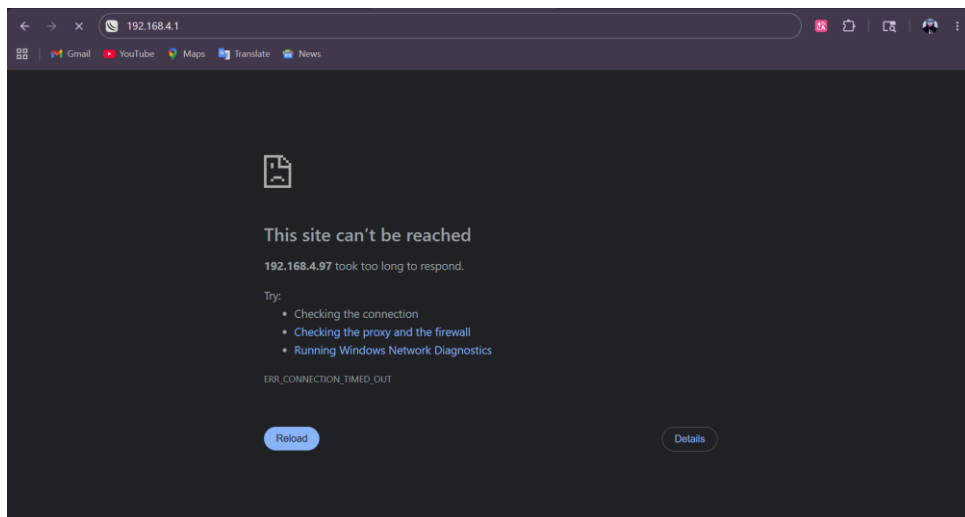
Klik open



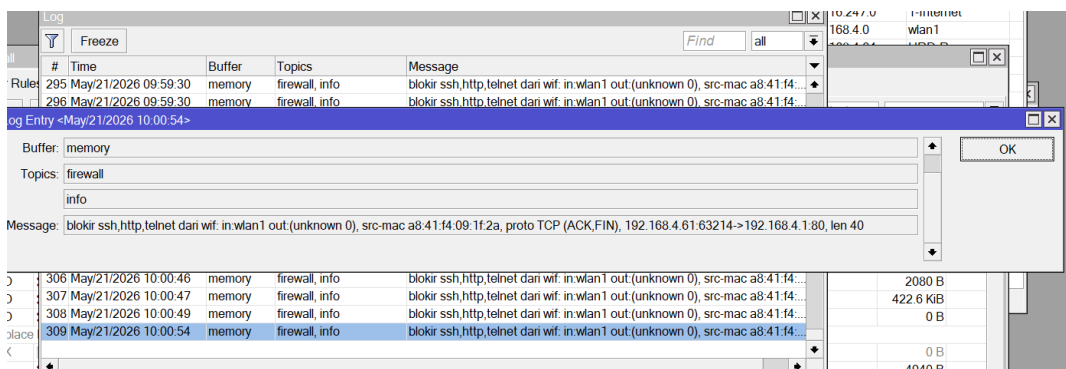
Pasti seperti ini, setelah itu buka winbox dan lihat lognya



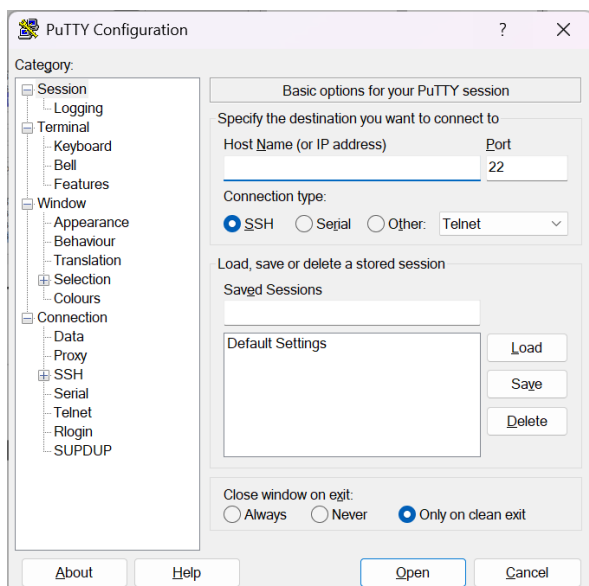
Selanjutnya gw mau block http pakai ip wlan



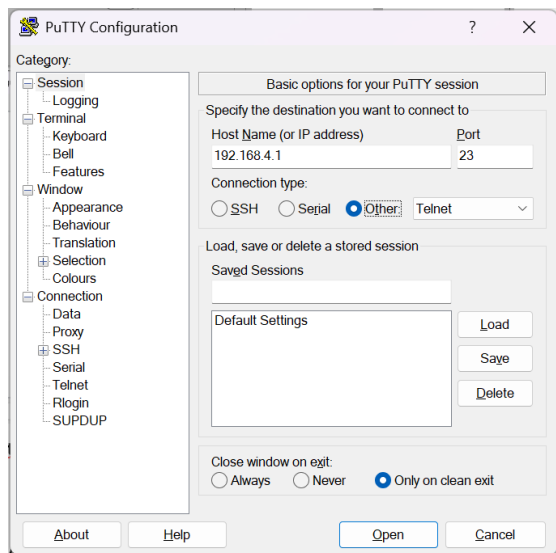
Buka browser dan ketik ip wlan nya



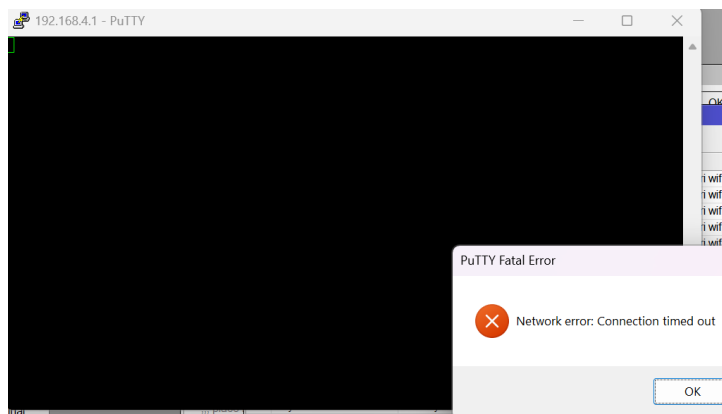
Selanjutnya saya mau block telnet pakai ip wlan



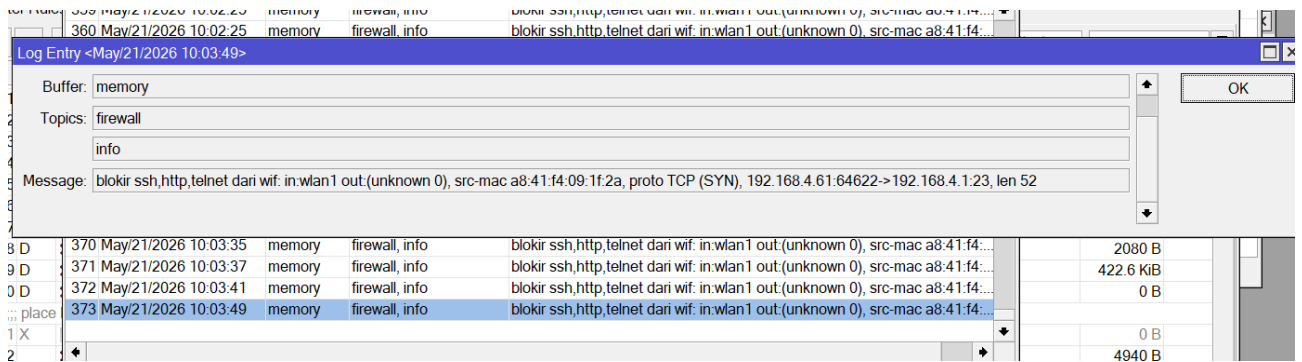
Buka putty, ketik ip wlan dan ganti connection type nya jadi other (telnet)



Seperti itu dan klik open



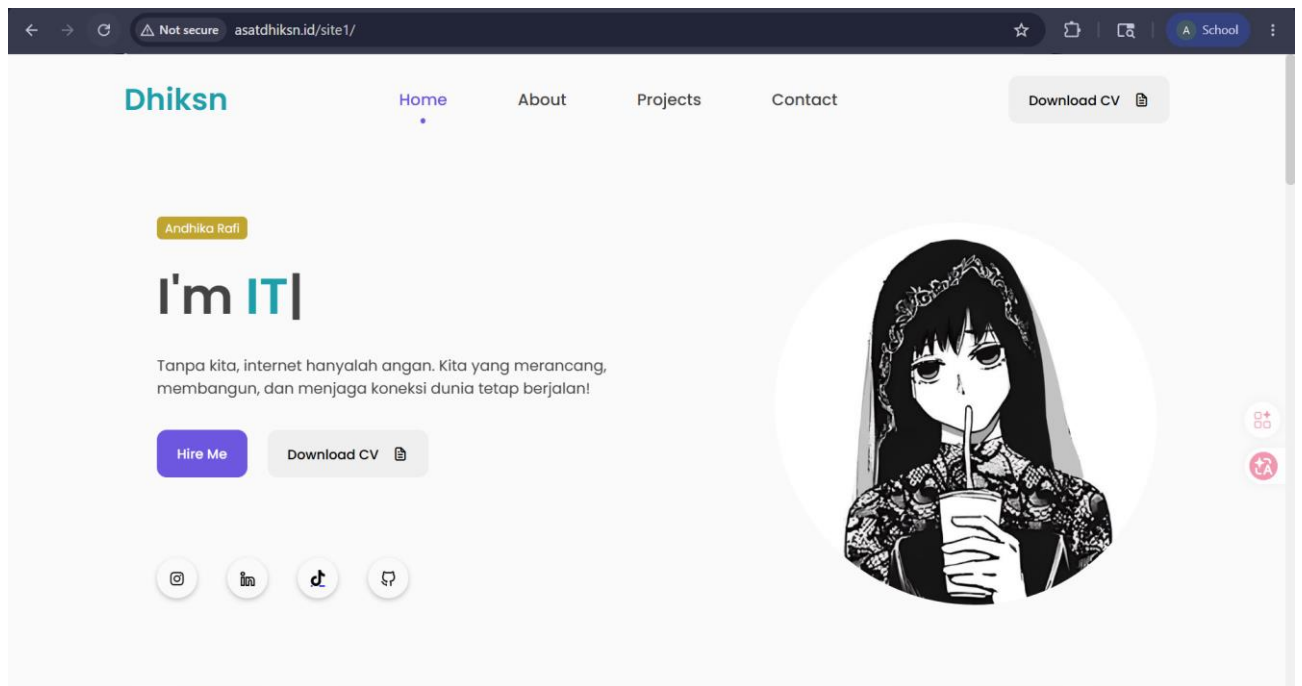
Lanjutnya lihat log di winbox



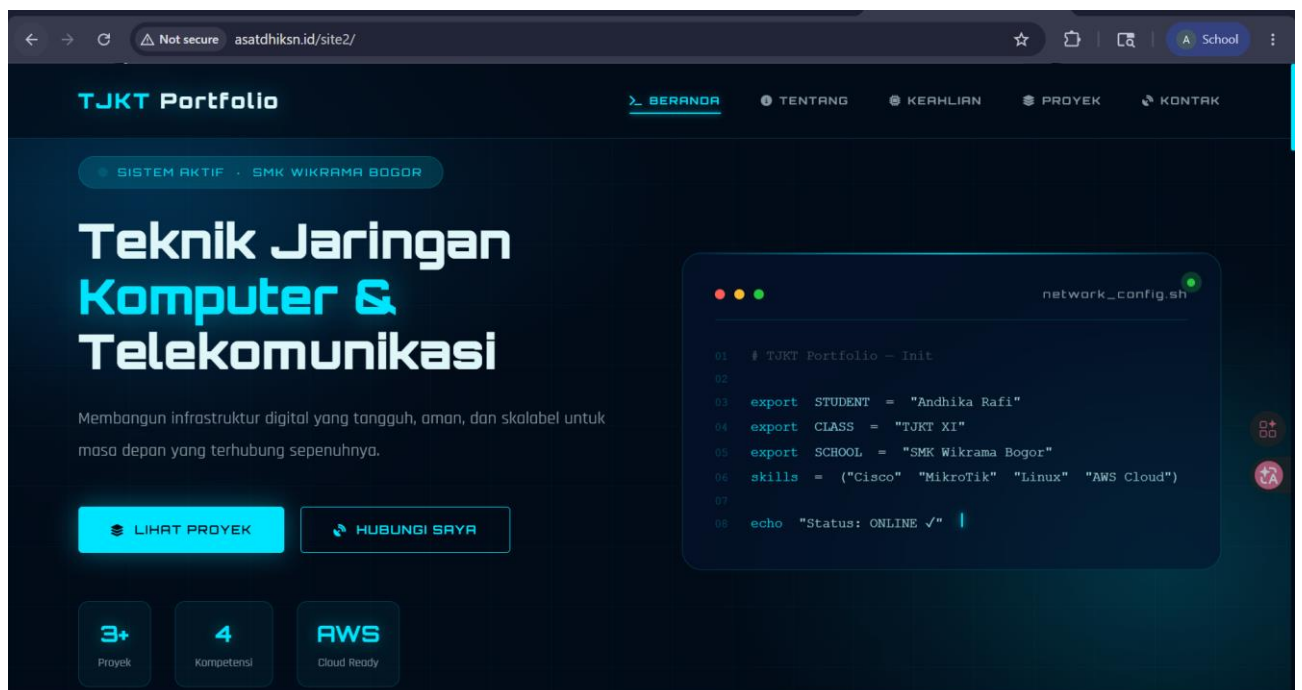
Ini hasil block telnet dari wlan

6. PENGUJIAN AWS

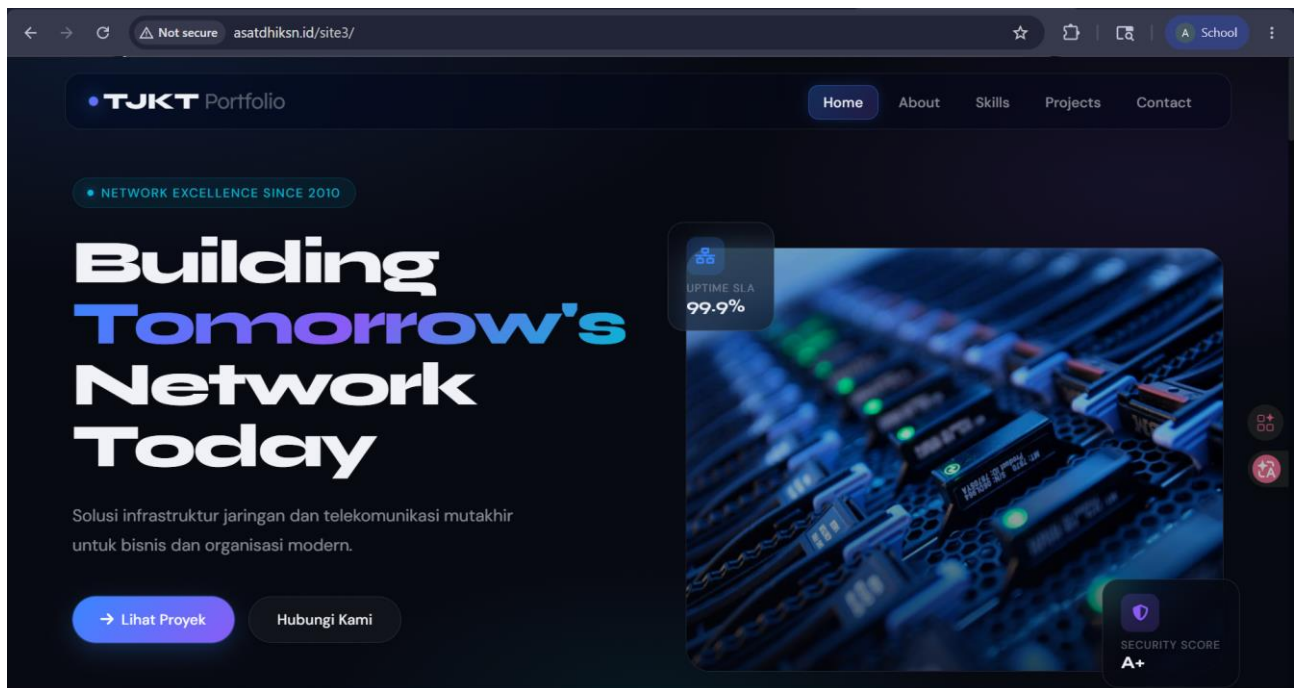
Site1



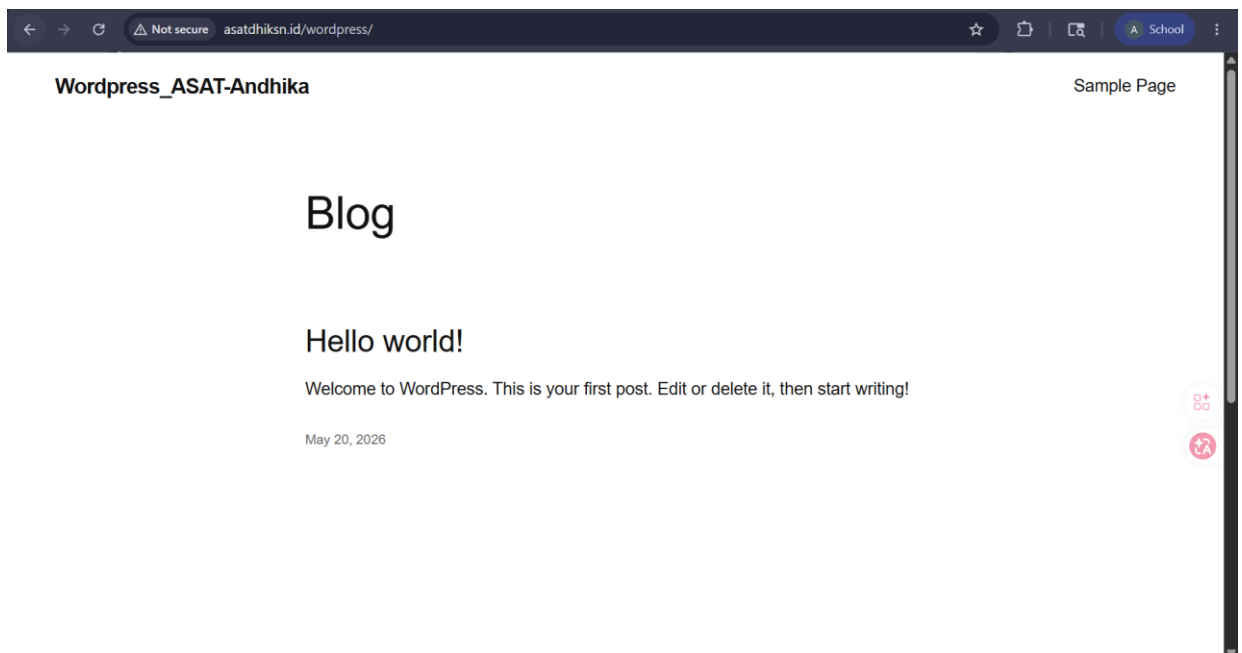
Site2



Site3



Wordpress



Lakukan pengujian sedetail mungkin untuk mendapatkan hasil ujicoba yang lengkap dan membuktikan bahwa semua hasil konfigurasi sudah berhasil. Dokumentasikan dan berikan keterangan/narasi yang jelas.

PENUTUP

1. KESIMPULAN

Berdasarkan hasil pelaksanaan proyek Asesmen Sumatif Akhir Tahun (ASAT) Praktik Konsentrasi Keahlian TJKT Tahun 2026, dapat disimpulkan bahwa seluruh tahapan perancangan, implementasi, konfigurasi, hingga pengujian jaringan dan layanan cloud computing telah berhasil dilakukan dengan baik.

Proyek ini berhasil mengimplementasikan jaringan komputer menggunakan perangkat Mikrotik yang meliputi konfigurasi IP Address, DHCP Server, DNS, VLAN, wireless, hotspot login, firewall, NAT, serta manajemen bandwidth sesuai dengan kebutuhan perusahaan. Selain itu, implementasi cloud server menggunakan Amazon Web Services (AWS) juga berhasil dilakukan melalui pembuatan EC2 Instance, konfigurasi security group, deploy website, instalasi database MariaDB, serta instalasi WordPress.

Hasil pengujian menunjukkan bahwa:

- Seluruh client dapat memperoleh IP Address secara otomatis.
- Jaringan antar divisi dapat saling terhubung sesuai aturan yang telah dibuat.
- Layanan hotspot berjalan dengan baik menggunakan sistem login user.
- Website yang di-deploy pada AWS dapat diakses menggunakan DNS lokal.
- Firewall dan keamanan layanan berhasil diterapkan sesuai kebutuhan.
- Server dapat diakses menggunakan remote SSH dengan konfigurasi keamanan tambahan.

Melalui proyek ini, penulis memperoleh pengalaman dan pemahaman mengenai proses pembangunan infrastruktur jaringan komputer dan cloud server secara langsung mulai dari tahap analisis kebutuhan hingga tahap pengujian sistem. Selain meningkatkan kemampuan teknis, proyek ini juga melatih kemampuan analisis, troubleshooting, dokumentasi, dan penerapan keamanan jaringan sesuai kebutuhan dunia kerja dan industri.

2. Saran

Agar sistem jaringan dan layanan server yang telah dibangun dapat menjadi lebih baik dan optimal, terdapat beberapa saran yang dapat dipertimbangkan, antara lain:

- Menambahkan sistem monitoring jaringan untuk memantau trafik dan performa jaringan secara real-time.
- Melakukan backup konfigurasi router dan backup database secara berkala untuk menghindari kehilangan data.
- Menambahkan sistem keamanan jaringan yang lebih lanjut seperti IDS/IPS dan VPN.
- Menggunakan domain publik dan sertifikat SSL agar website lebih aman dan profesional.
- Menambahkan manajemen user dan autentikasi yang lebih baik pada layanan hotspot dan server.
- Melakukan pemeliharaan dan update sistem secara berkala untuk menjaga keamanan dan stabilitas layanan.

Dengan adanya pengembangan tersebut, diharapkan sistem jaringan yang telah dibangun dapat menjadi lebih aman, stabil, dan mampu memenuhi kebutuhan pengguna secara lebih optimal di masa mendatang.